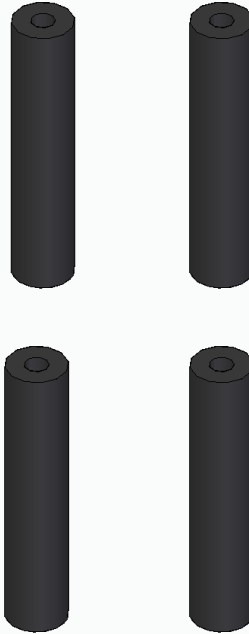
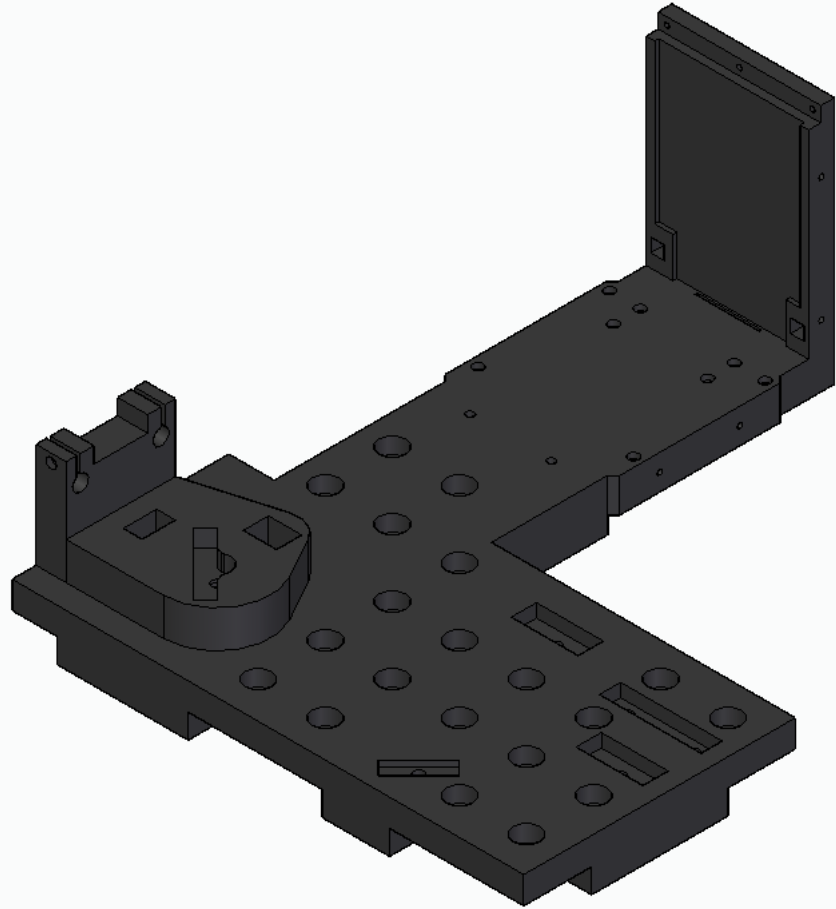


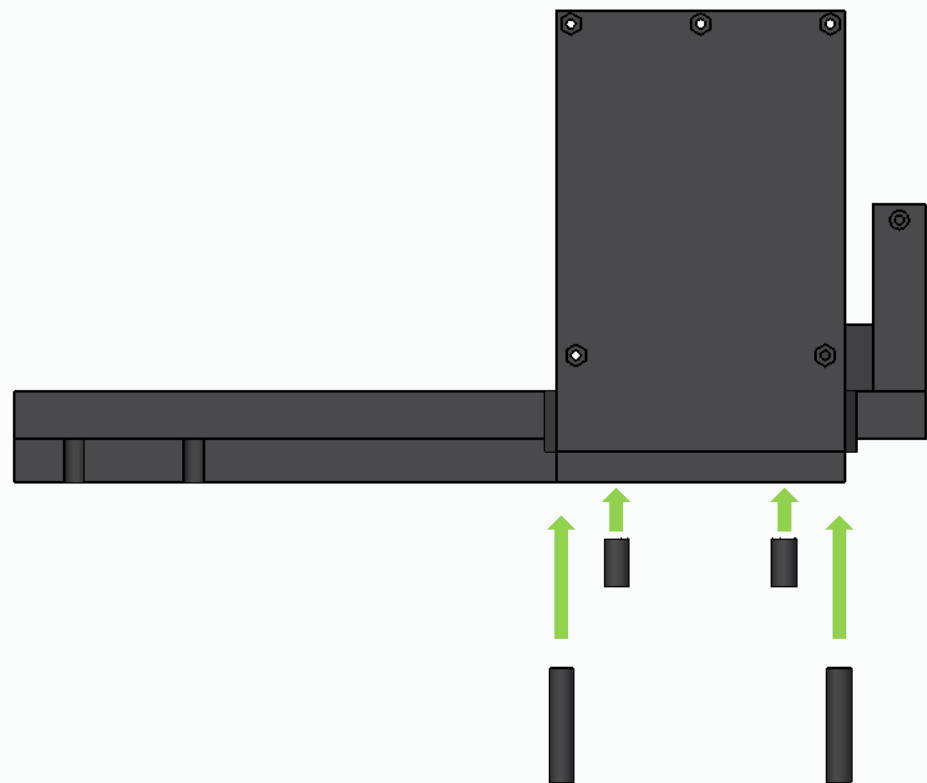
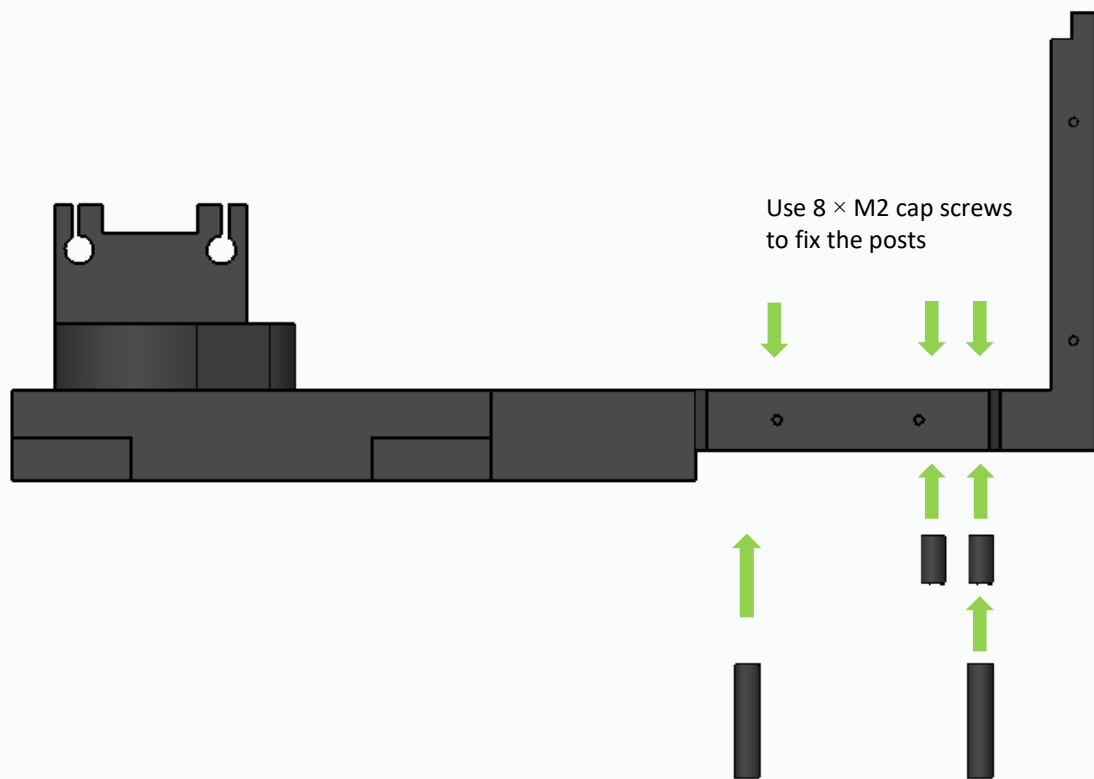
Fixation-stimulus channel (FiSt)

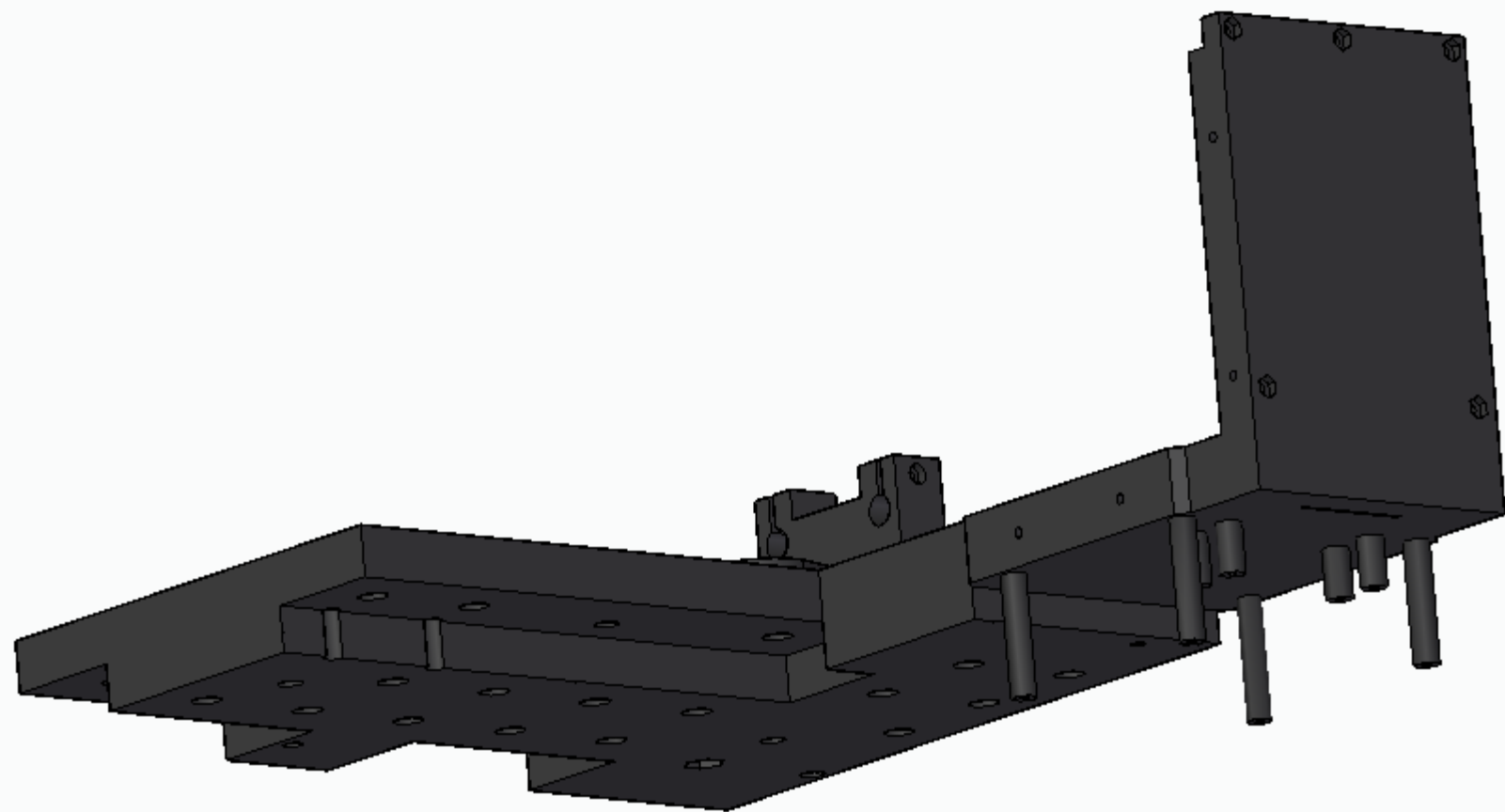
Assembly instructions

v1.0

1 × 01_FDML_stimfix_mainboard
4 × 02_FDML_stimfix_electronics_post_24mm
4 × 03_FDML_stimfix_electronics_post_10mm

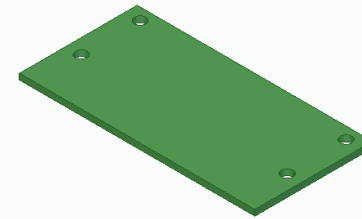
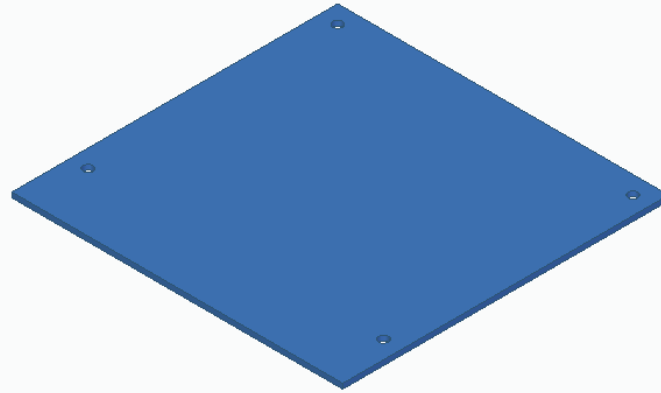
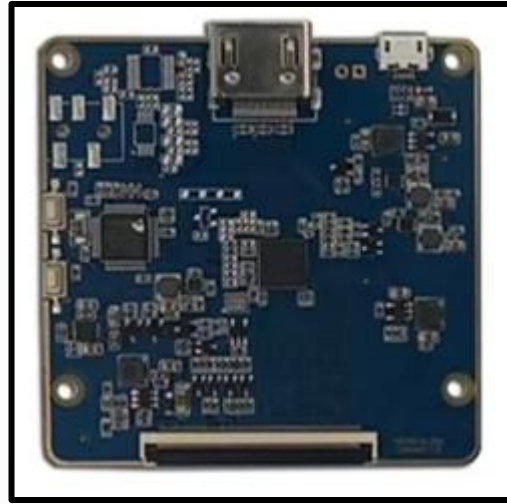
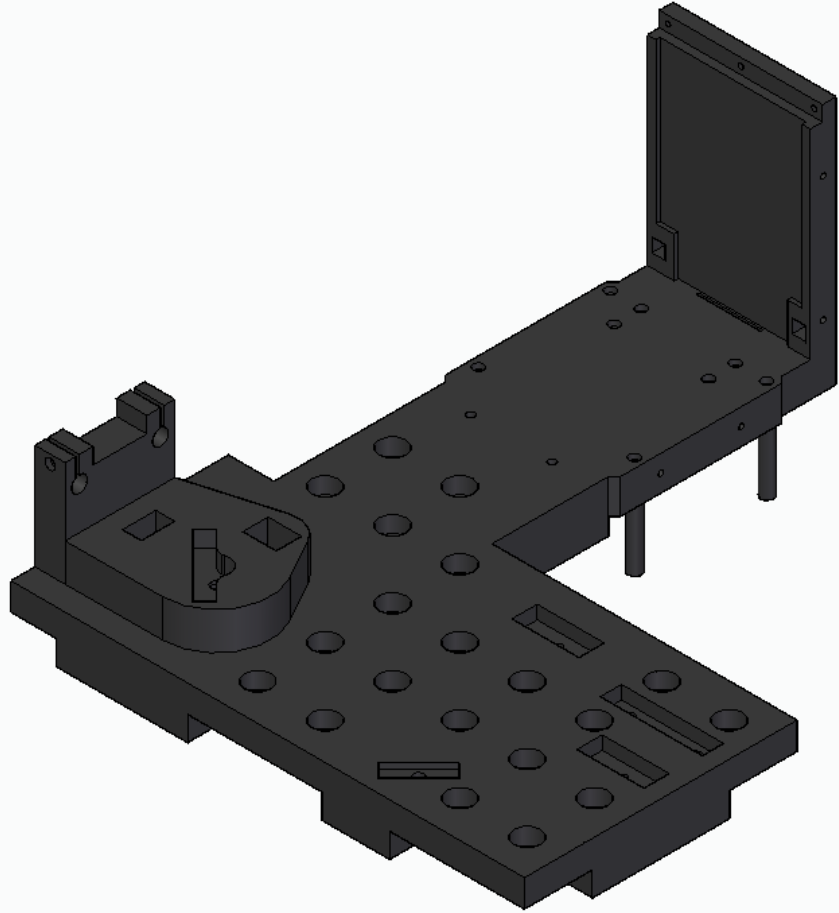


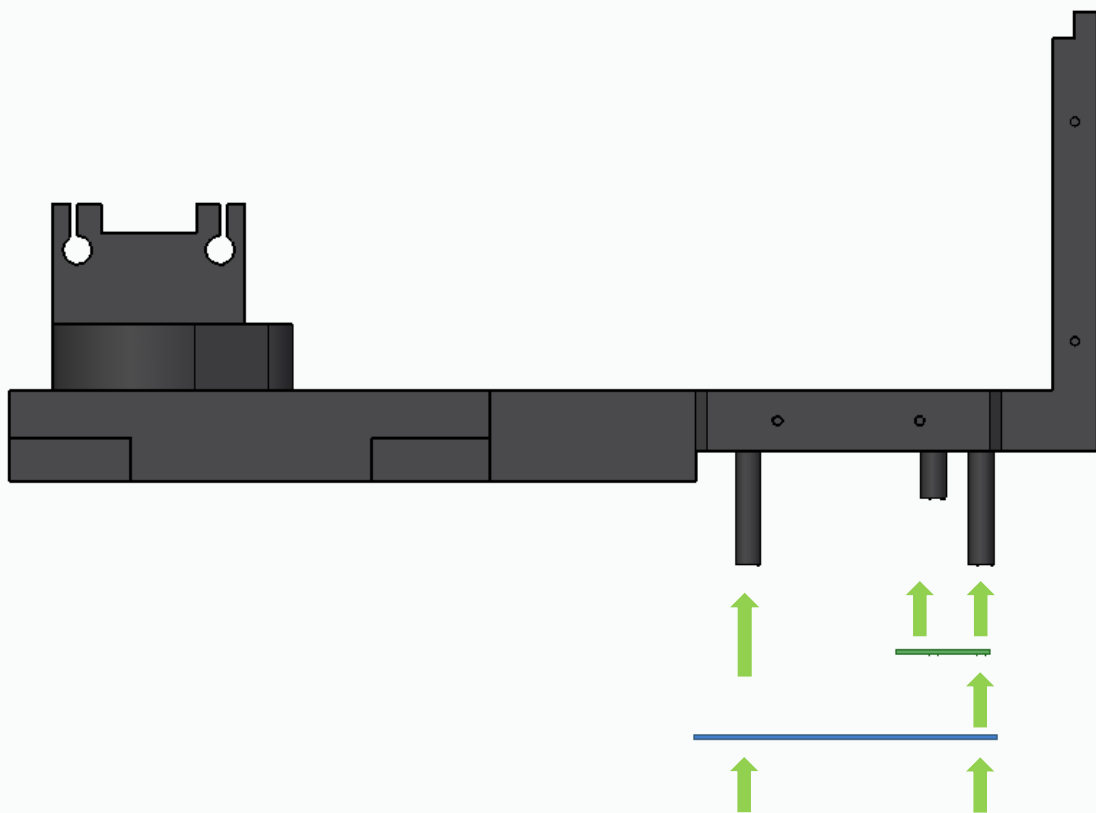




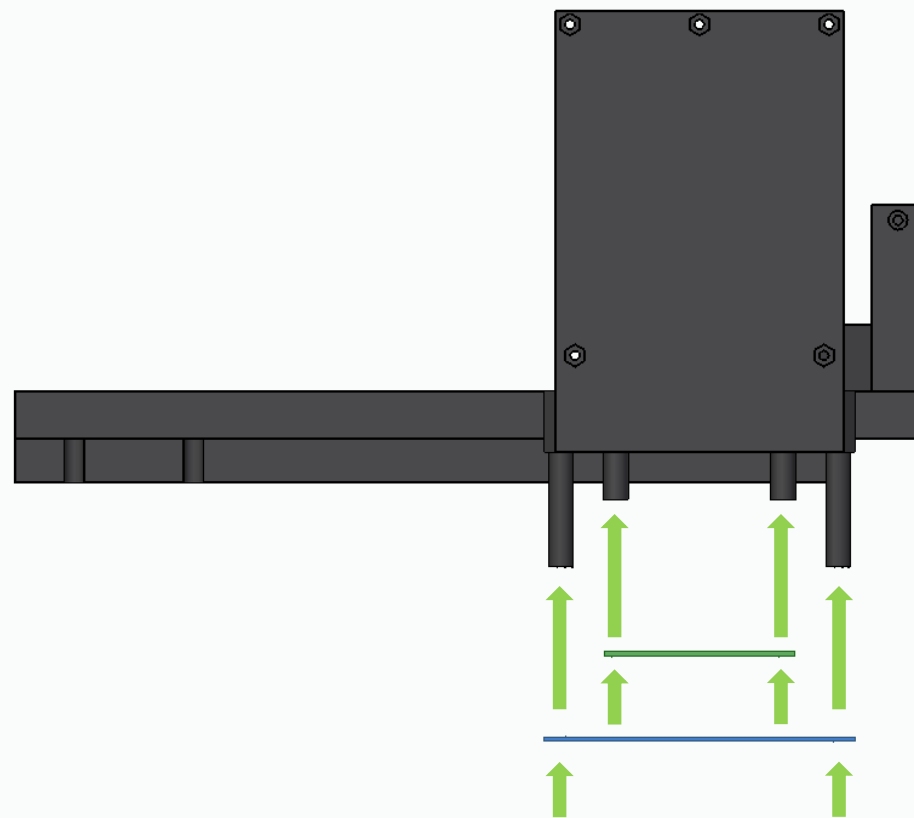
1 × PCB drive board small (comes with LCD screen)

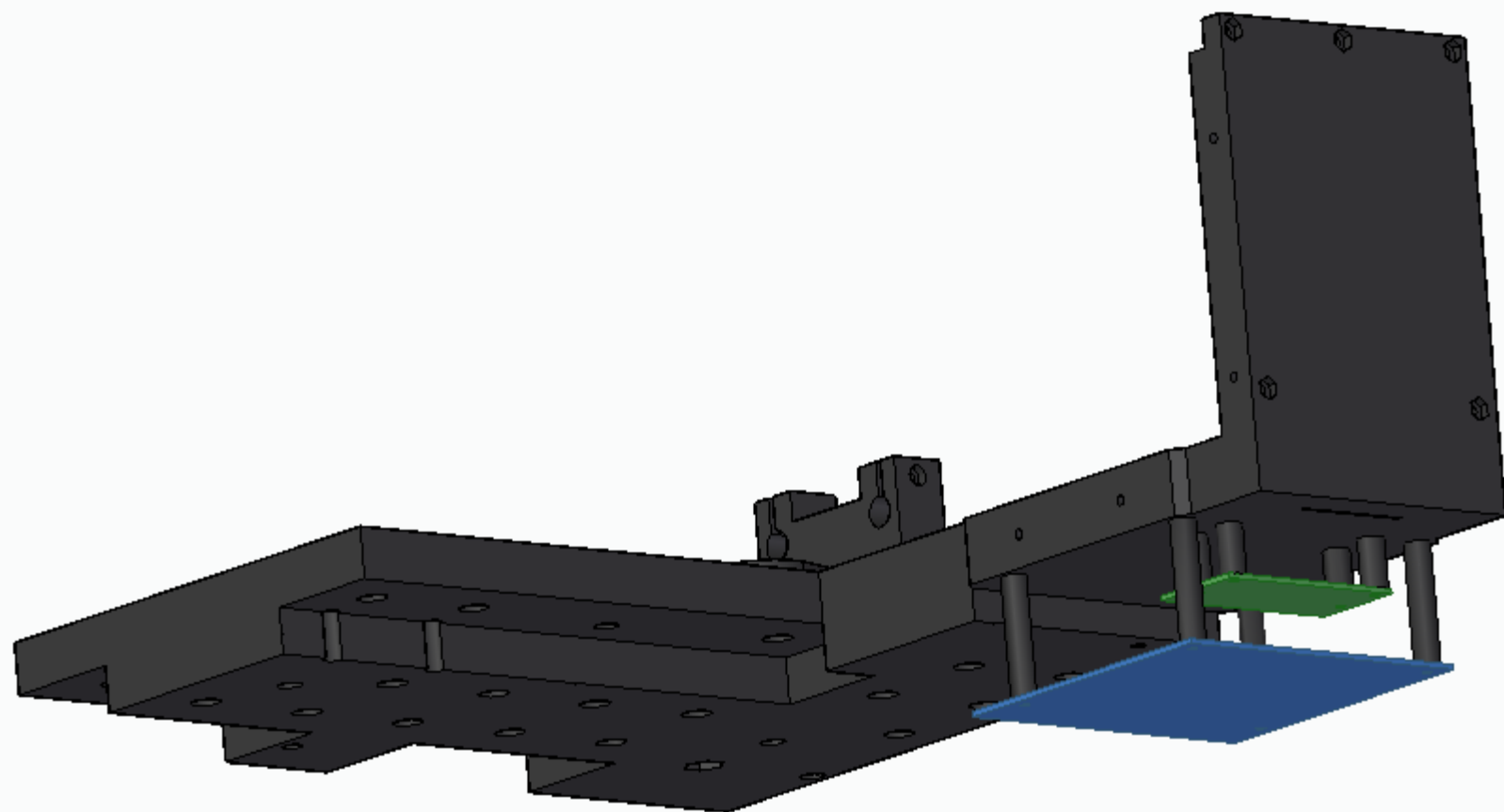
1 × PCB drive board (comes with LCD screen)



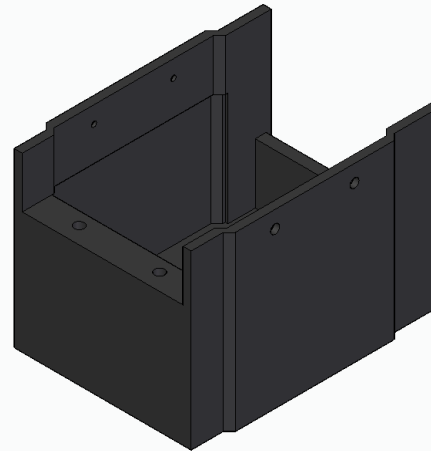
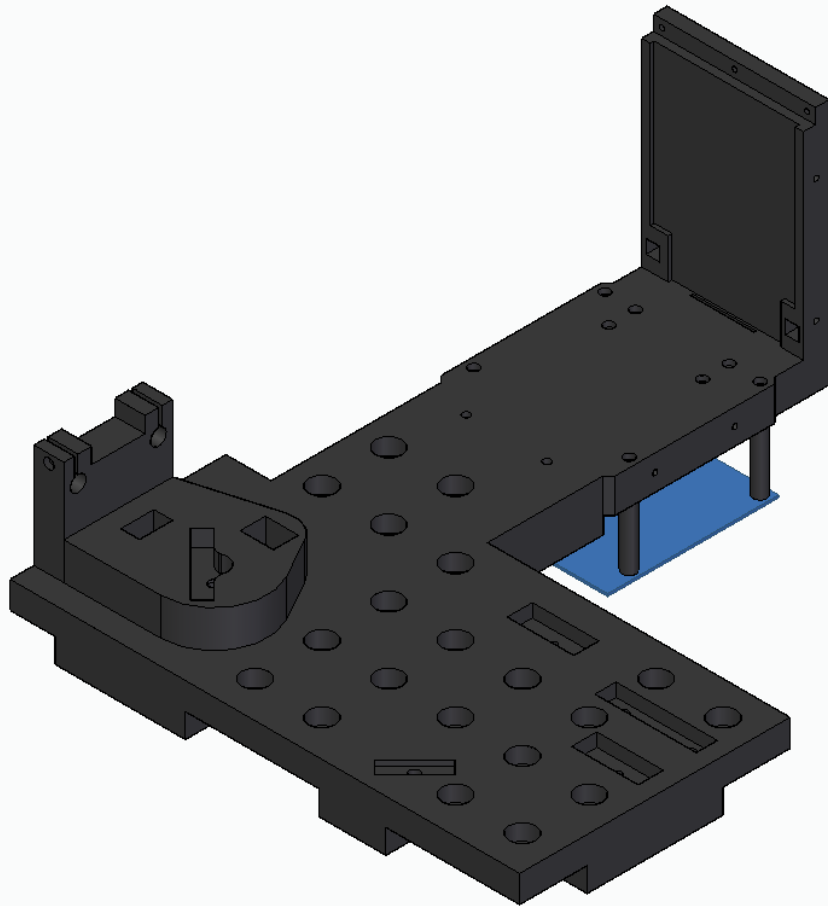


Use 8 × M2 cap screws to
fix both PCB boards to the
posts

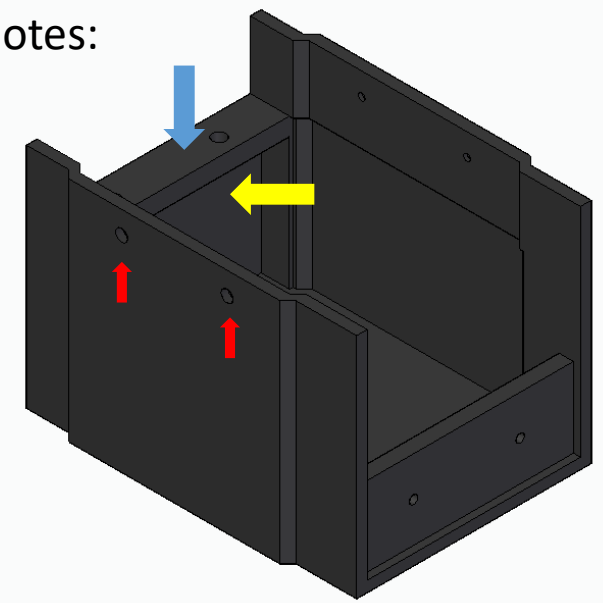




1 × 04_FDML_stimfix_PCB_boards_housing



Side notes:



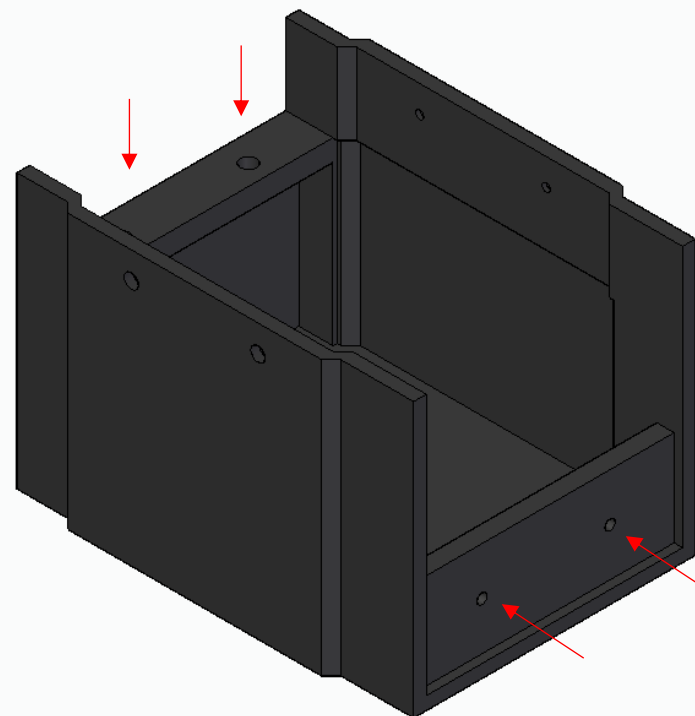
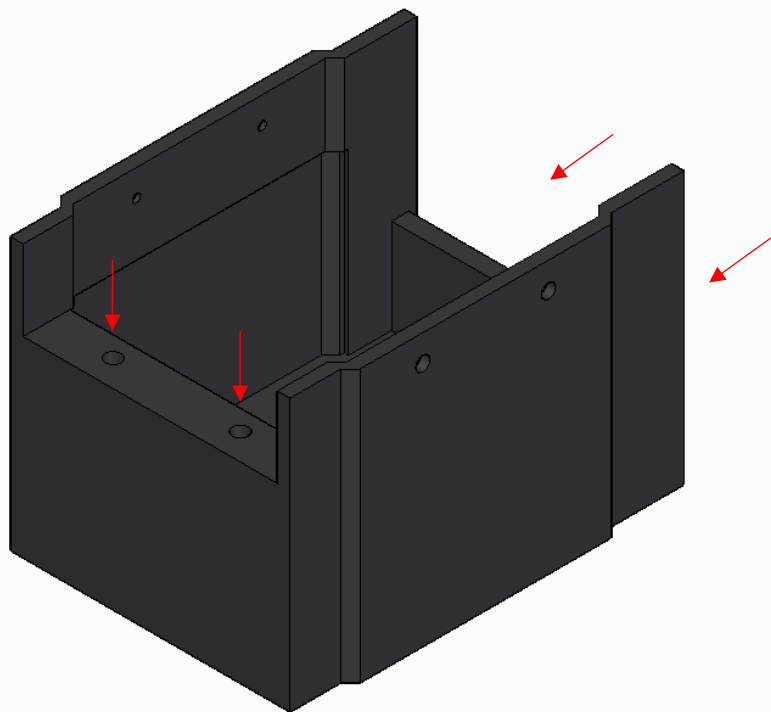
The surface indicated by the blue arrow aligns with the bottom of the main board. Its purpose is to secure the enclosure, preventing light leakage from the PCB board LEDs.

The spacing between the holes should be increased to create a slot for routing the pitch flexible flat cable from the PCB blue board. The current spacing is too tight, making it difficult to install this component without risking damage or disconnection of the cable.

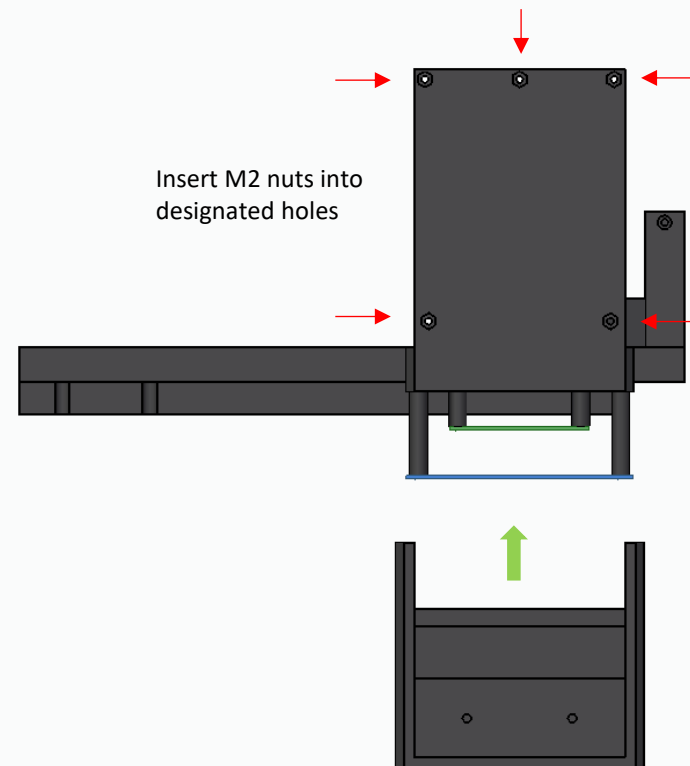
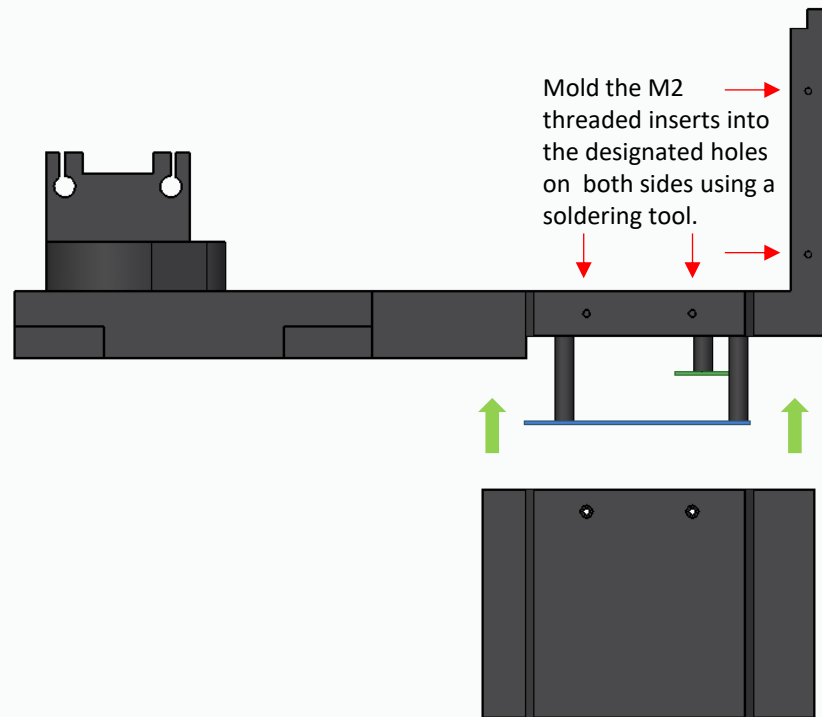
Additionally, counterbores need to be added to the main board. These should be applied to the holes designed as inserts for the cap screws that secure the section indicated by the blue arrow.

Securing the housing at the locations indicated by the red arrows prevents its removal once the Badal system is installed. Consider an alternative mounting solution to allow easier disassembly or access without interference from the installed system.

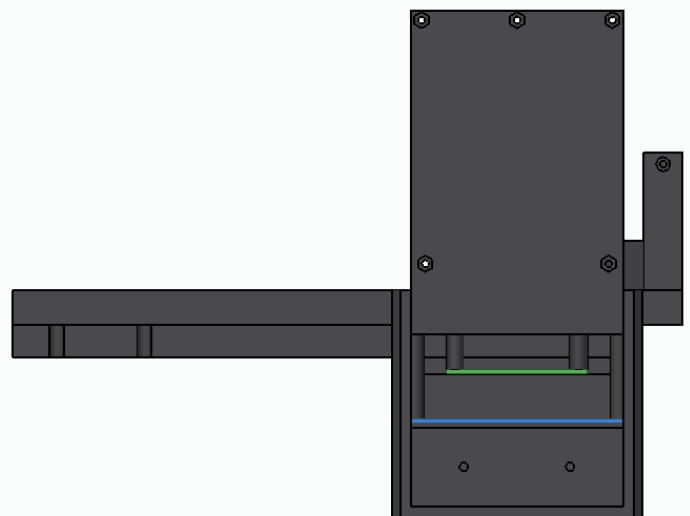
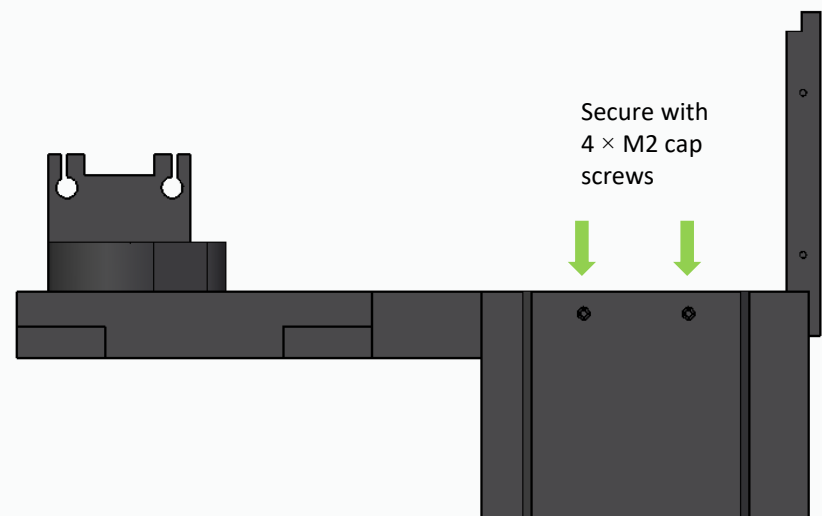
Mold the M3 threaded inserts into the designated holes on both sides using a soldering tool.



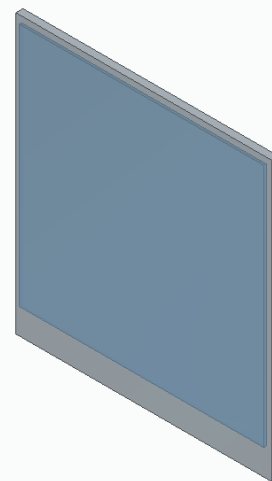
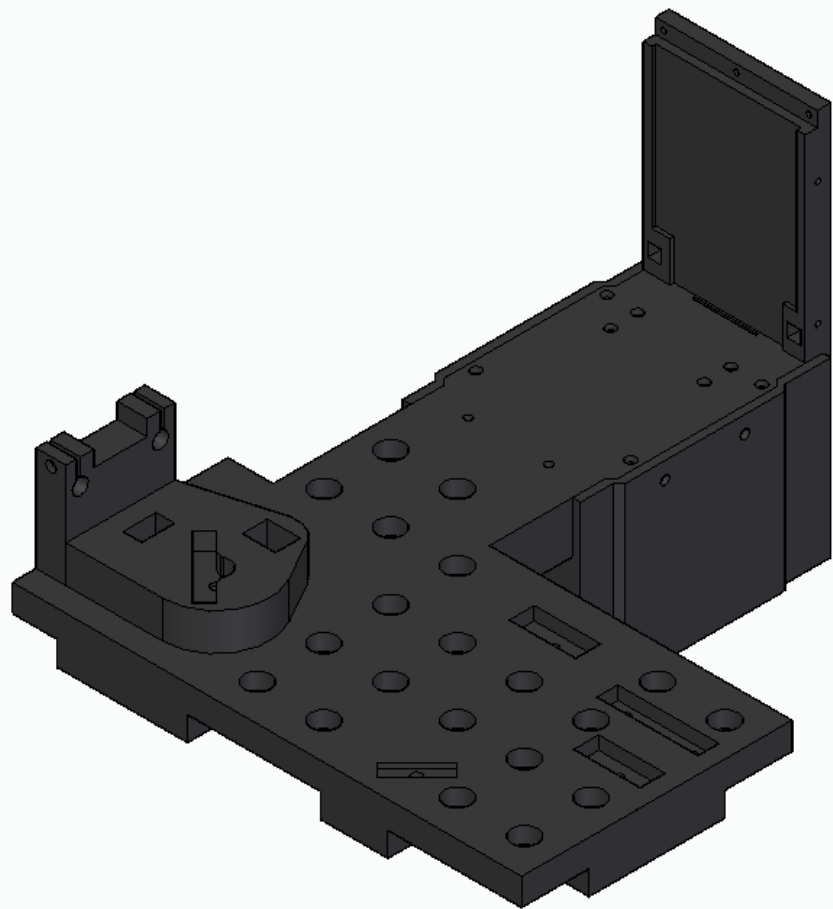
1



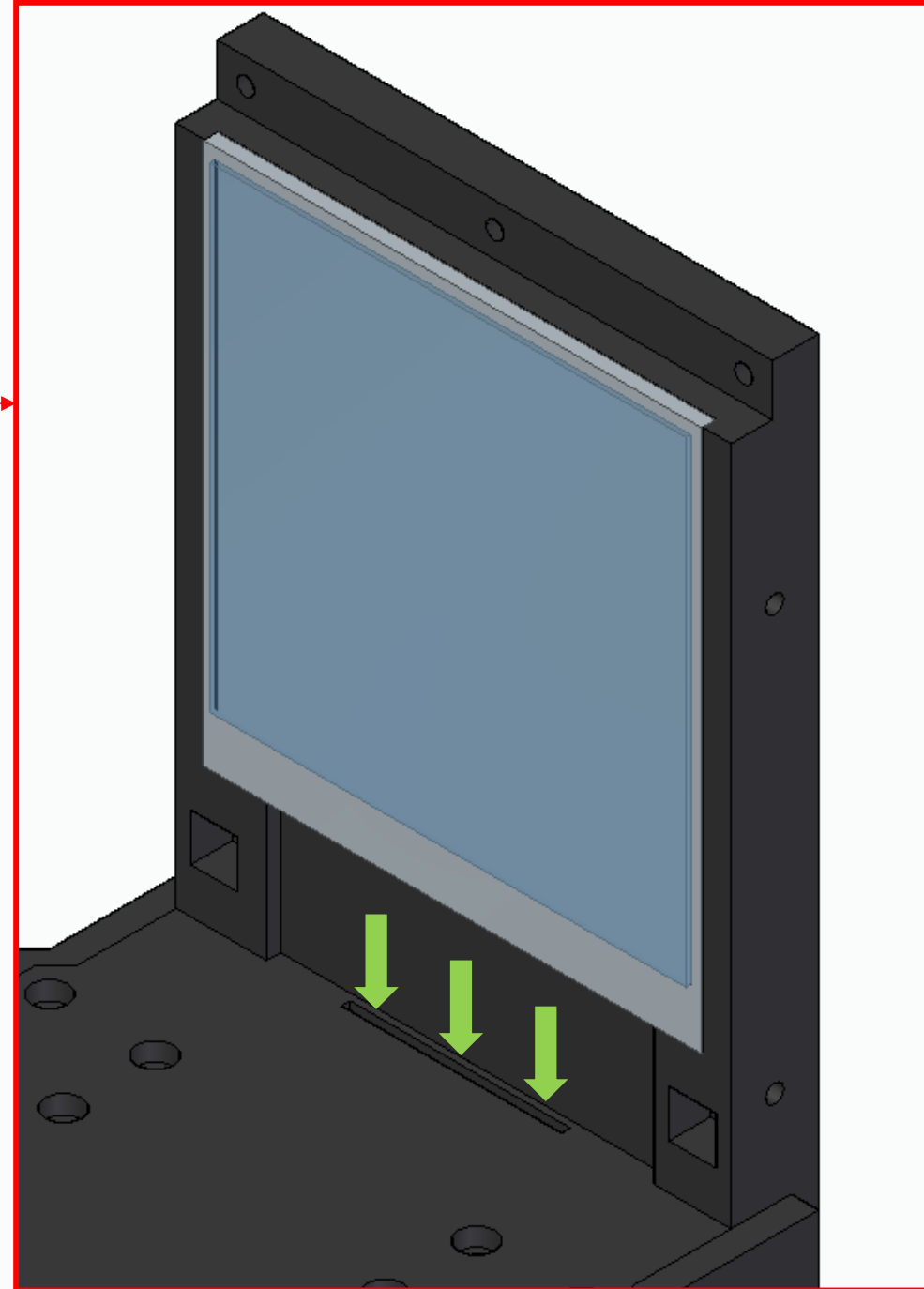
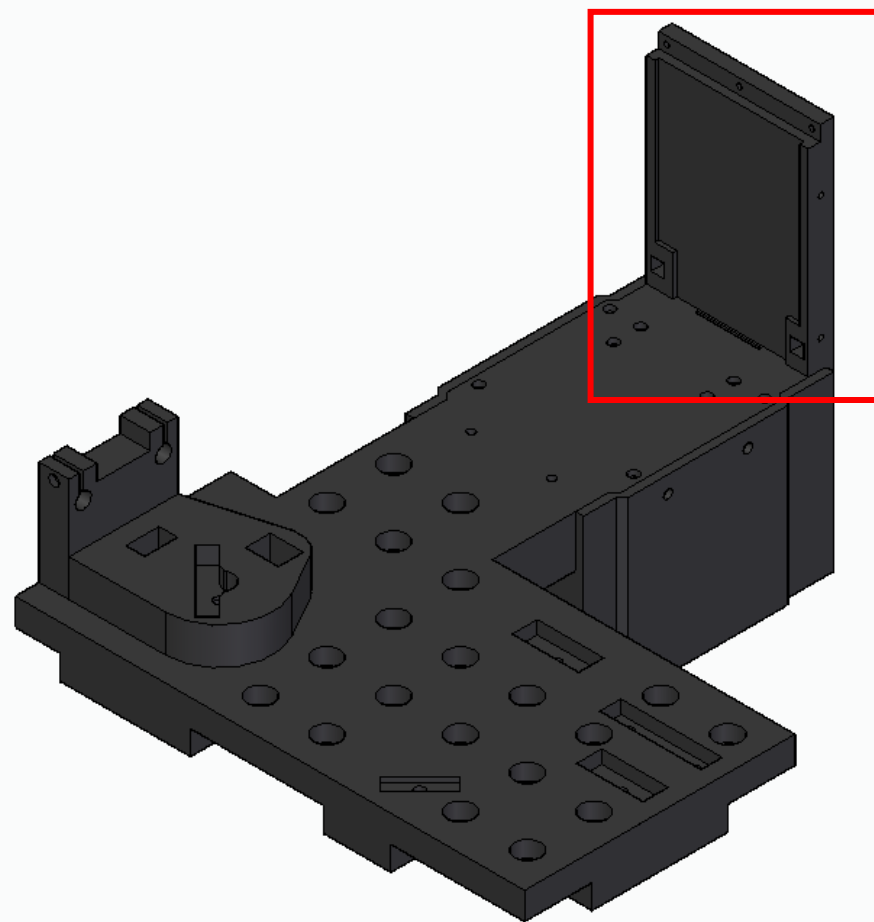
2



1 × LCD screen



1 × LCD screen

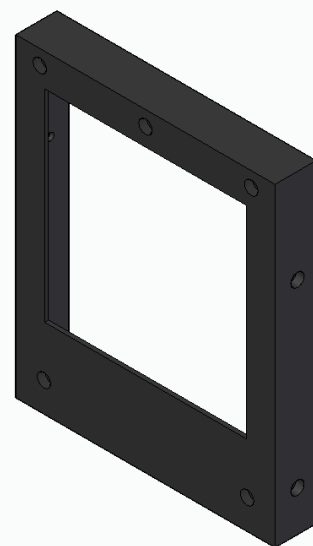
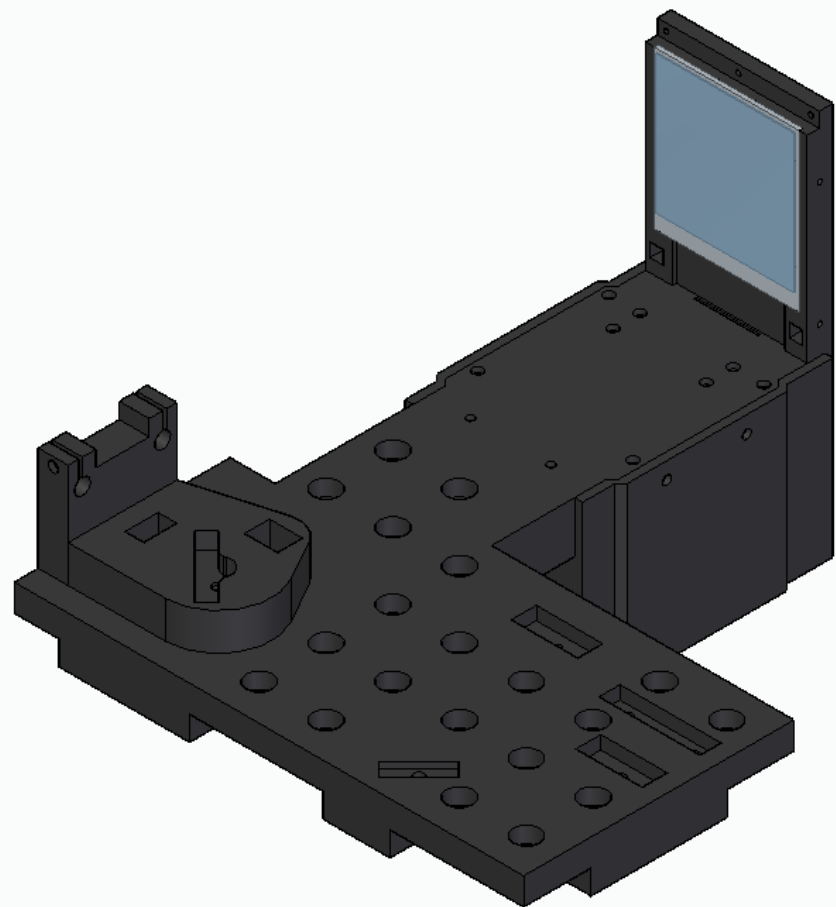


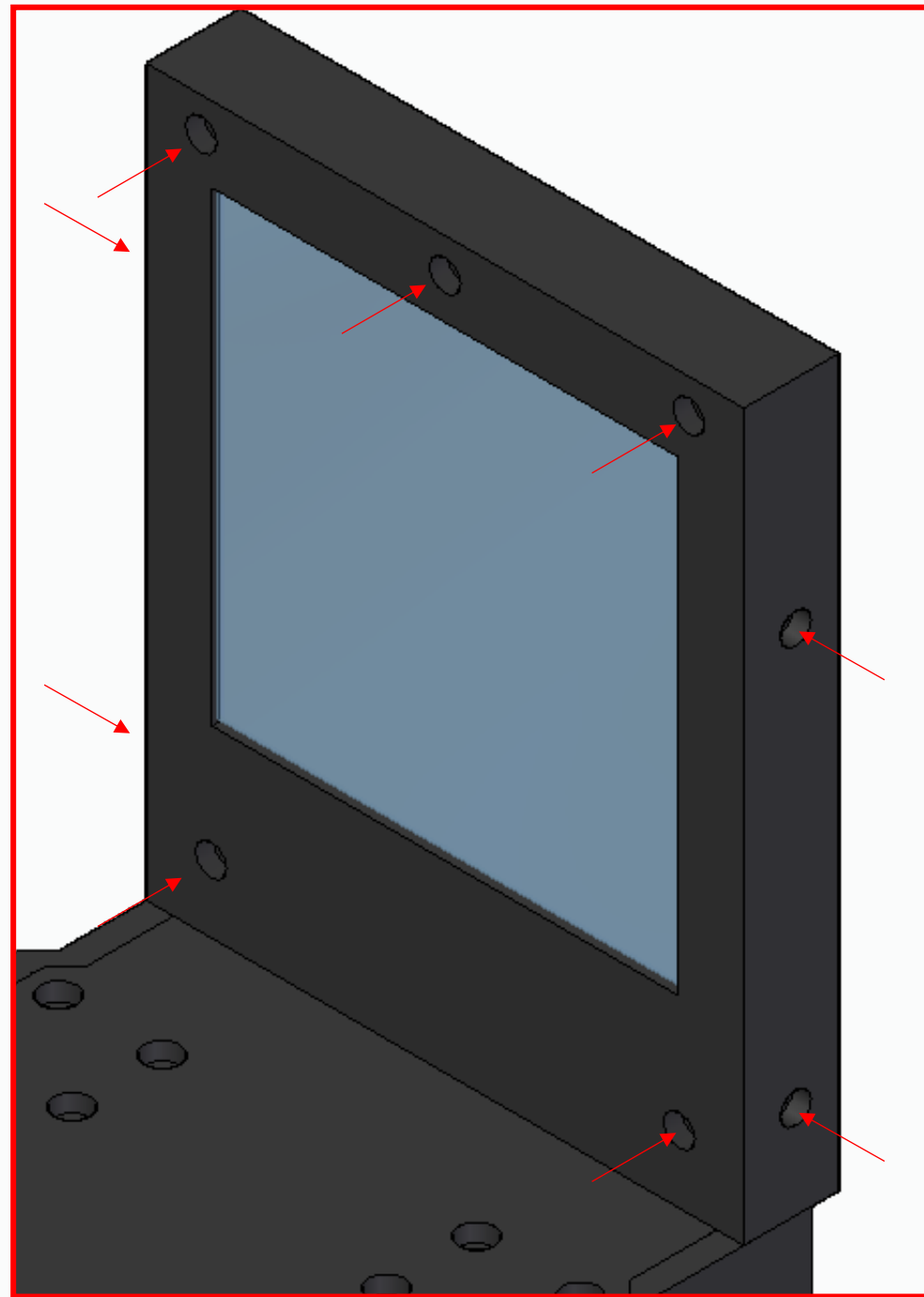
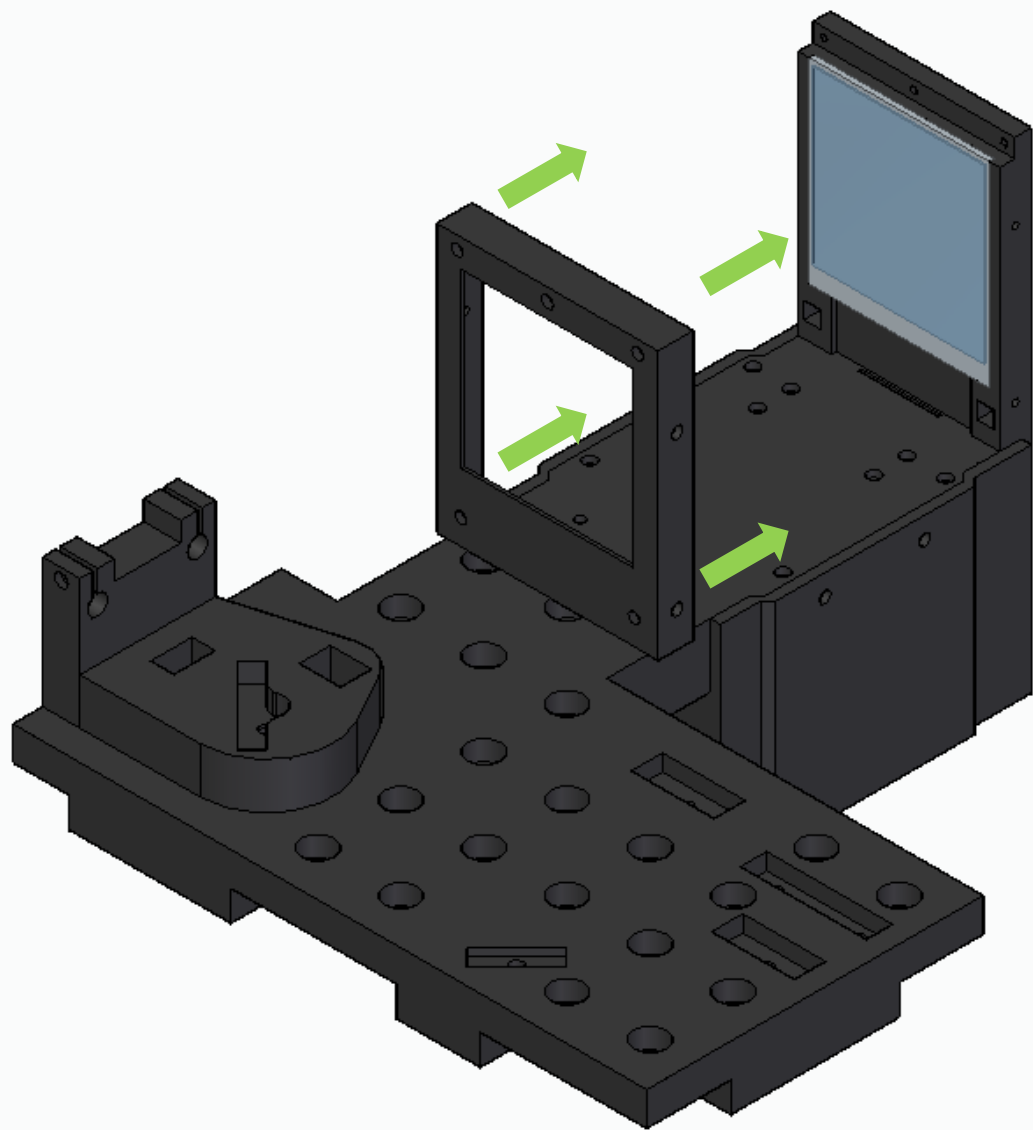
Insert the screen into designated slot.

Pass the flat cable (not shown in the model) through the designated slot as indicated by green arrows.

Connect the cable to the green PCB board

1 × 05_FDML_stimfix_LCD_screen_front_panel





Secure with
12 × M2
cap screws

1 × Thorlabs_BP207 (a 3D printed alternative holder could also be used)



<https://www.thorlabs.com/thorproduct.cfm?partnumber=BP207>

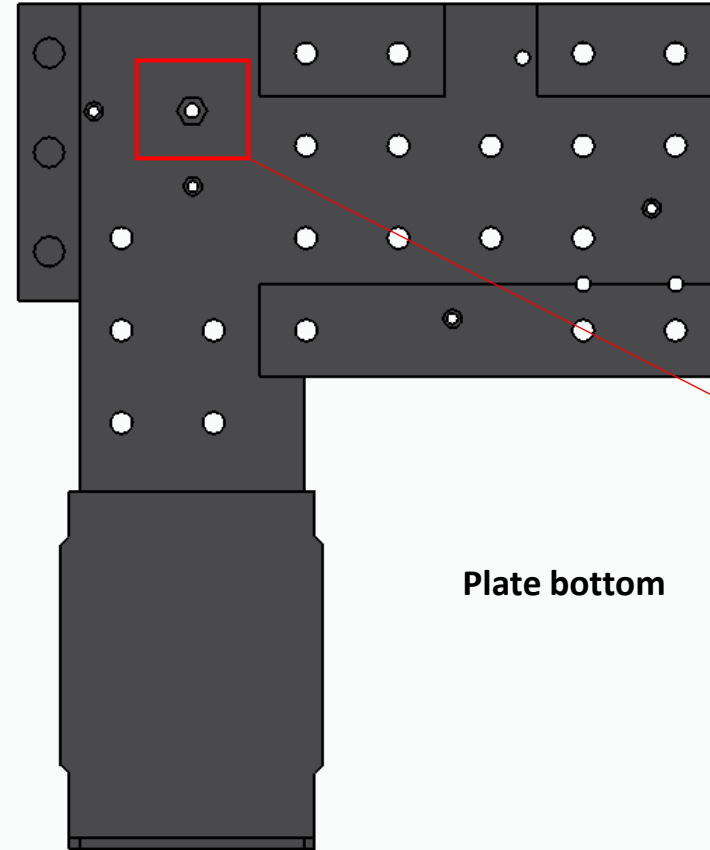
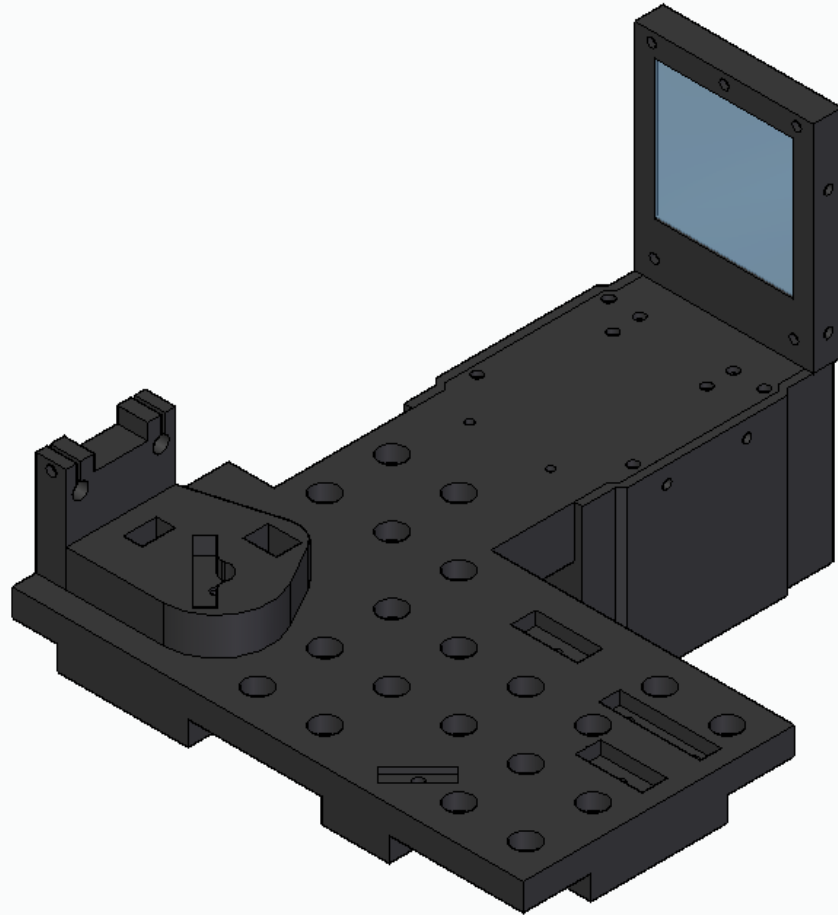
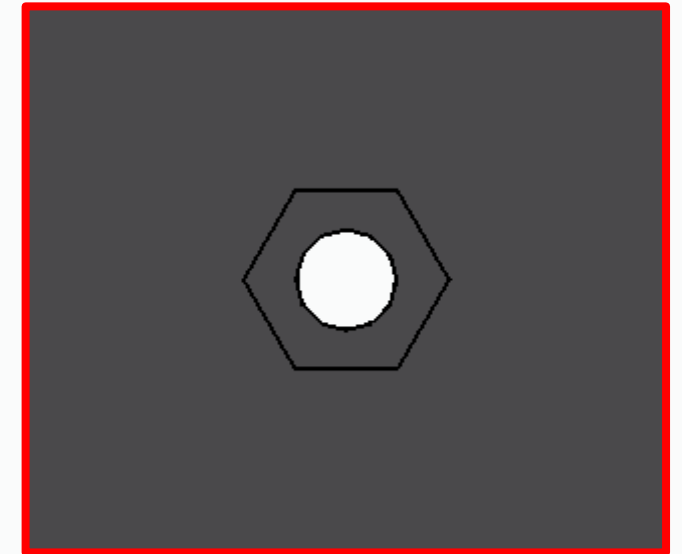
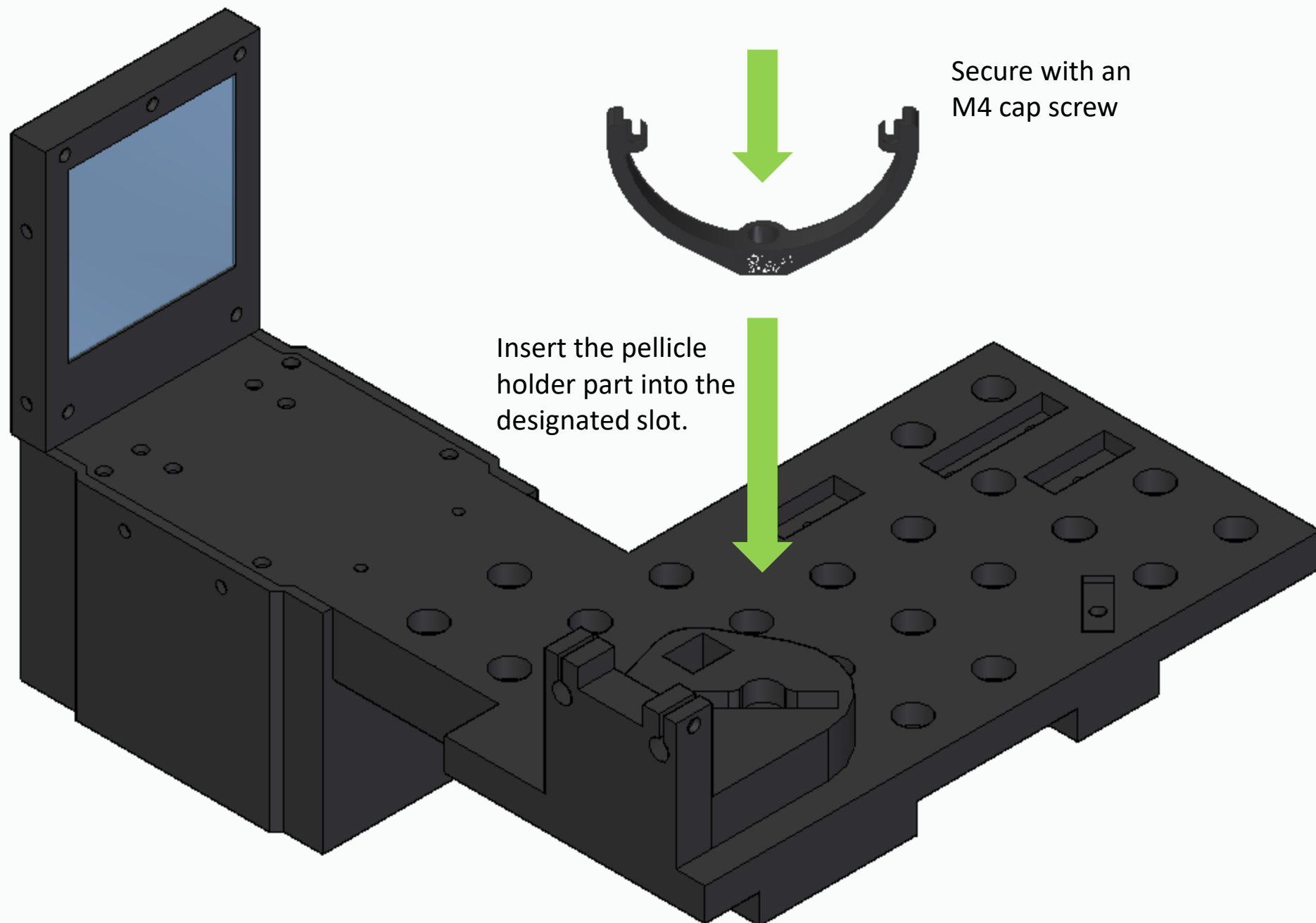


Plate bottom

Insert an M4 nut into designated hole





Secure with an
M4 cap screw

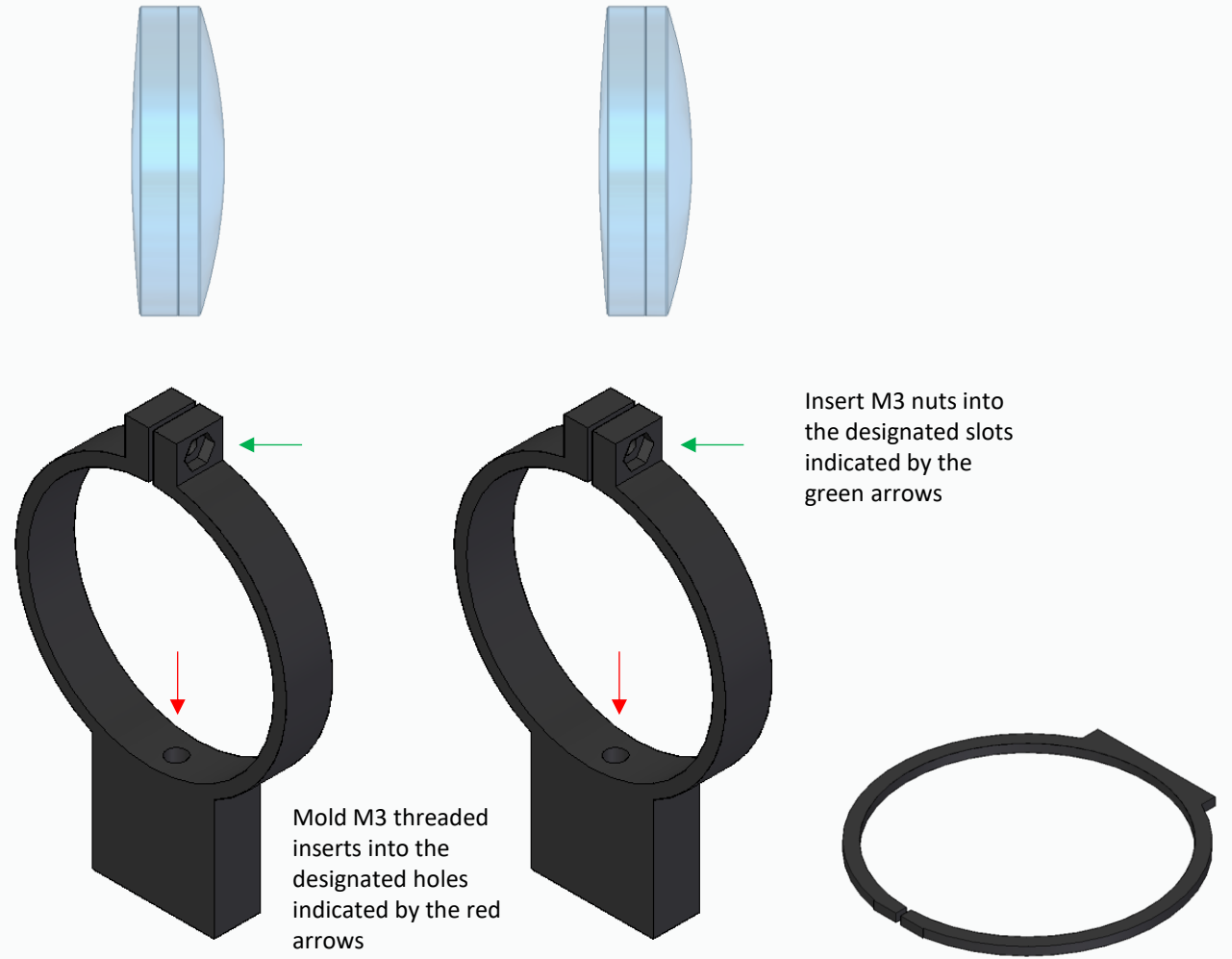
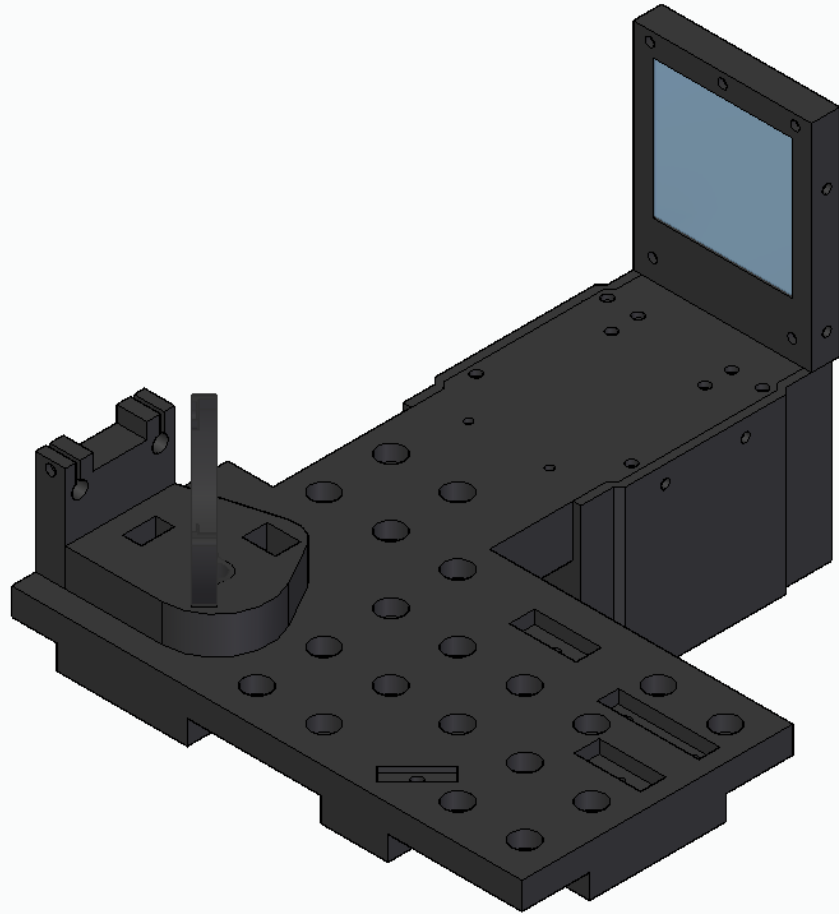
Insert the pellicle
holder part into the
designated slot.

2 × 06_FDML_stimfix_2in_lens_clamp_holder

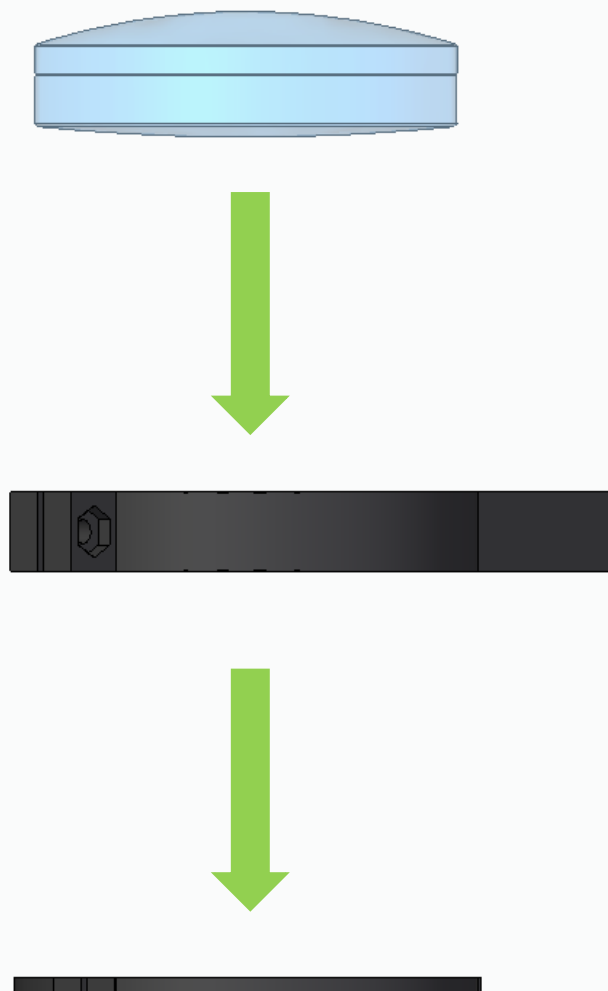
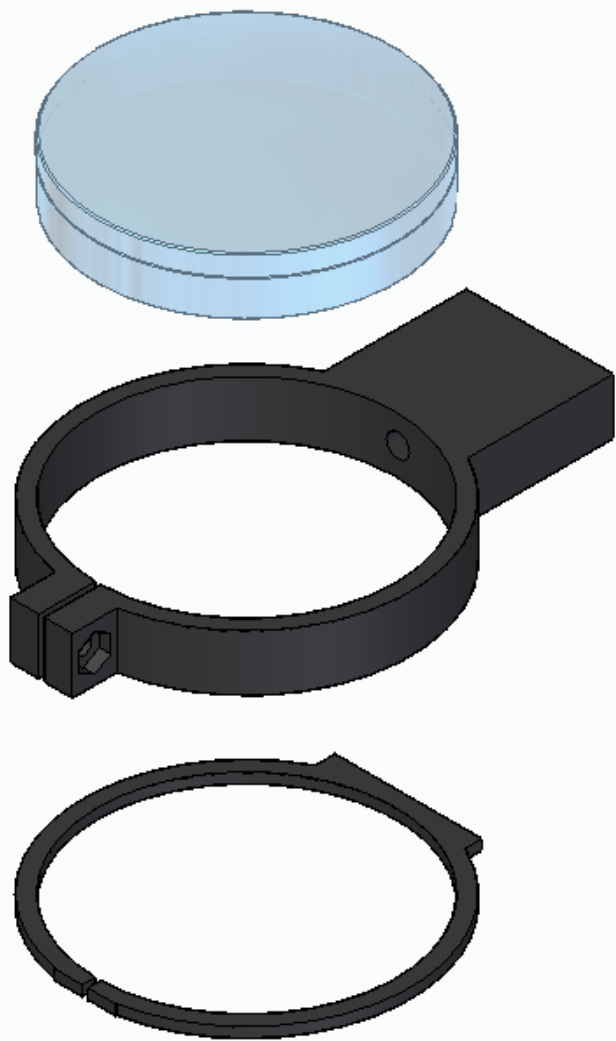
<https://www.thorlabs.com/thorproduct.cfm?partnumber=AC508-150-A>

2 × Thorlabs_AC508-150-A

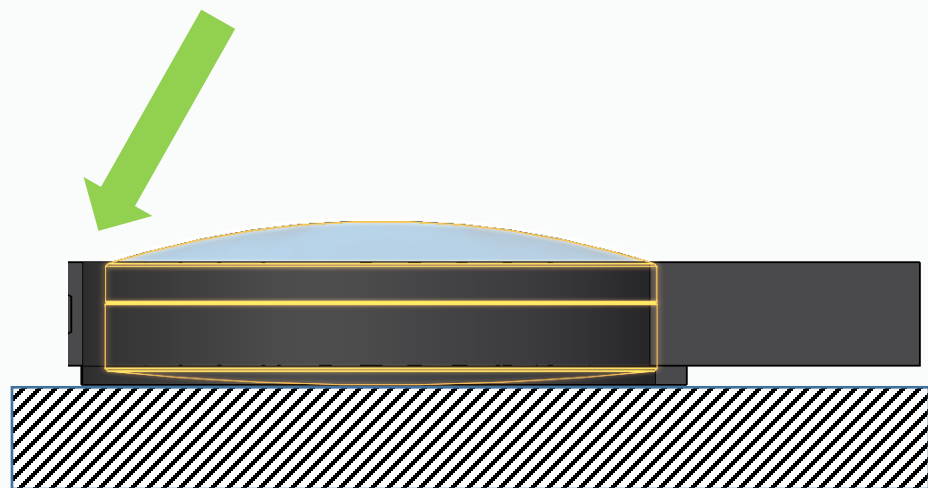
1 × extra_stimfix_2in_lens_clamp_holder_mounting_tool (optional)

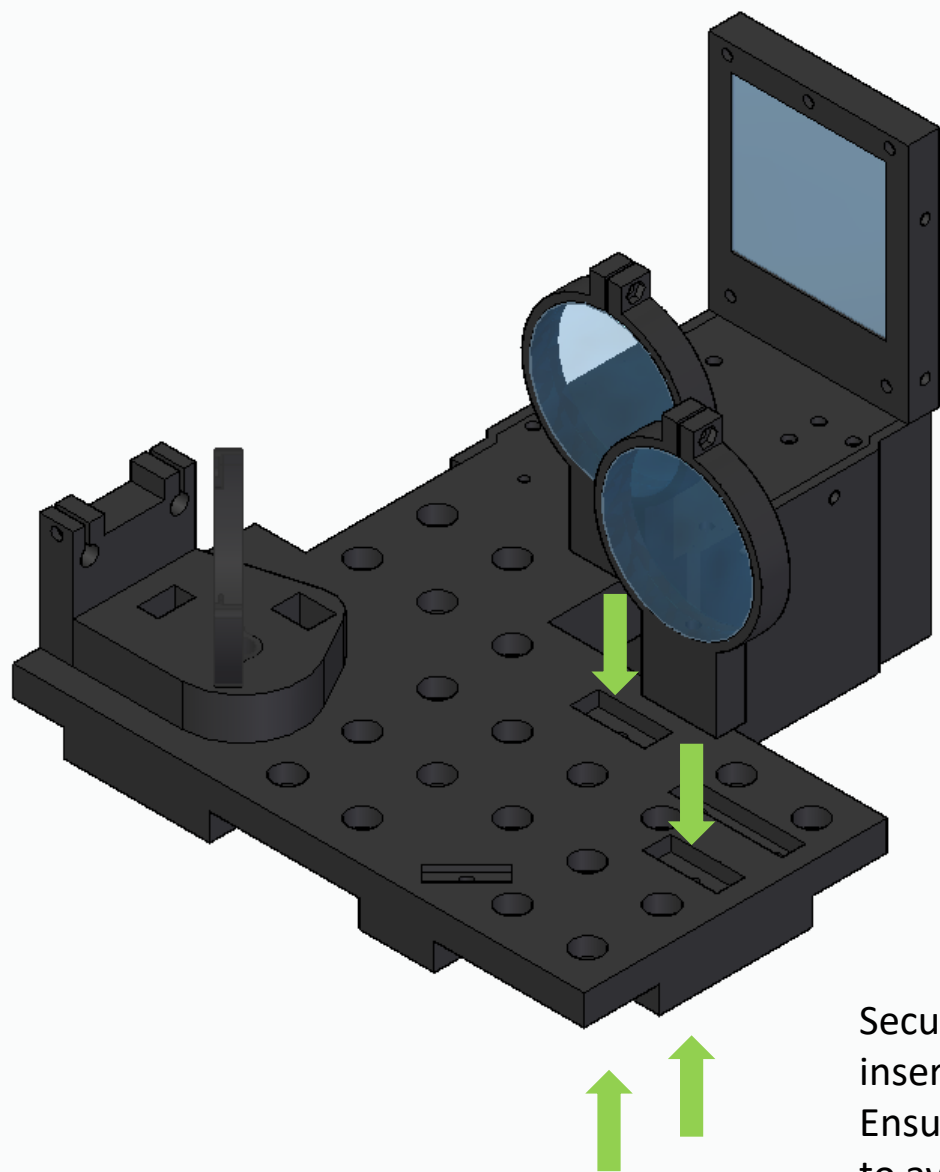


2 ×

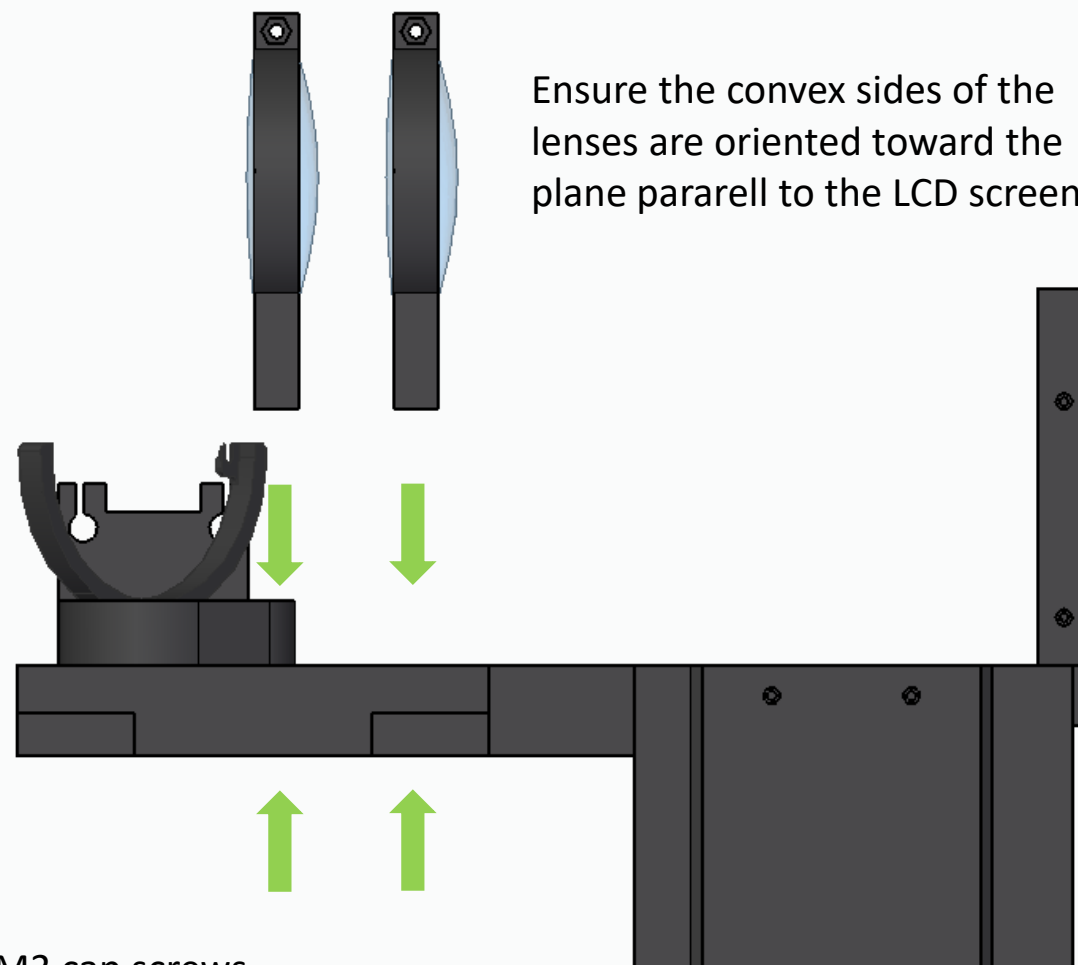


Once in the proper position, secure the lens using an M3 cap screw





Secure using M3 cap screws inserted from the bottom side. Ensure the screws are **not too long** to avoid drilling into the lens



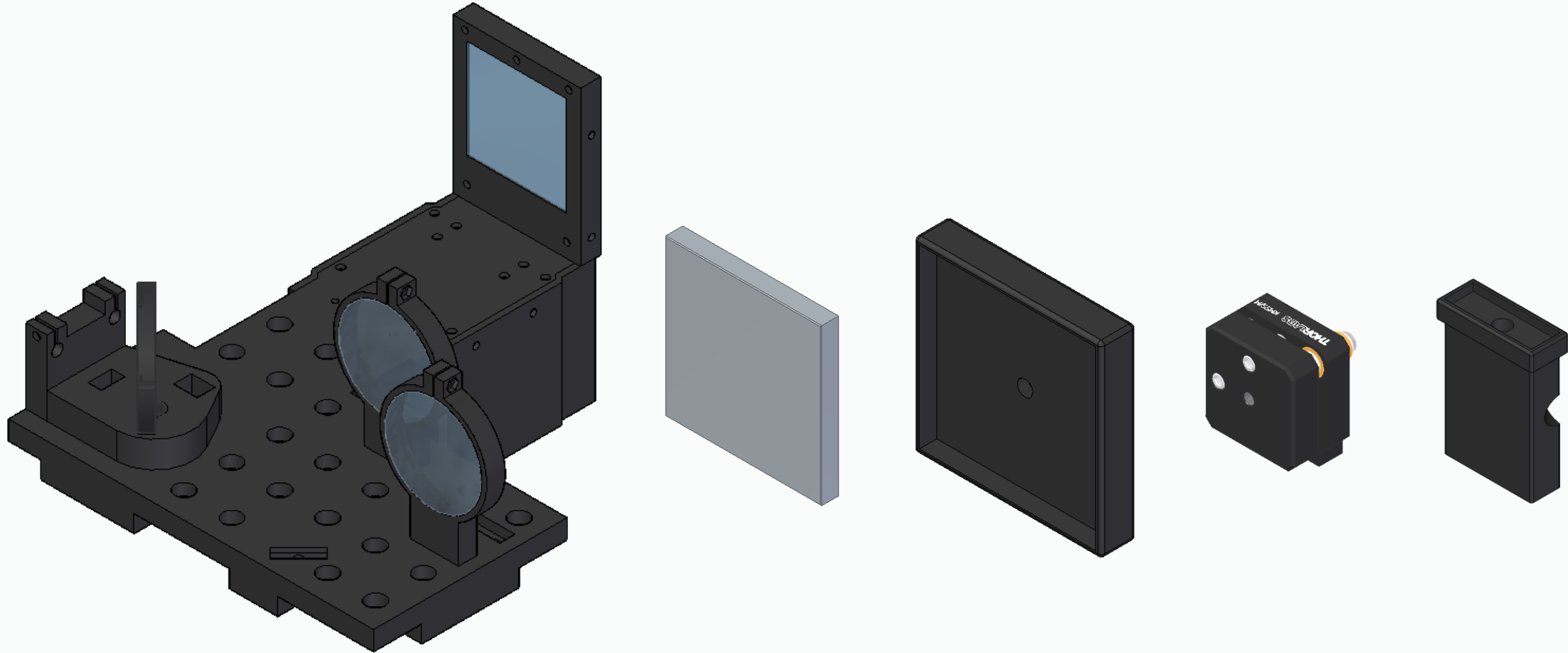
Ensure the convex sides of the lenses are oriented toward the plane parallel to the LCD screen

1 × Thorlabs_PFSQ20-03-P01 (<https://www.thorlabs.com/thorproduct.cfm?partnumber=PFSQ20-03-P01>)

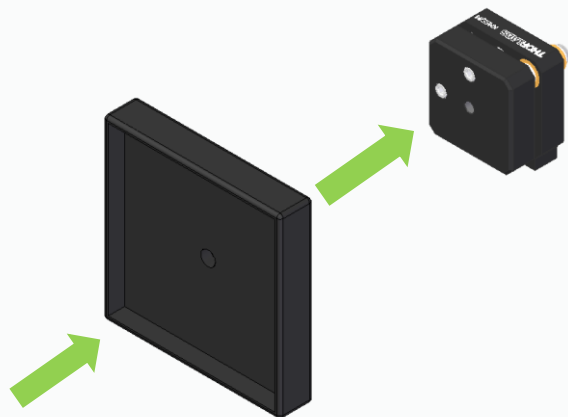
1 × 07_FDML_stimfix_square_mirror_holder_small

1 × Thorlabs_KMSS_M (<https://www.thorlabs.com/thorproduct.cfm?partnumber=KMSS/M>)

1 × 08_FDML_stimfix_square_mirror_support_with_6mm_rod_cutout

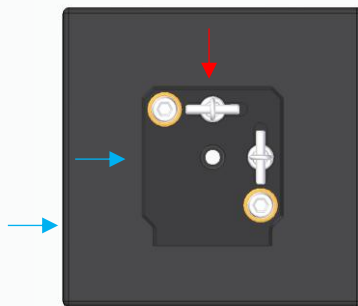


1

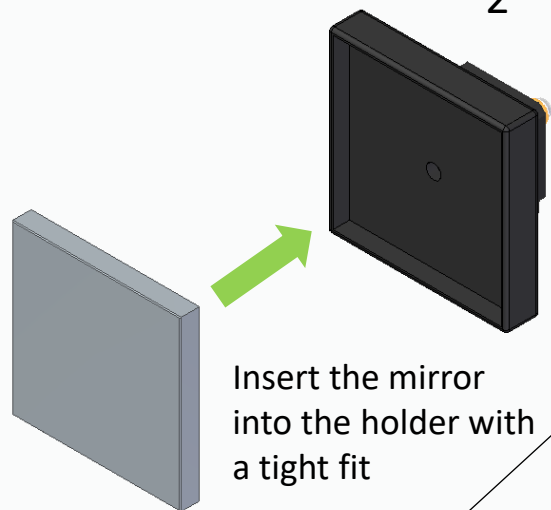


Secure
using 4-40
cap screws

Ensure the corresponding
surfaces are as parallel
as possible

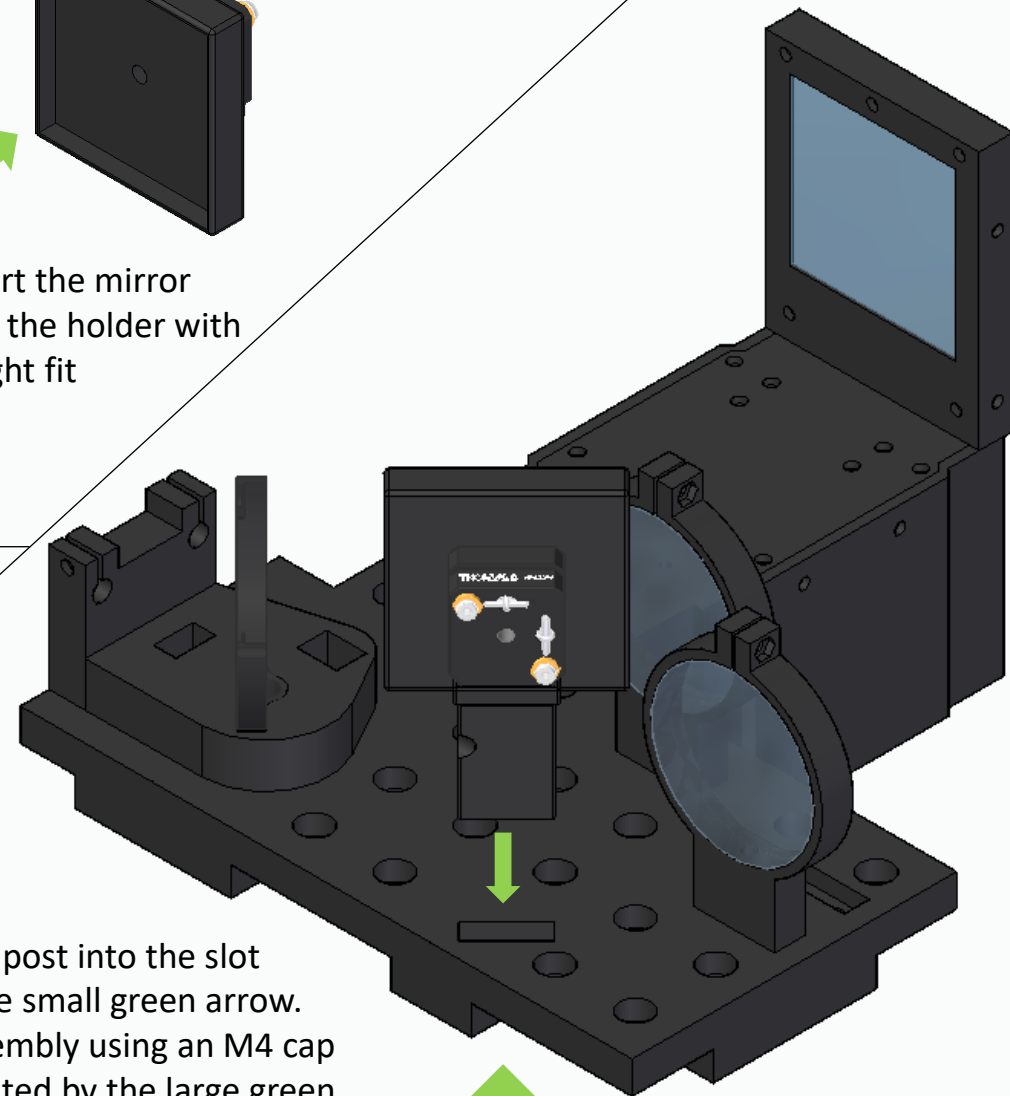


2



Insert the mirror
into the holder with
a tight fit

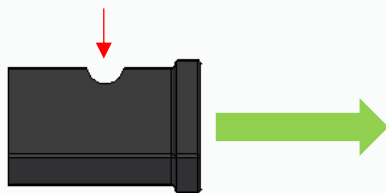
4



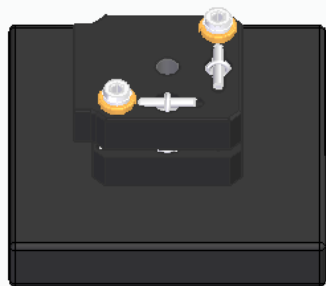
Fit the support post into the slot
indicated by the small green arrow.
Secure the assembly using an M4 cap
screw, as indicated by the large green
arrow. The screw should pass through
from the bottom of the main board
to the kinematic mount.

3

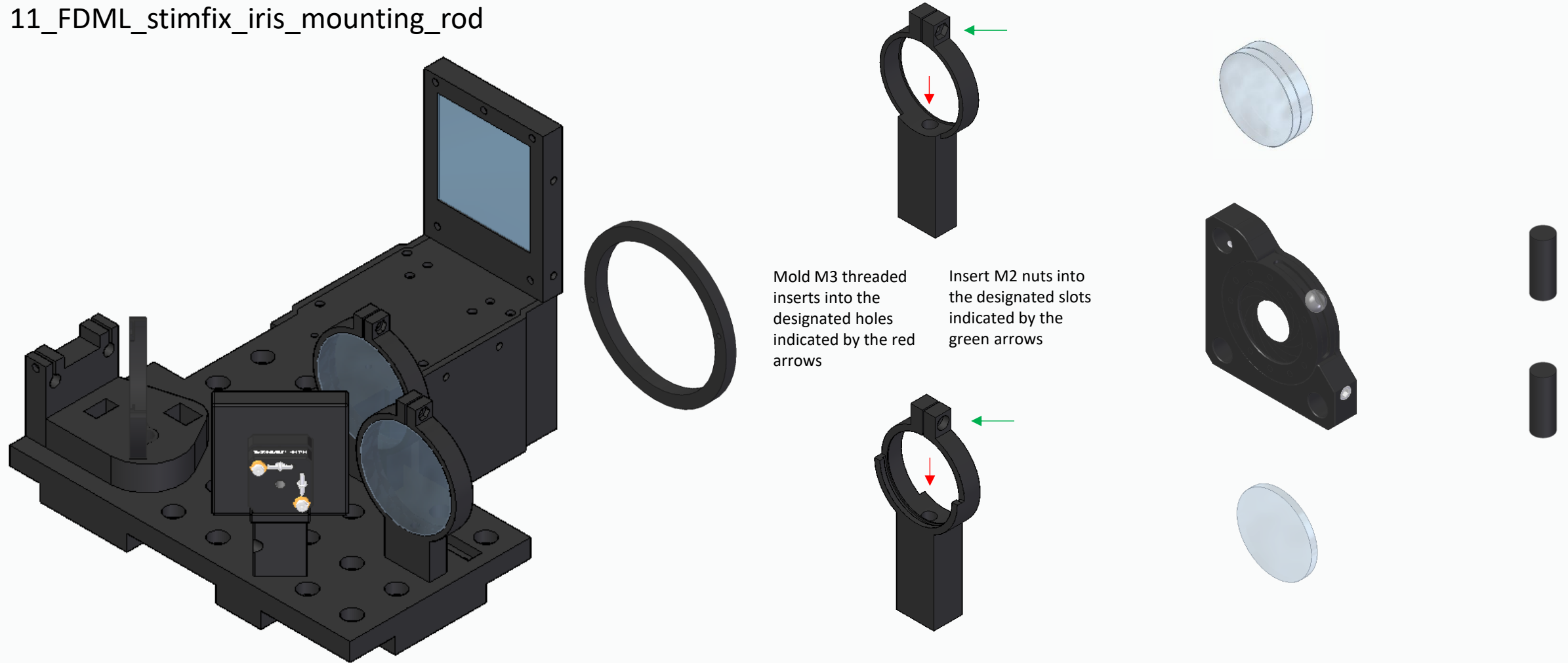
Ensure the position of the
cutout is as indicated



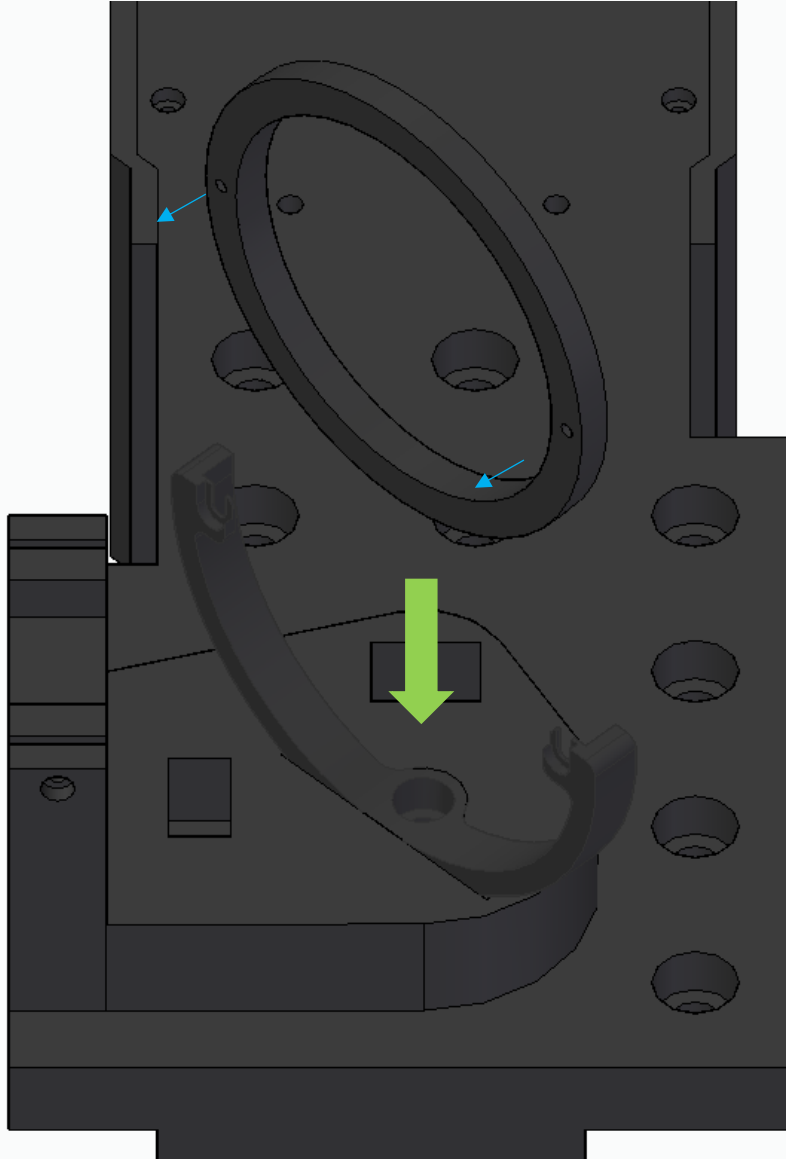
Slide the kinematic mount
into the post holder as tight
as possible. Be careful as it
may fall off – final fixing is
with the M4 cap screw in
the next step



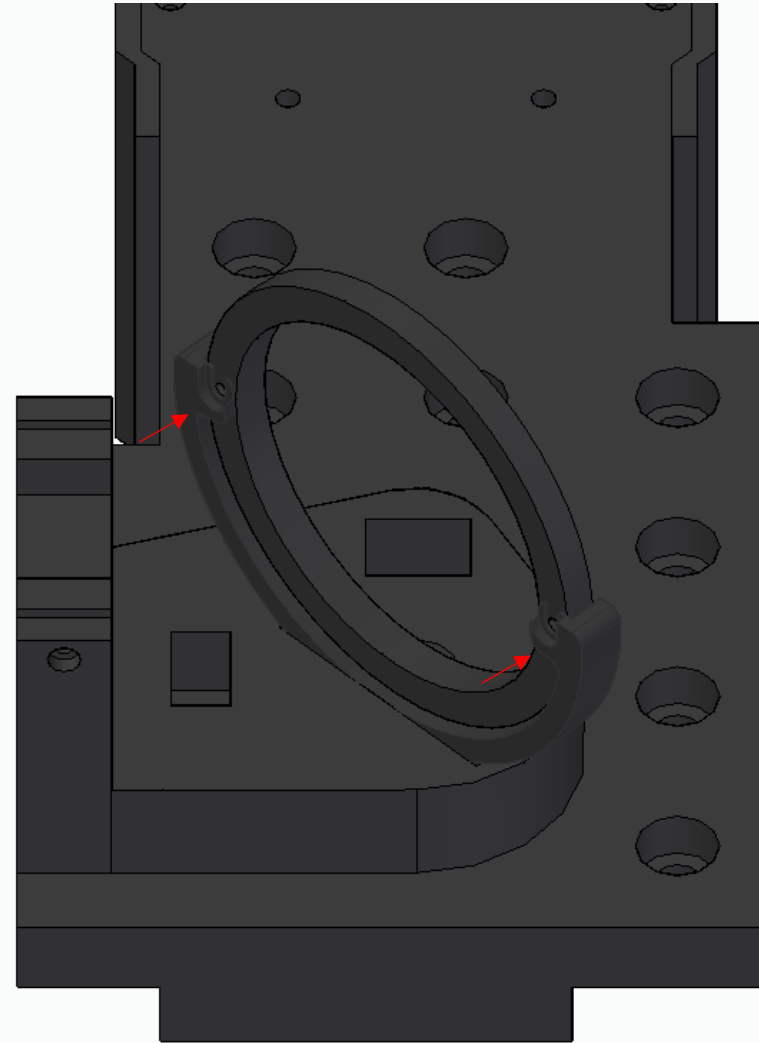
- 1 × Thorlabs_BP208
- 1 × 09_FDML_stimfix_1in_lens_clamp_holder_stim
- 1 × 10_FDML_stimfix_1in_lens_clamp_holder_fix
- 1 × Thorlabs_AC254-075-A (<https://www.thorlabs.com/thorproduct.cfm?partnumber=AC254-075-A>)
- 1 × Thorlabs_CP20D (<https://www.thorlabs.com/thorproduct.cfm?partnumber=CP20D>)
- 1 × Thorlabs_LB1294-A (<https://www.thorlabs.com/thorproduct.cfm?partnumber=LB1294-A>)
- 2 × 11_FDML_stimfix_iris_mounting_rod



Carefully insert the pellicle into the mounting fork.
Head screws should be facing the direction indicated by
the blue arrows and fitting the clearance slots



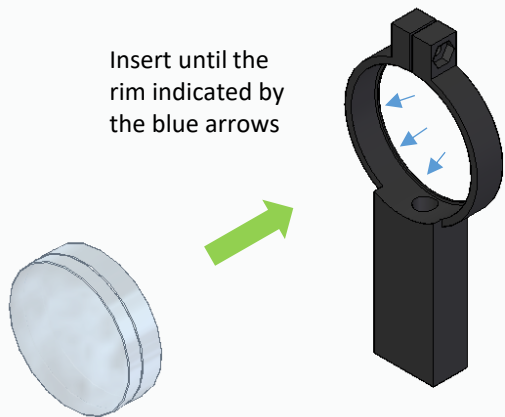
Secure with #2-56 button
head screws as indicated by
the red arrows



1

Insert until the rim indicated by the blue arrows

Convex side



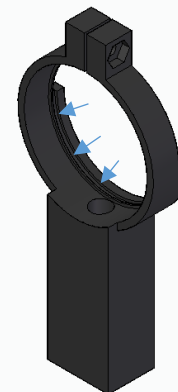
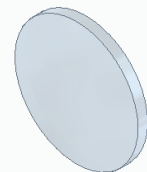
2

Tighten with an M2 cap screw



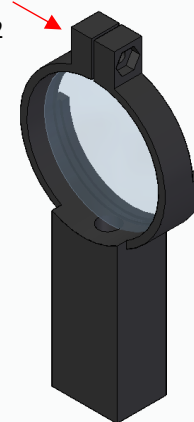
3

Insert until the rim indicated by the blue arrows



4

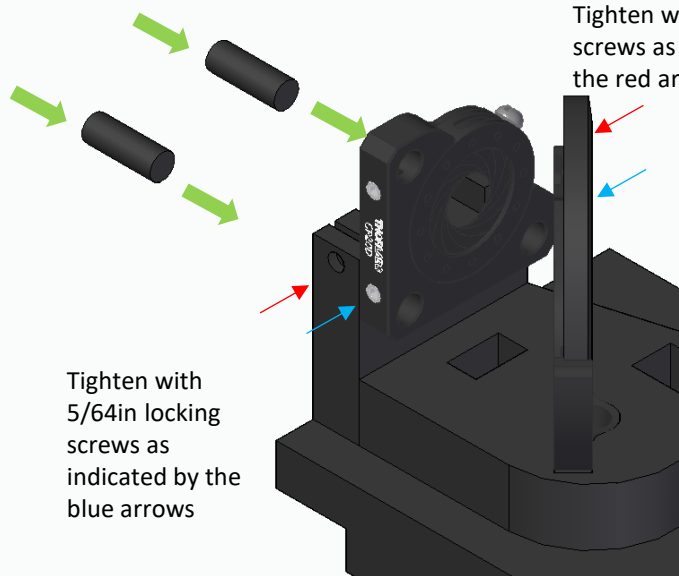
Tighten with an M2 cap screw



5

Tighten with an M2 cap screws as indicated by the red arrows

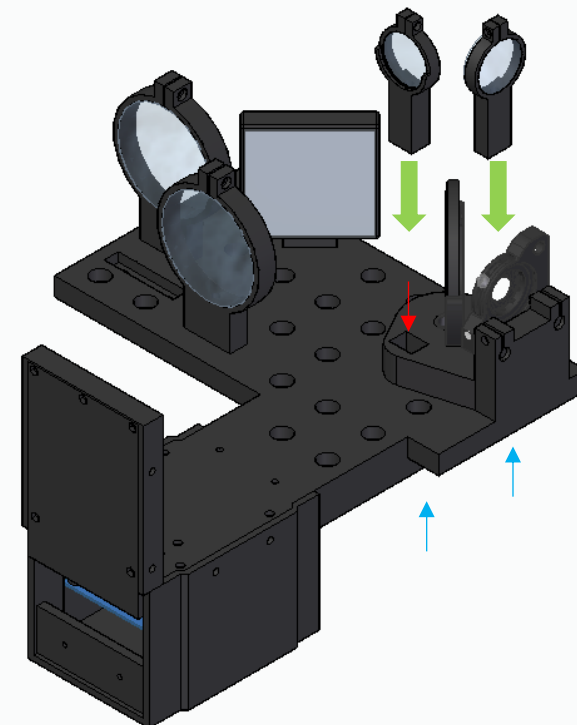
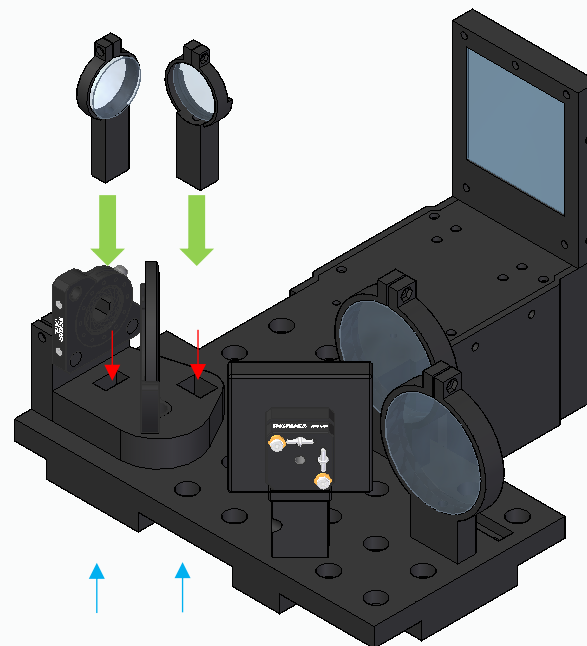
Tighten with 5/64in locking screws as indicated by the blue arrows



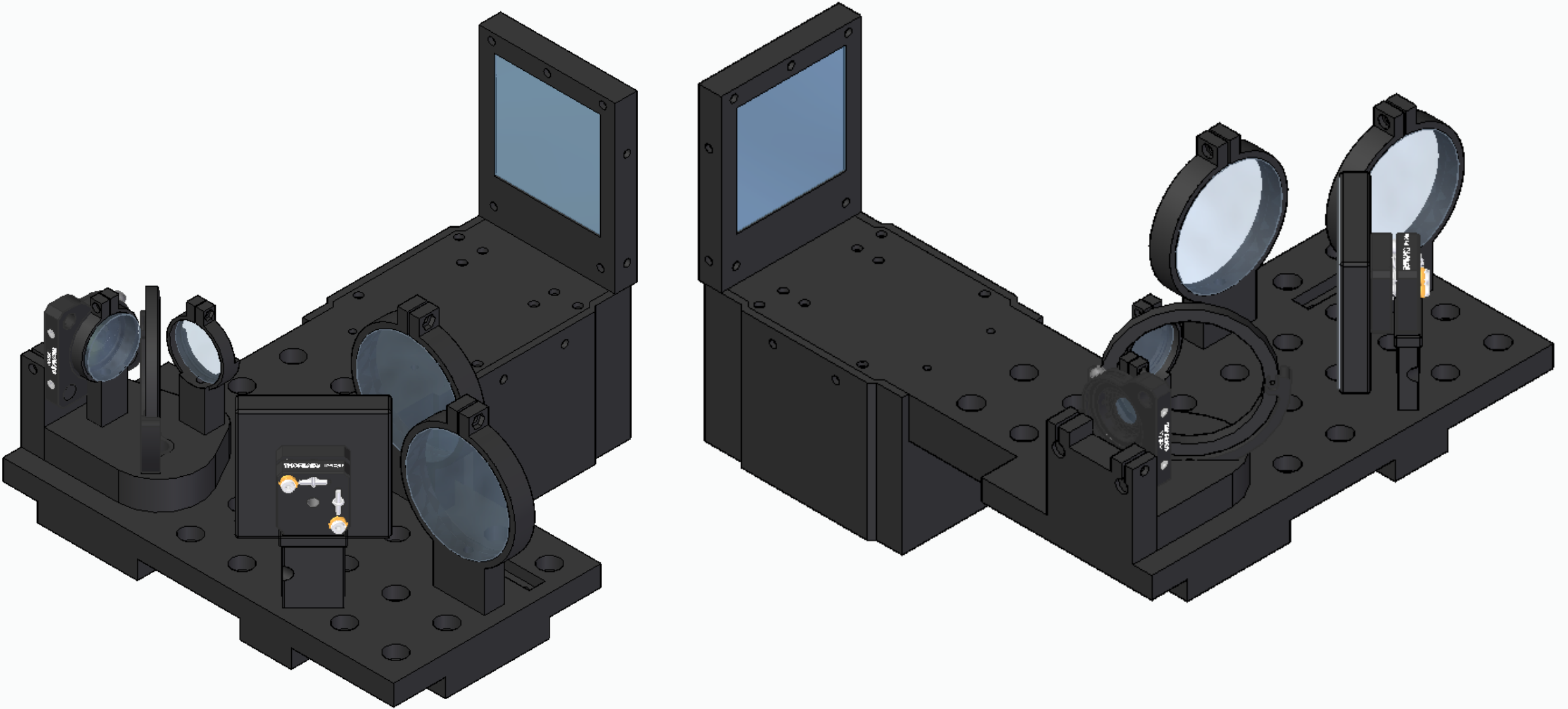
6

Insert both holders tightly into dedicated slots, paying attention to the orientation of the holder.

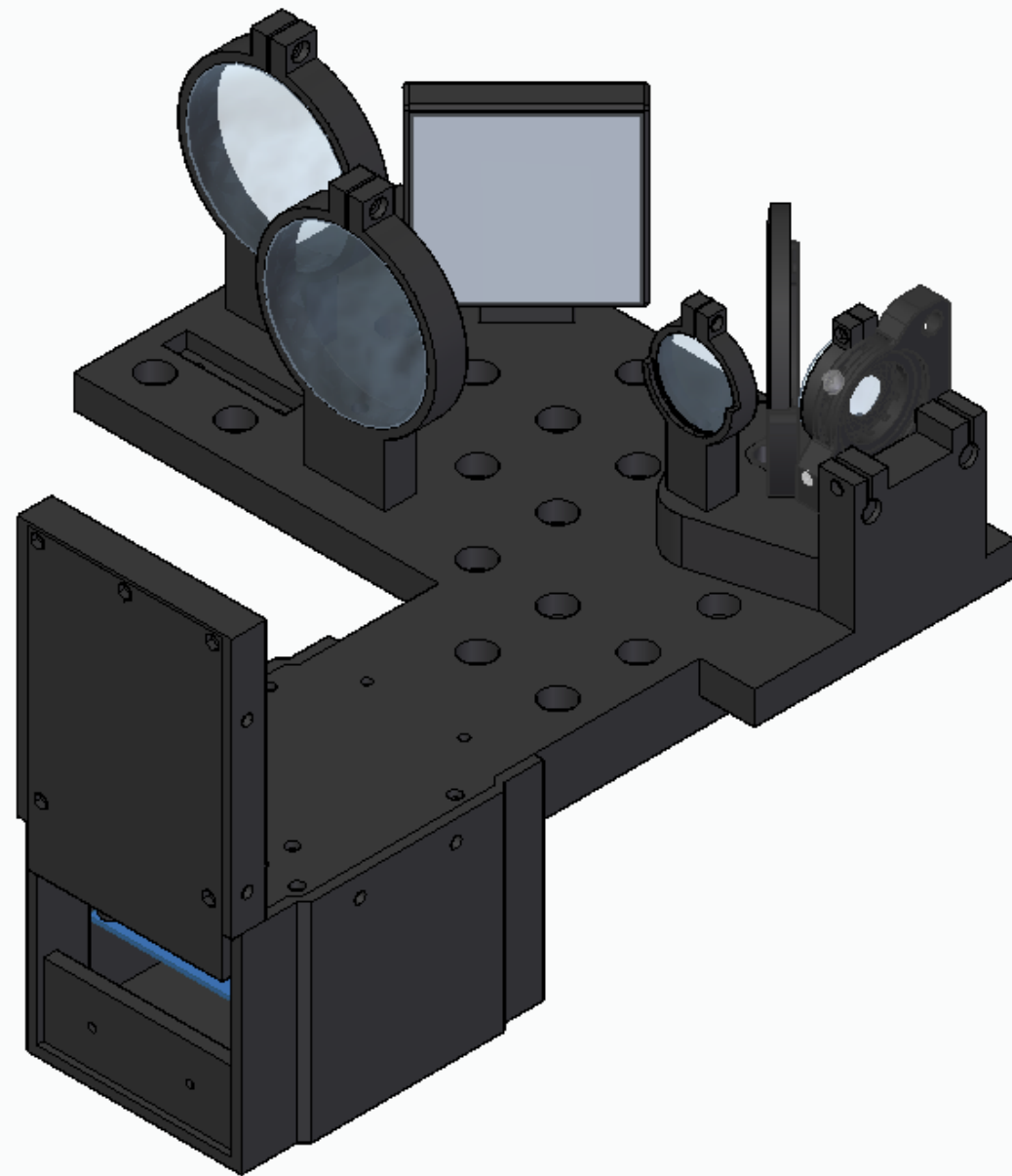
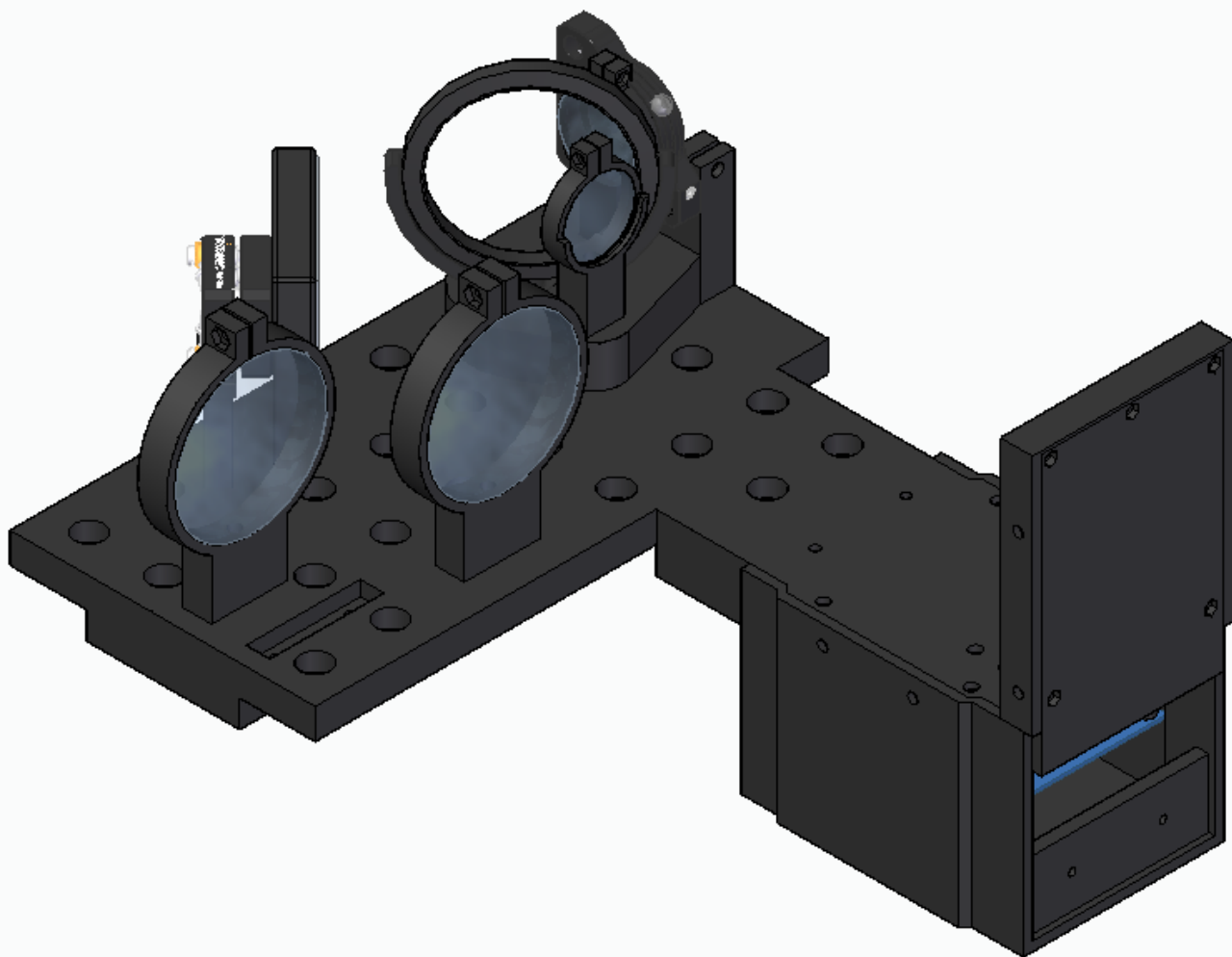
Secure with M3 cap screws inserted from the bottom as indicated by the blue arrows.



STAGE 1 COMPLETE

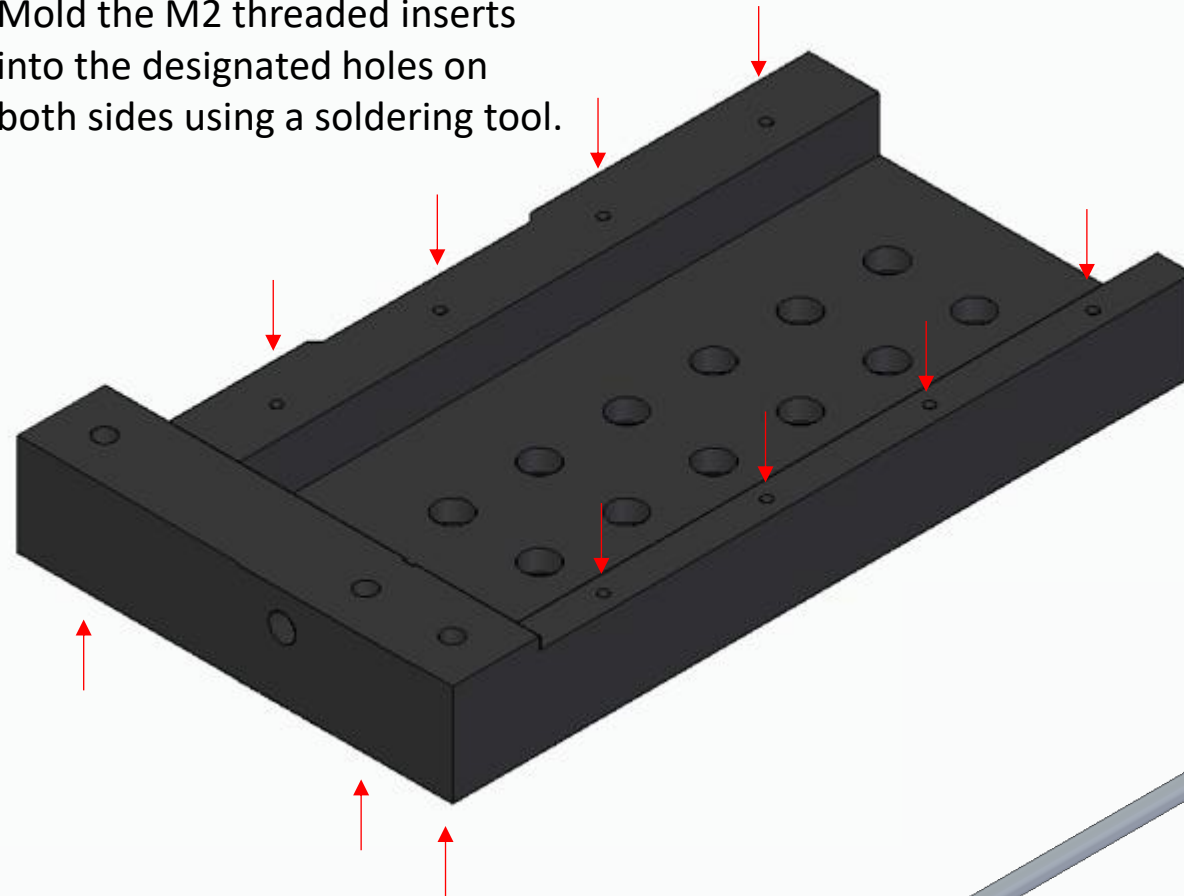


STAGE 1 COMPLETE

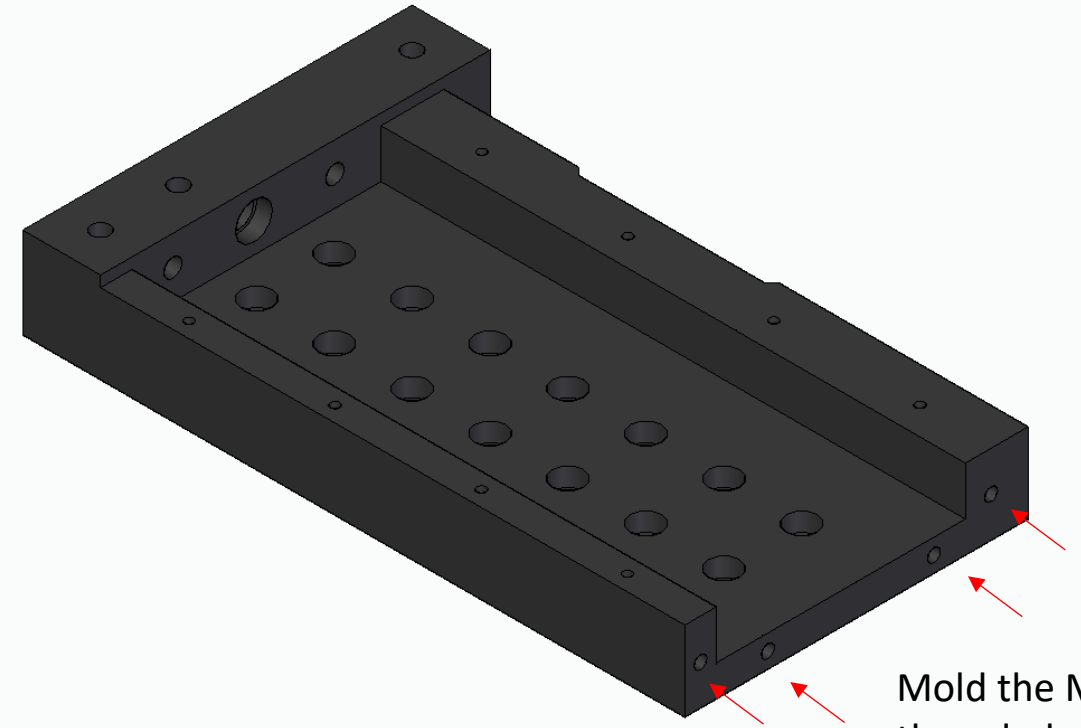


1 × 12_FDML_stimfix_Badal_stage_bottom
2 × model_04_stimfix_Badal_stage_sliding_rod_200mm

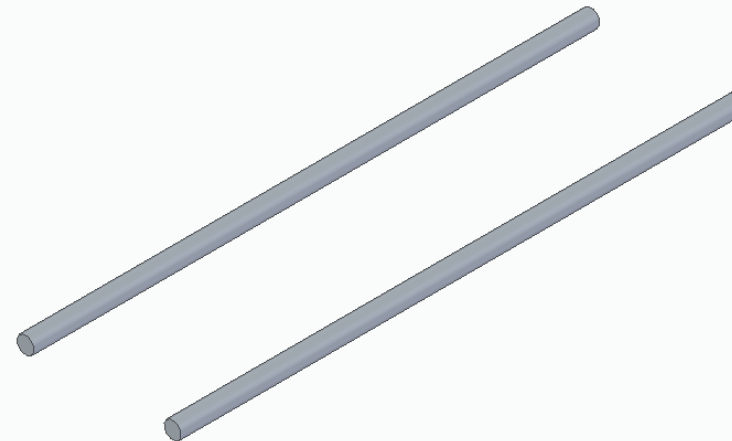
Mold the M2 threaded inserts into the designated holes on both sides using a soldering tool.



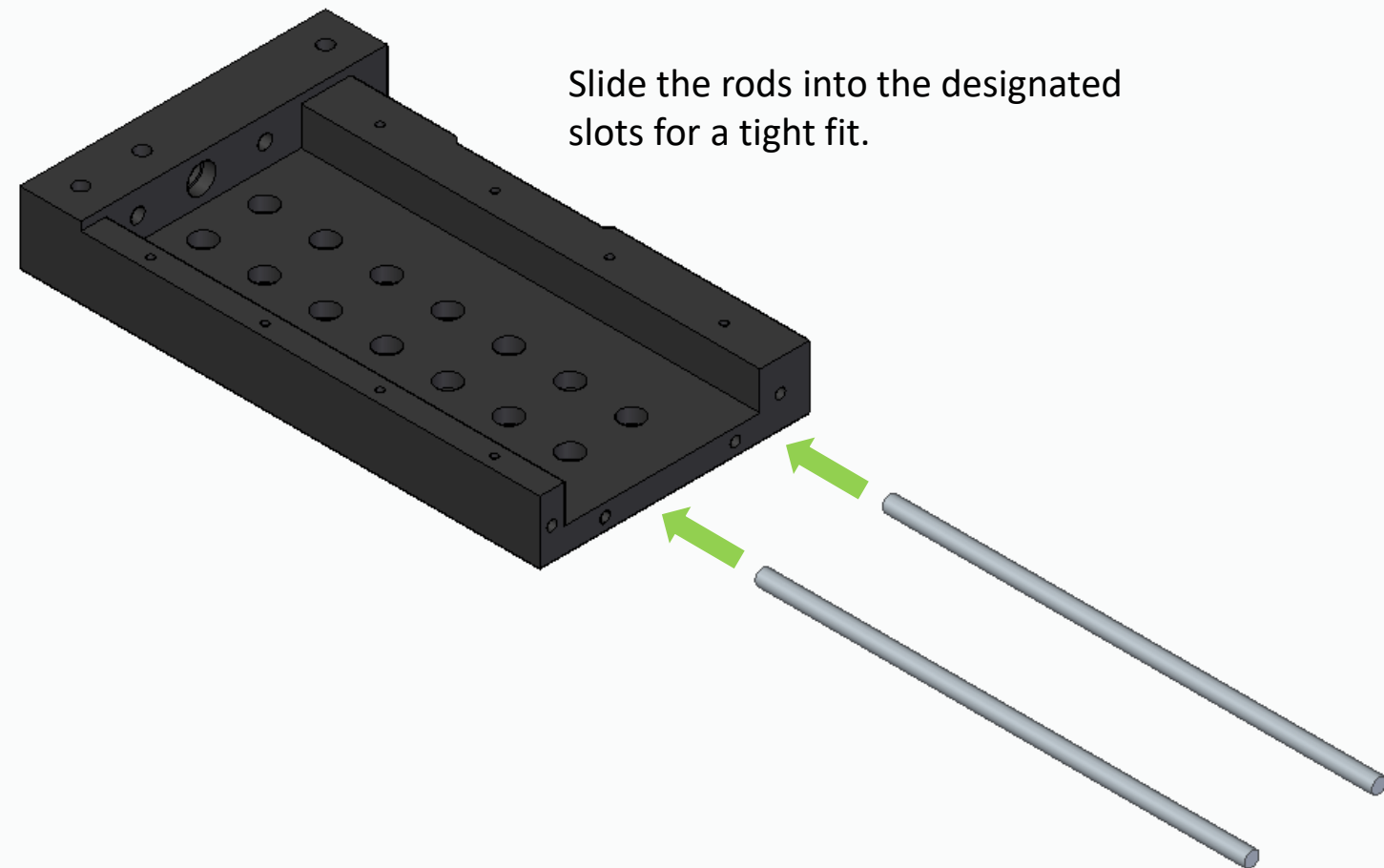
Mold the M6 threaded inserts into the designated holes using a soldering tool.



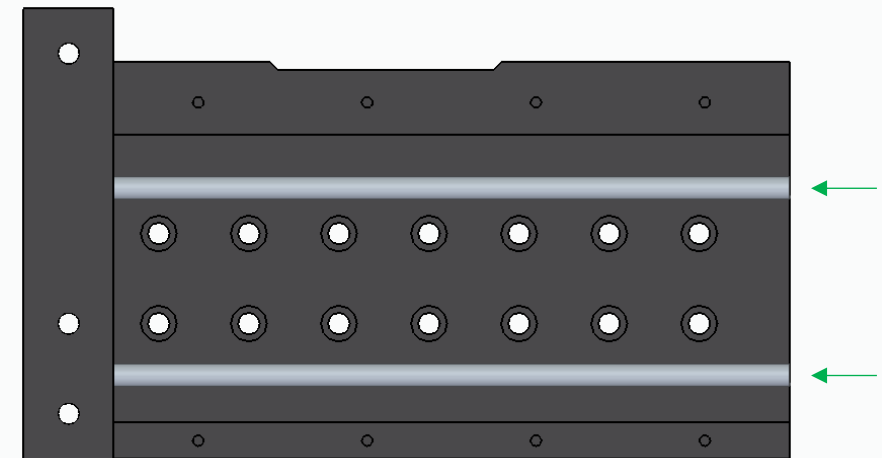
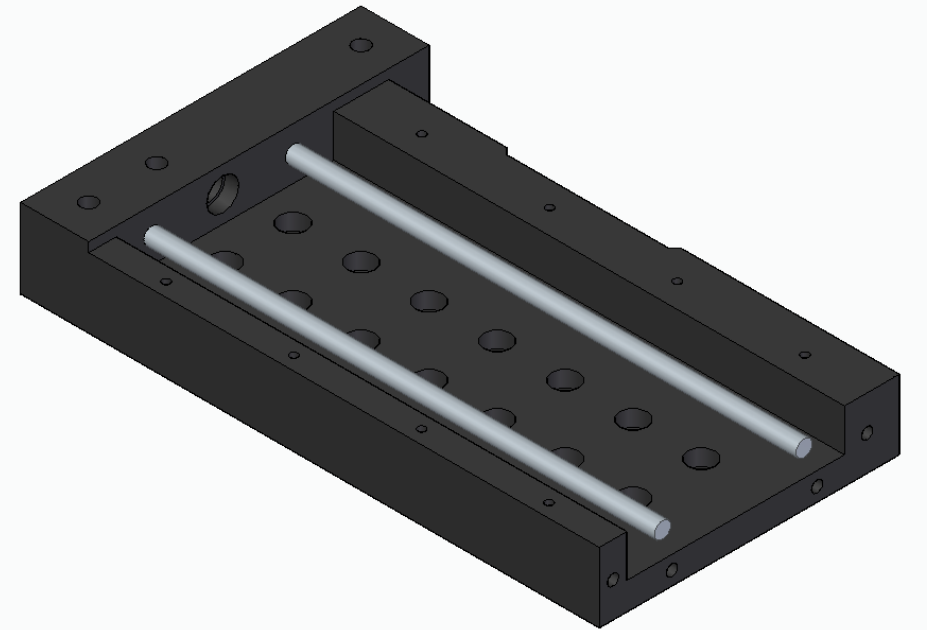
Mold the M3 threaded inserts into the designated holes using a soldering tool.



1

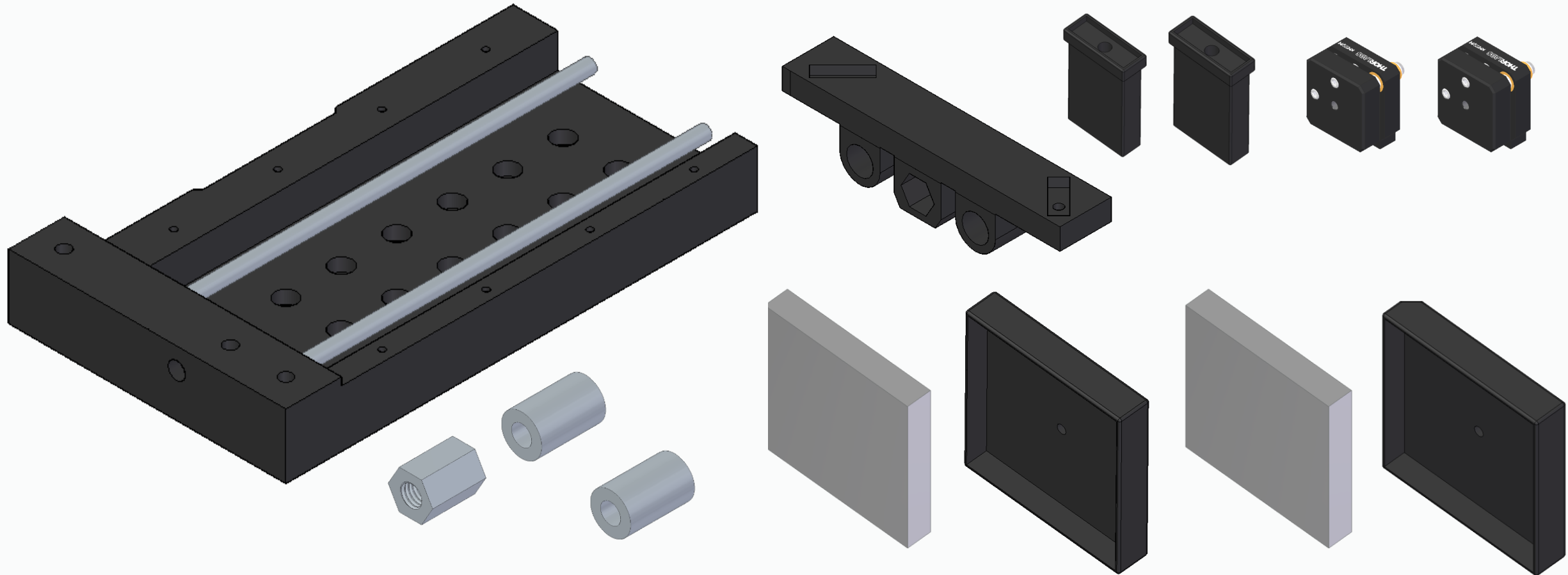


2

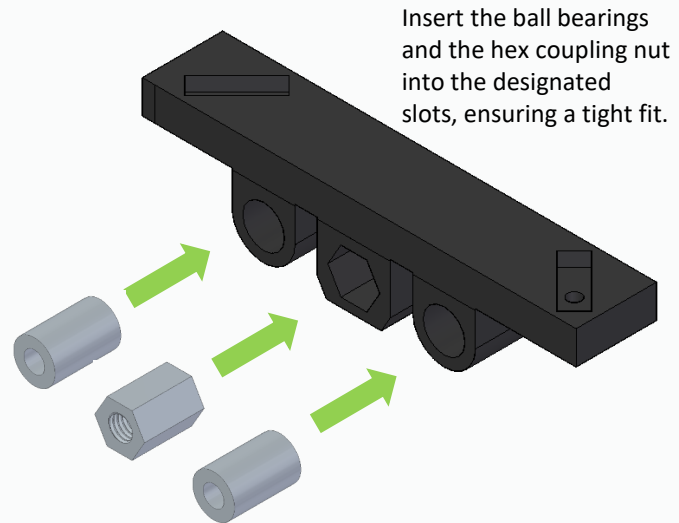


Ensure the end tips of the rods are flush with the encasing

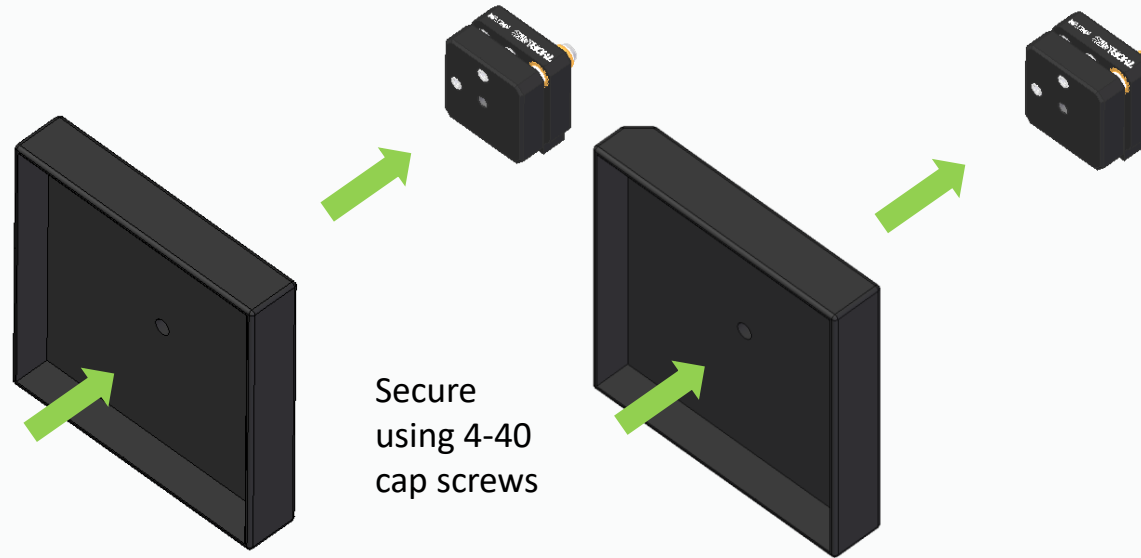
- 1 × 13_FDML_stimfix_Badal_stage_top
- 2 × 14_FDML_stimfix_square_mirror_support_Badal_stage
- 2 × Thorlabs_KMSS_M (<https://www.thorlabs.com/thorproduct.cfm?partnumber=KMSS/M>)
- 1 × 15_FDML_stimfix_square_mirror_holder_large
- 1 × 16_FDML_stimfix_square_mirror_holder_large_cut_corner
- 2 × EdmundOptics_88541 (<https://www.edmundoptics.com/p/75-x-75mm-protected-silver-lambda4-mirror/30799/>)
- 1 × model_05_hex_coupling_nut
- 2 × model_06_linear_ball_bearing_6mm



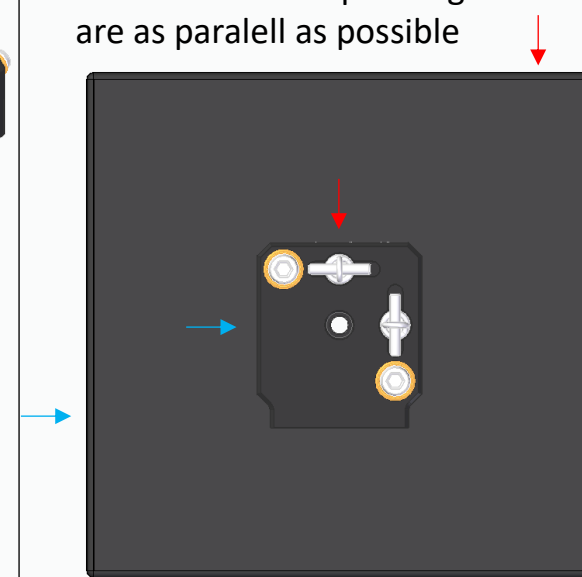
1



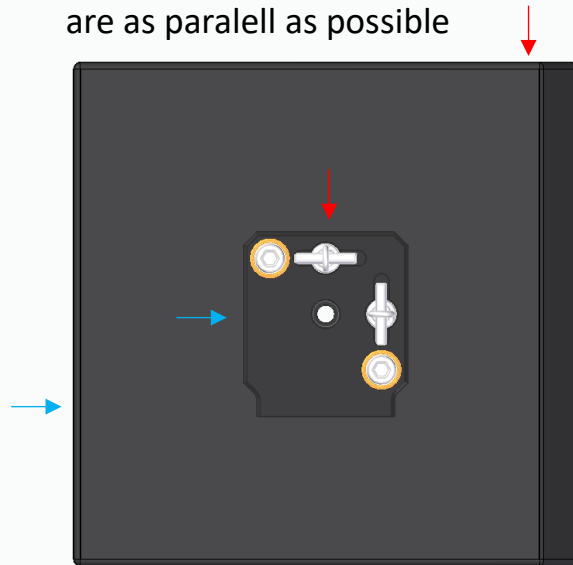
2



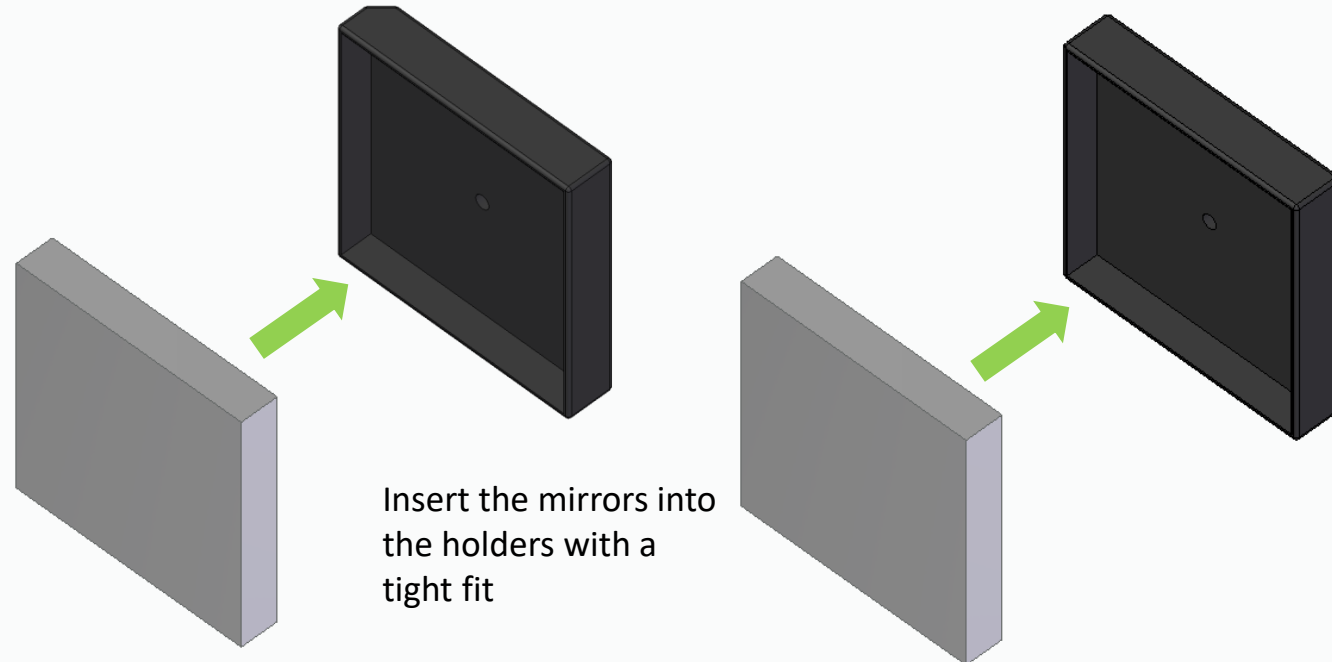
3 Ensure the corresponding surfaces are as parallel as possible



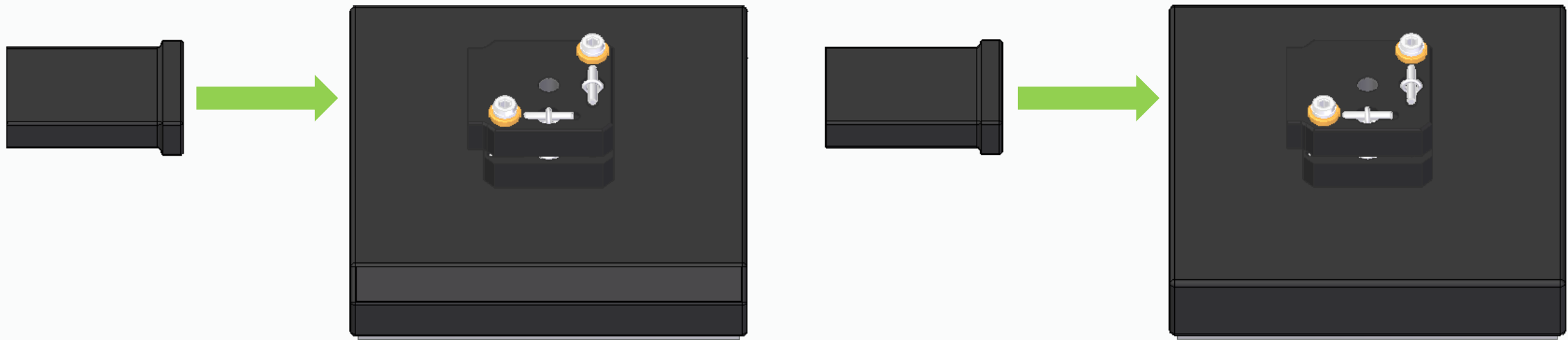
Ensure the corresponding surfaces are as parallel as possible



4

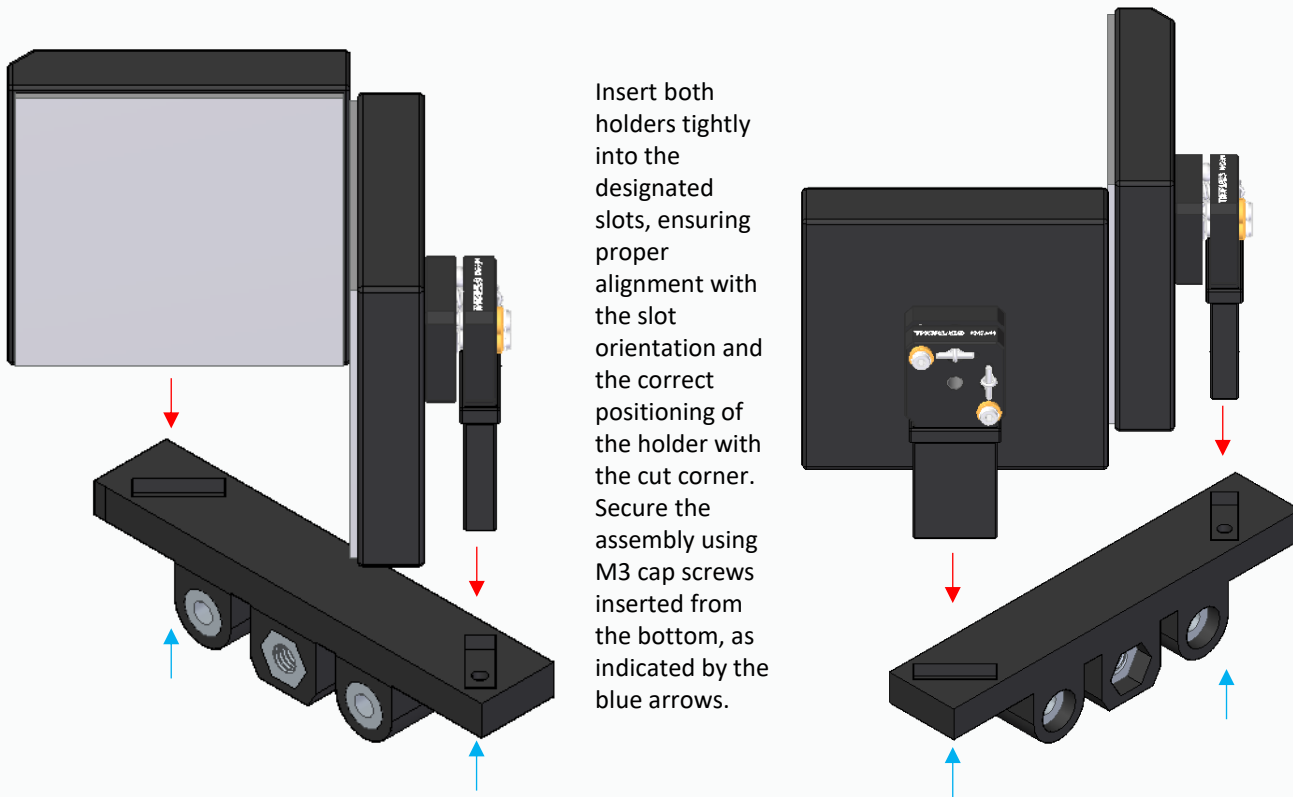


1

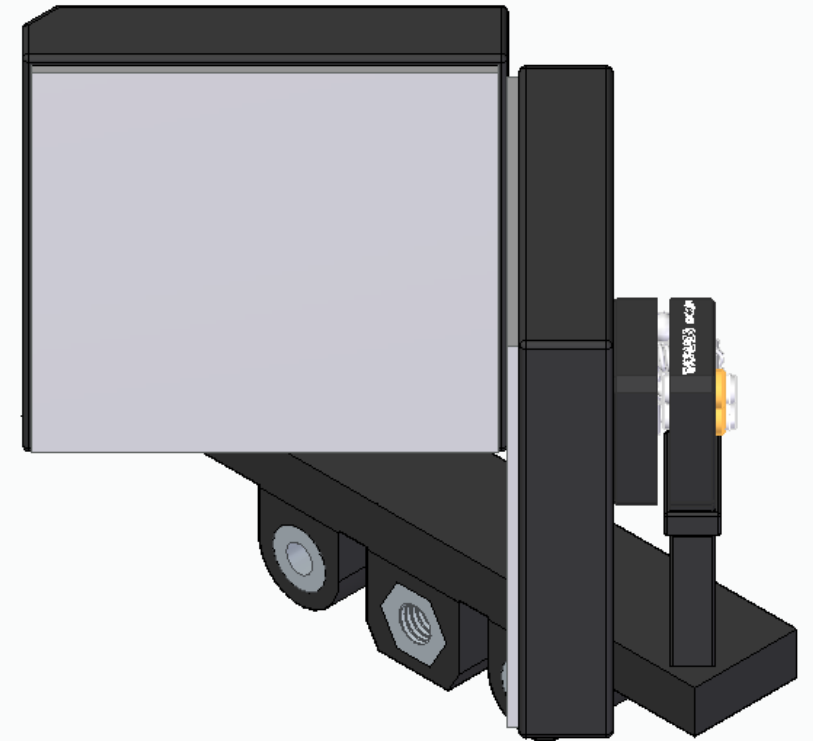


Slide the kinematic mounts into the post holders as tight as possible. Be careful as it may fall off – final fixing is with the M4 cap screw in the next step

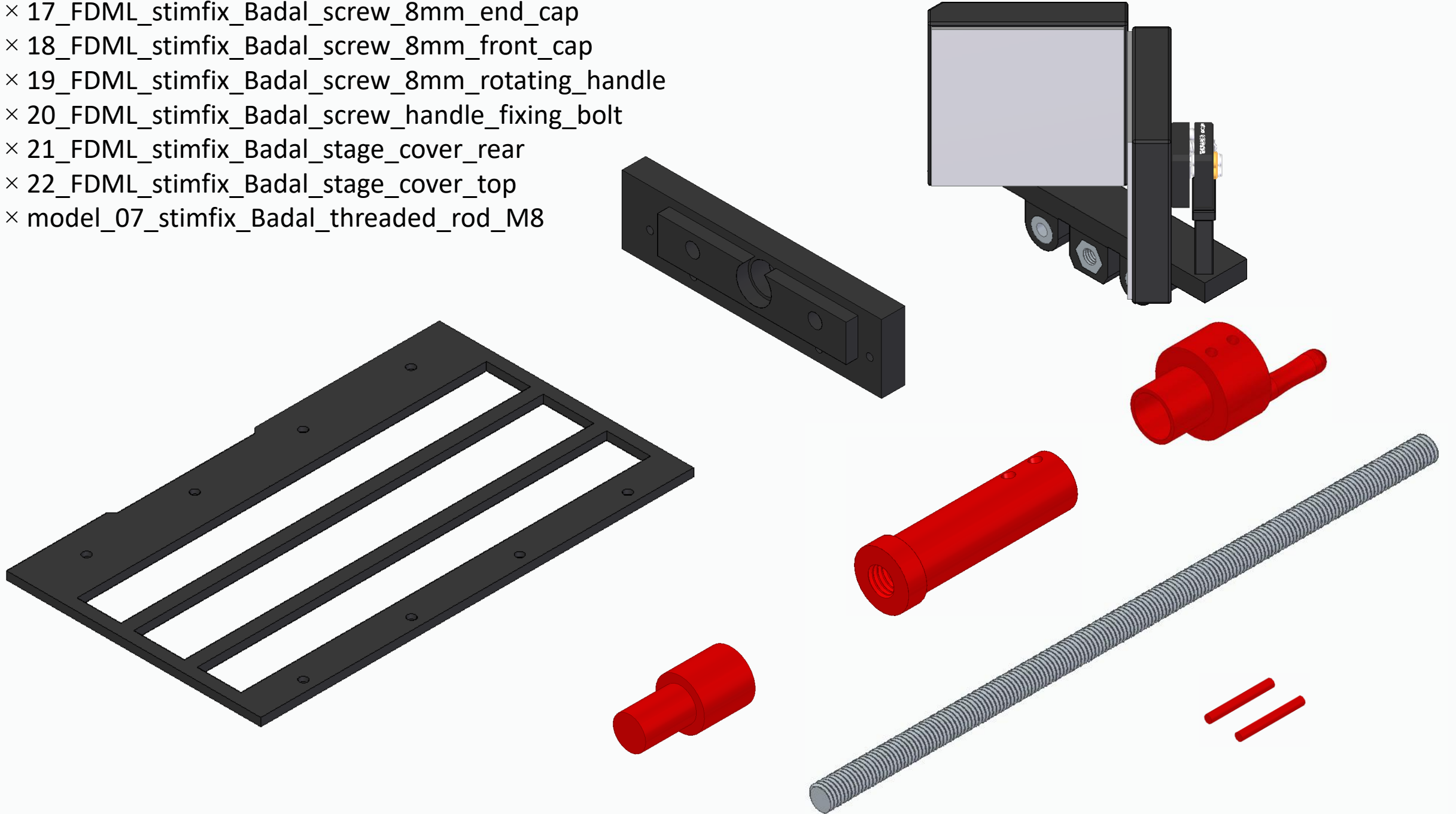
2



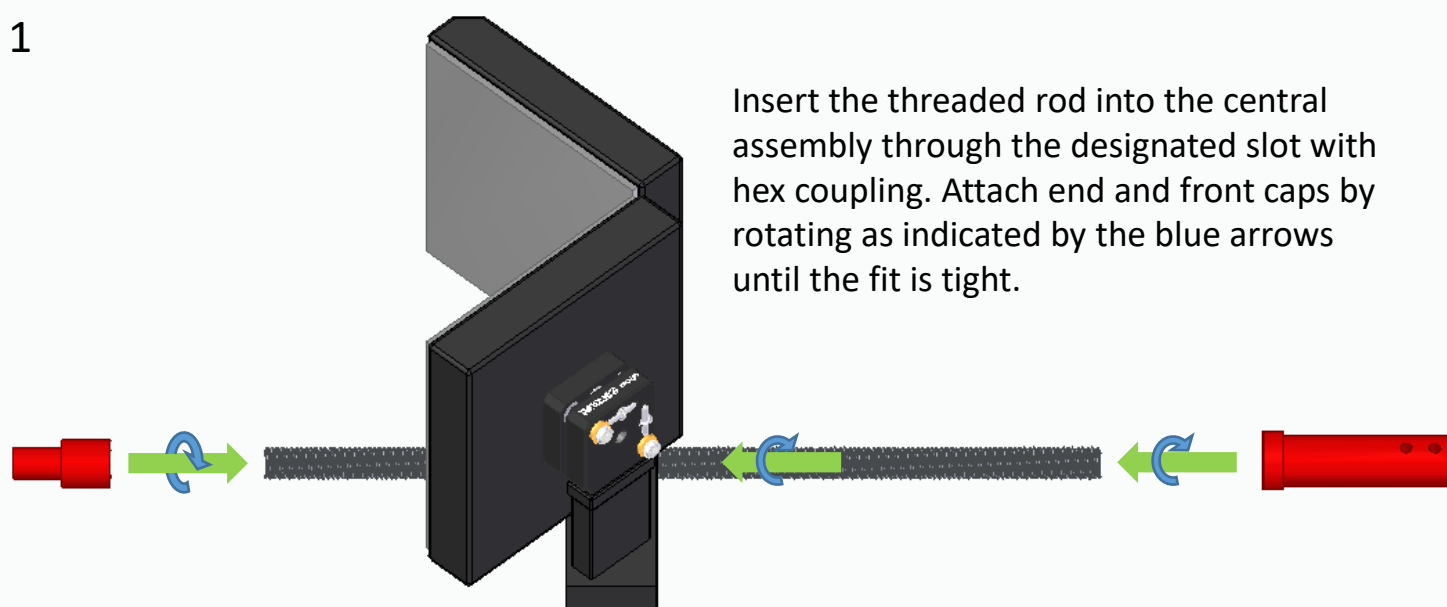
3



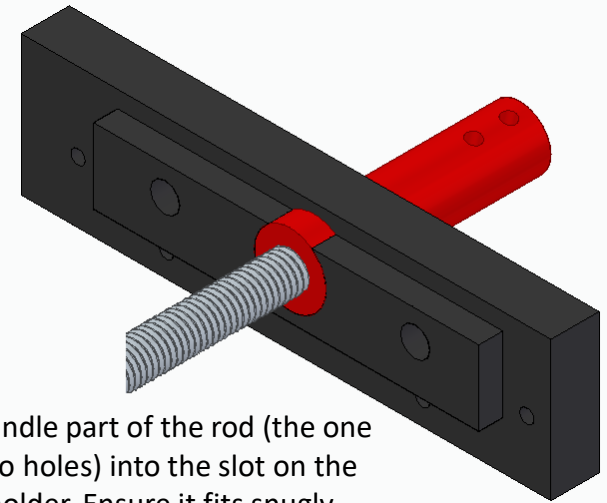
- 1 × 17_FDML_stimfix_Badal_screw_8mm_end_cap
- 1 × 18_FDML_stimfix_Badal_screw_8mm_front_cap
- 1 × 19_FDML_stimfix_Badal_screw_8mm_rotating_handle
- 2 × 20_FDML_stimfix_Badal_screw_handle_fixing_bolt
- 1 × 21_FDML_stimfix_Badal_stage_cover_rear
- 1 × 22_FDML_stimfix_Badal_stage_cover_top
- 1 × model_07_stimfix_Badal_threaded_rod_M8



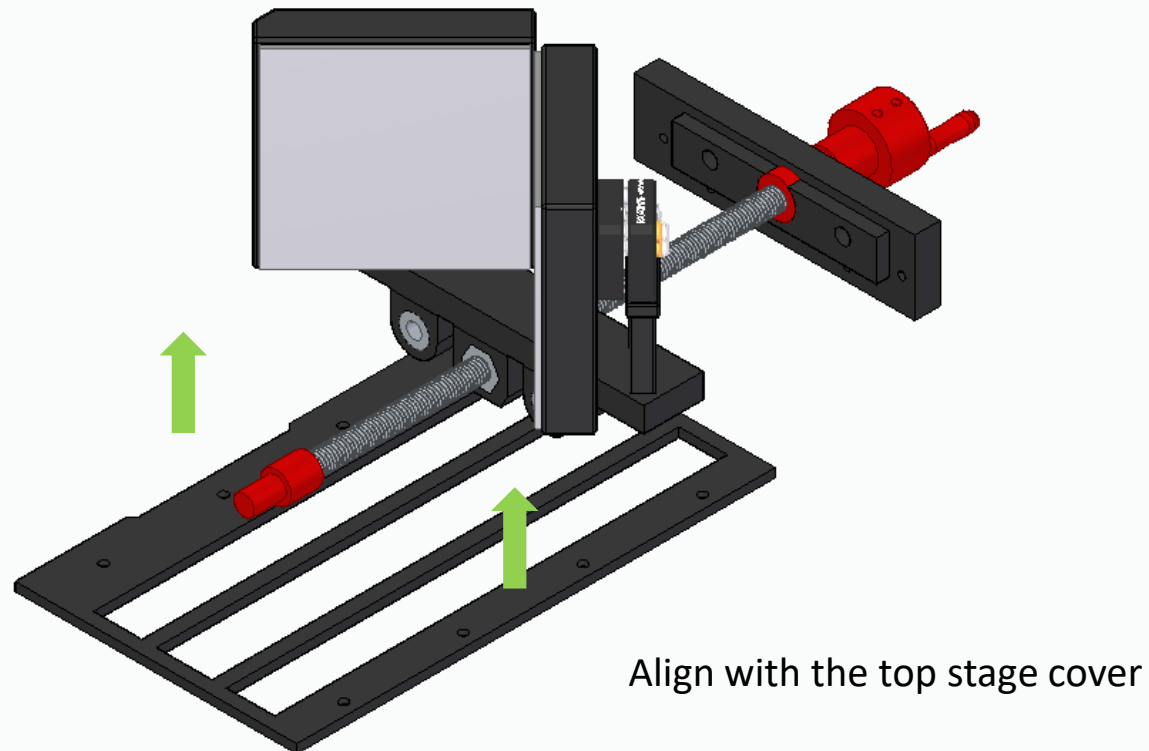
1



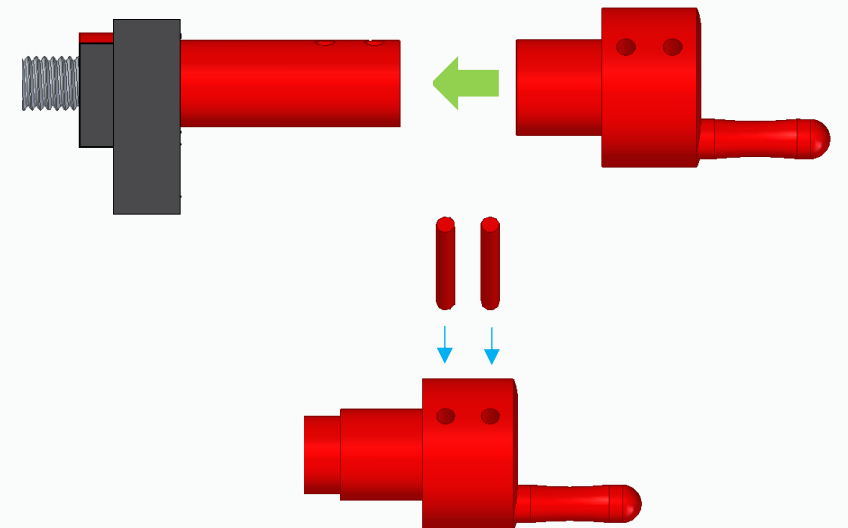
2



4



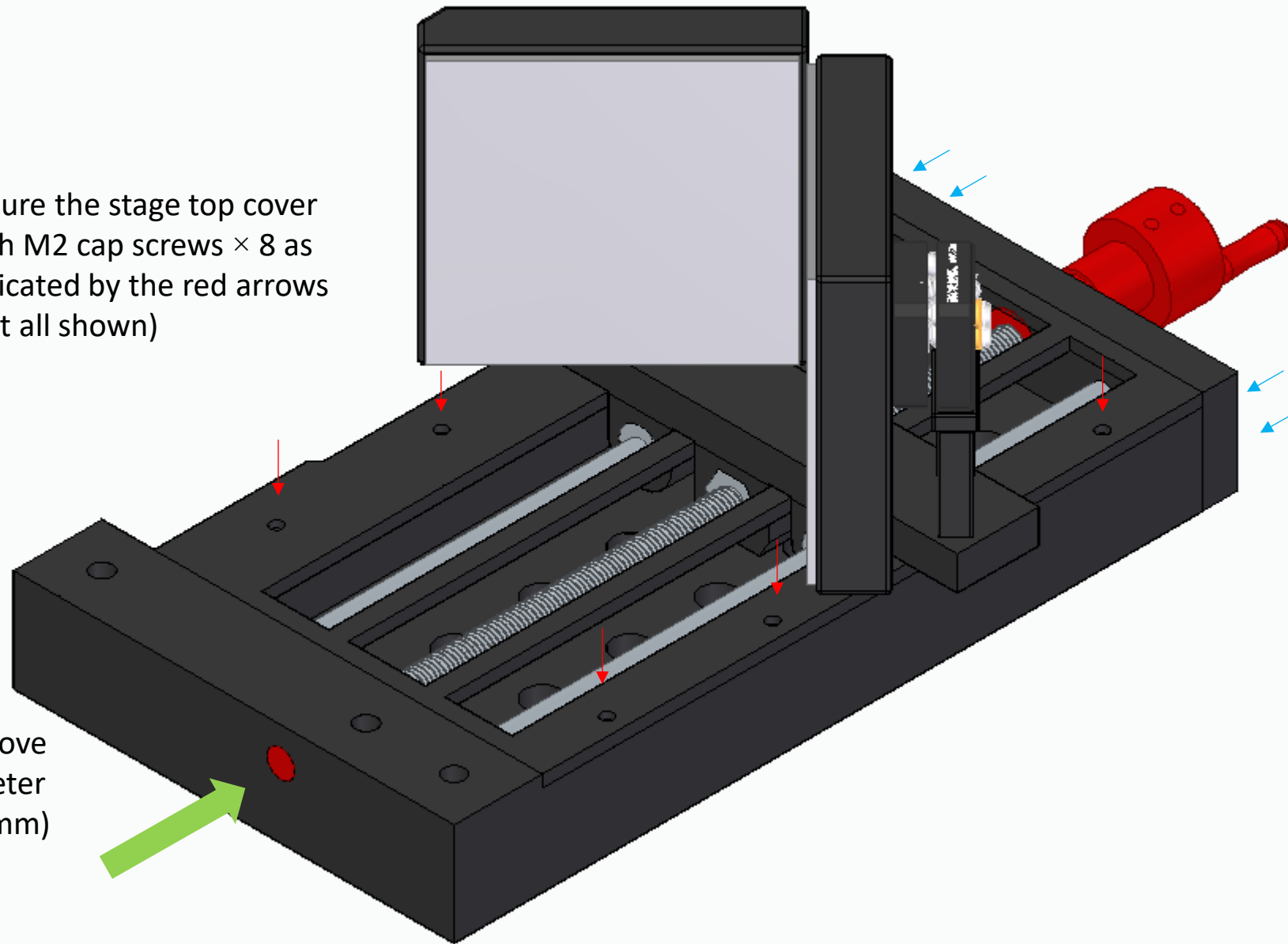
3



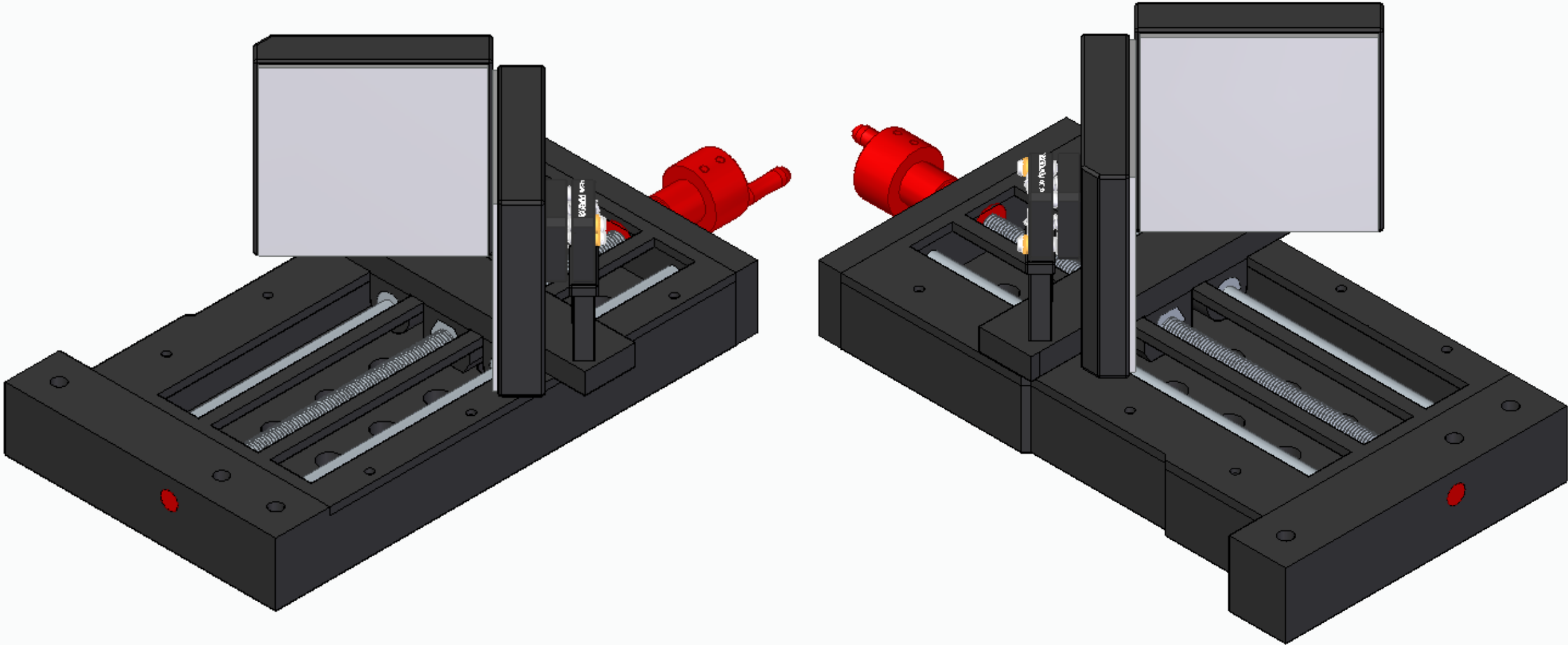
Secure the stage top cover
with M2 cap screws $\times 8$ as
indicated by the red arrows
(not all shown)

Secure the stage
rear cover with
M3 cap screws $\times 4$
as indicated by the
blue arrows

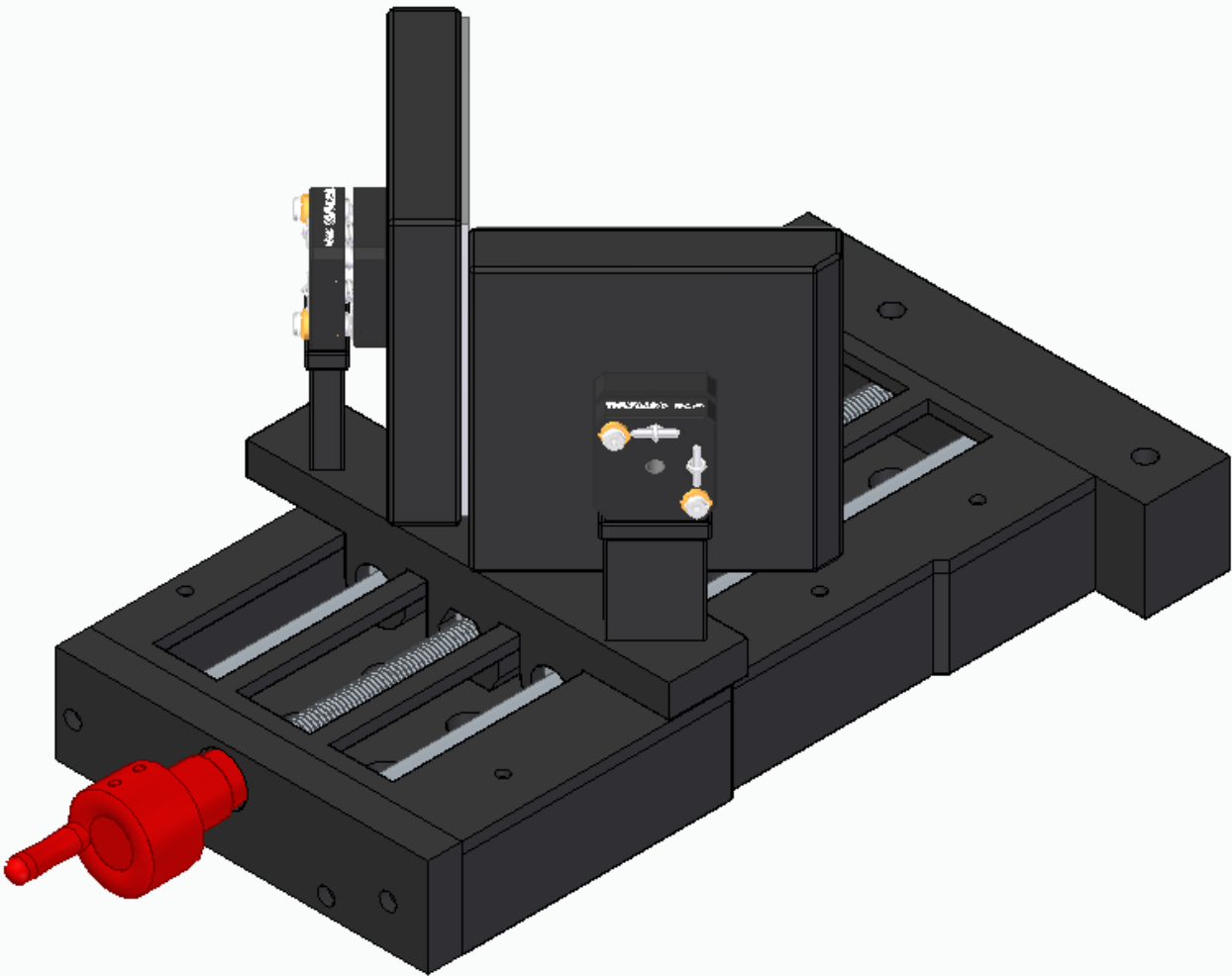
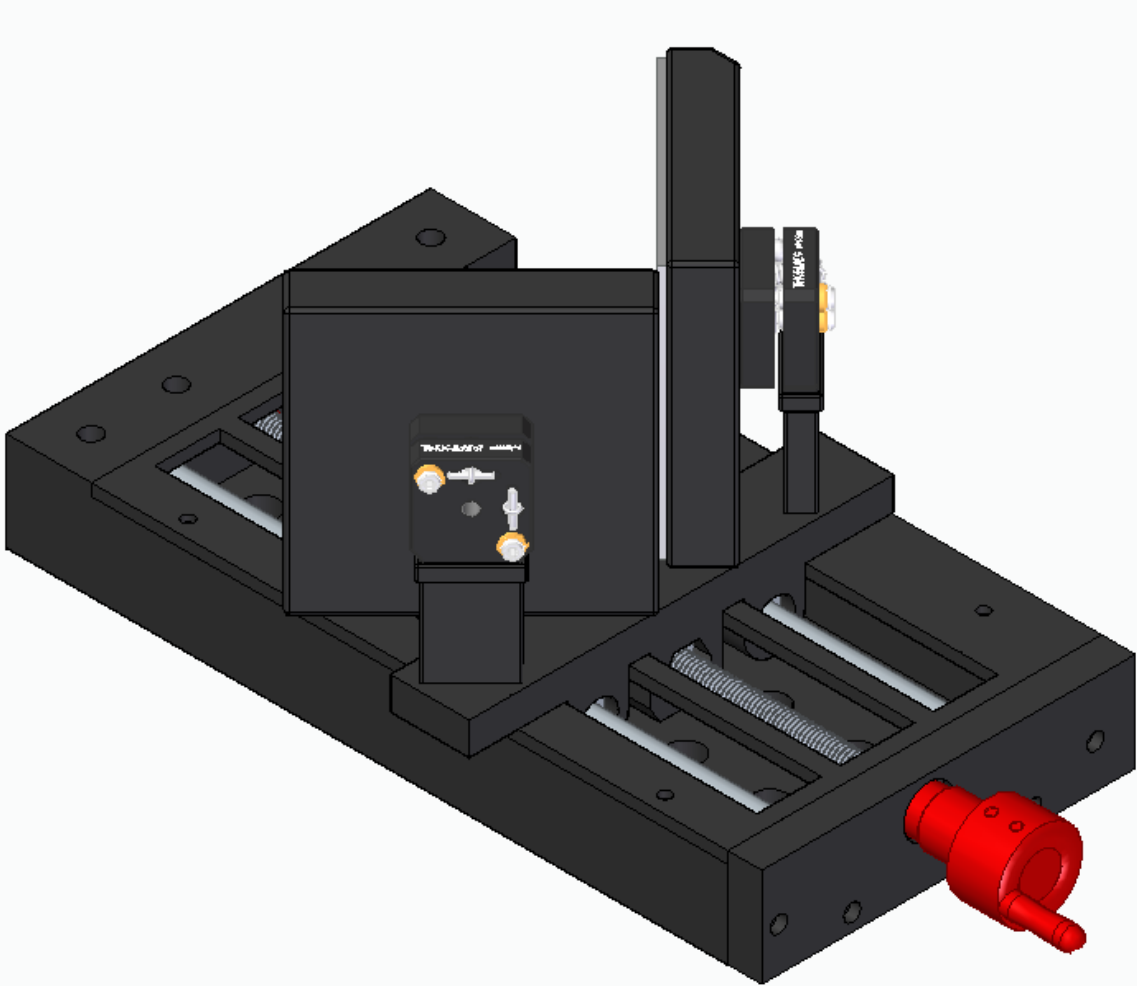
Insert and secure deep groove
ball bearings (inner diameter
8mm, outer diameter 12mm)



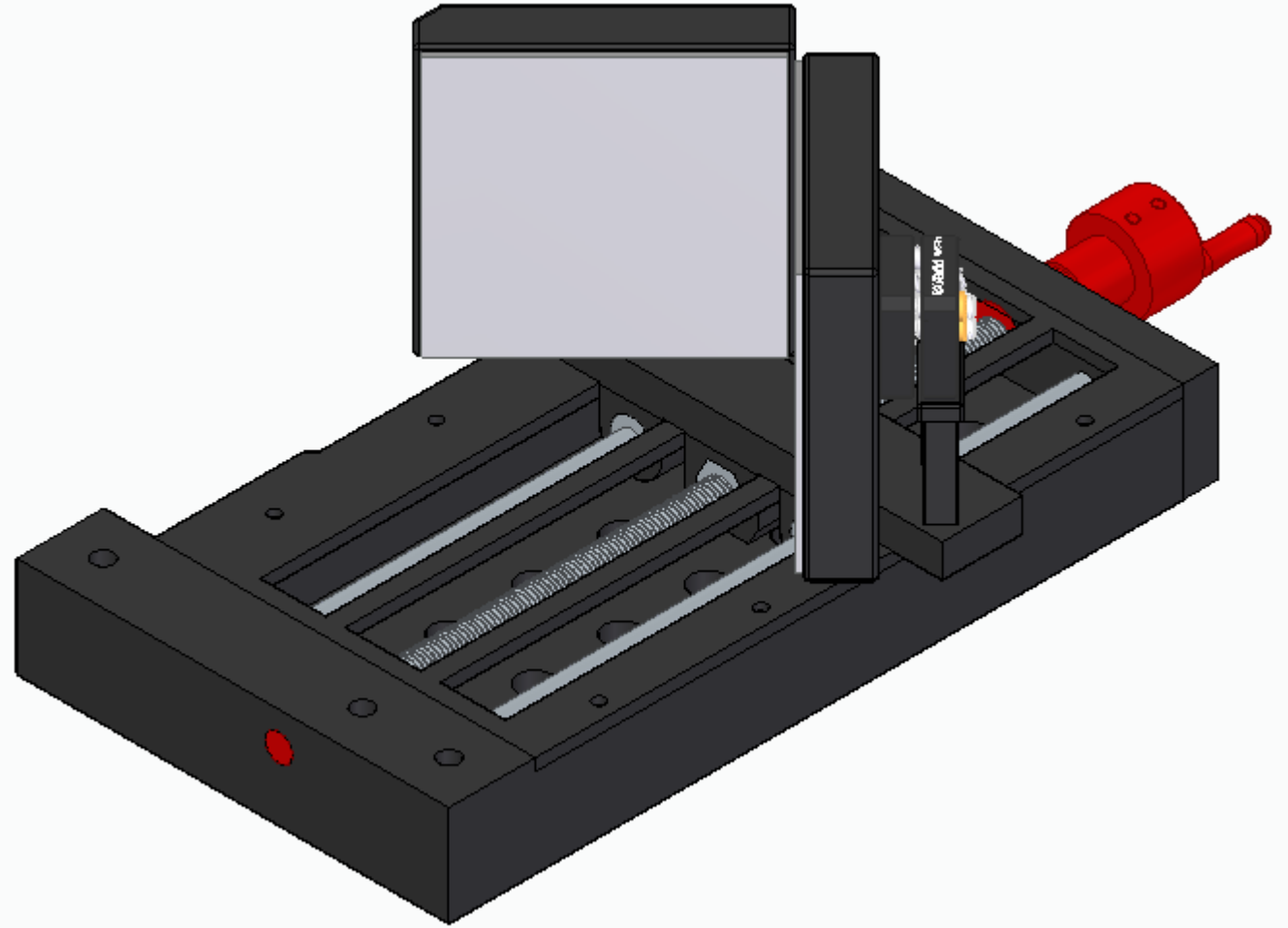
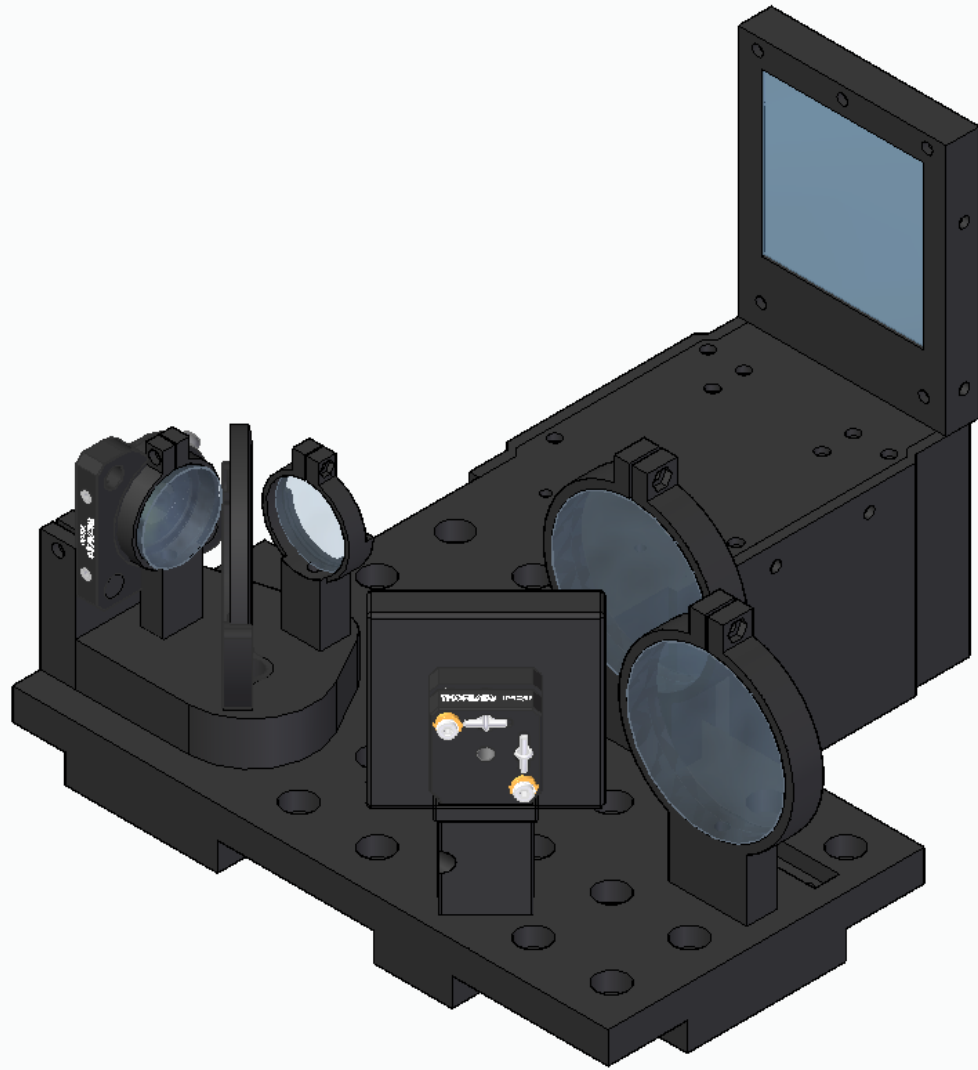
STAGE 2 COMPLETE

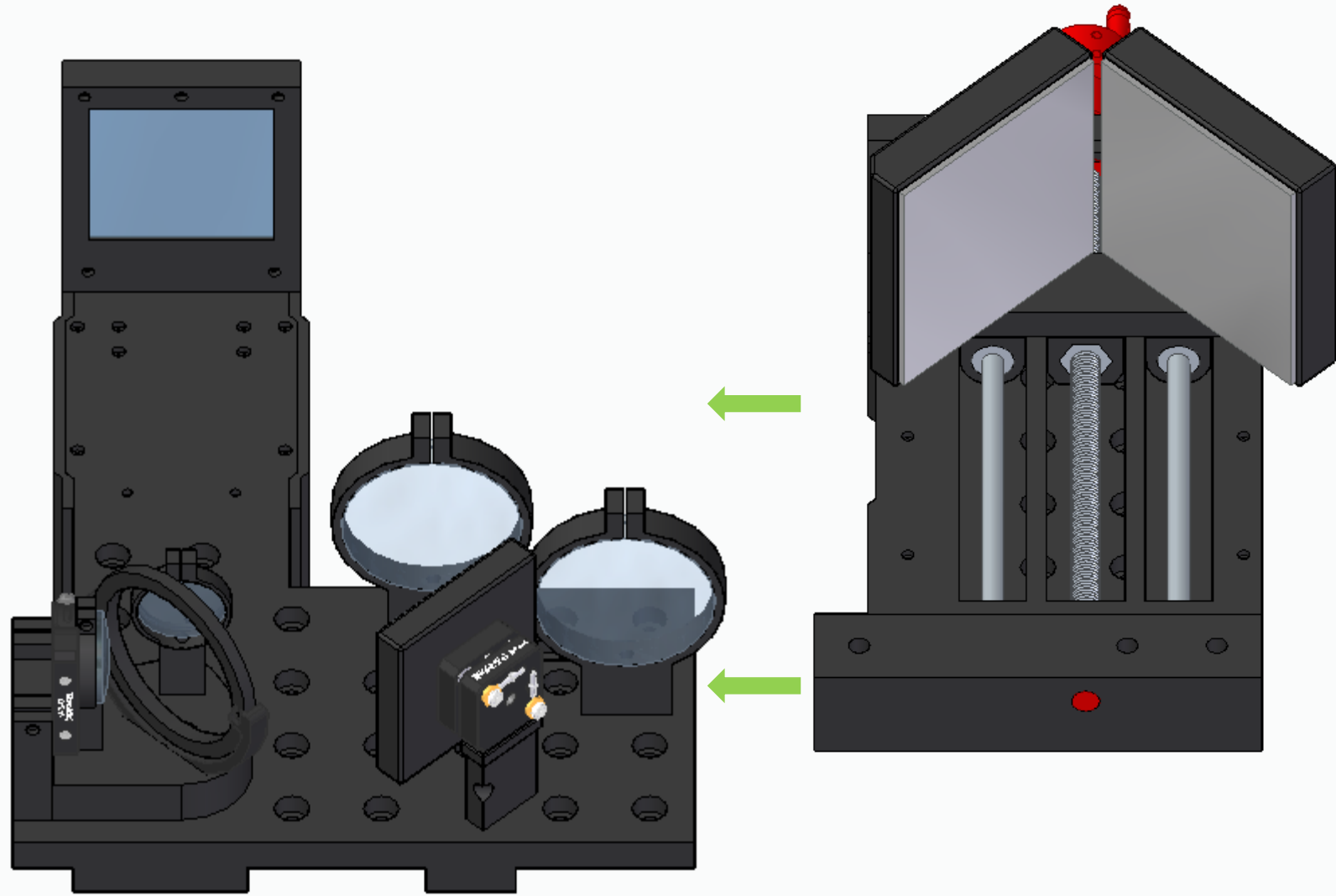


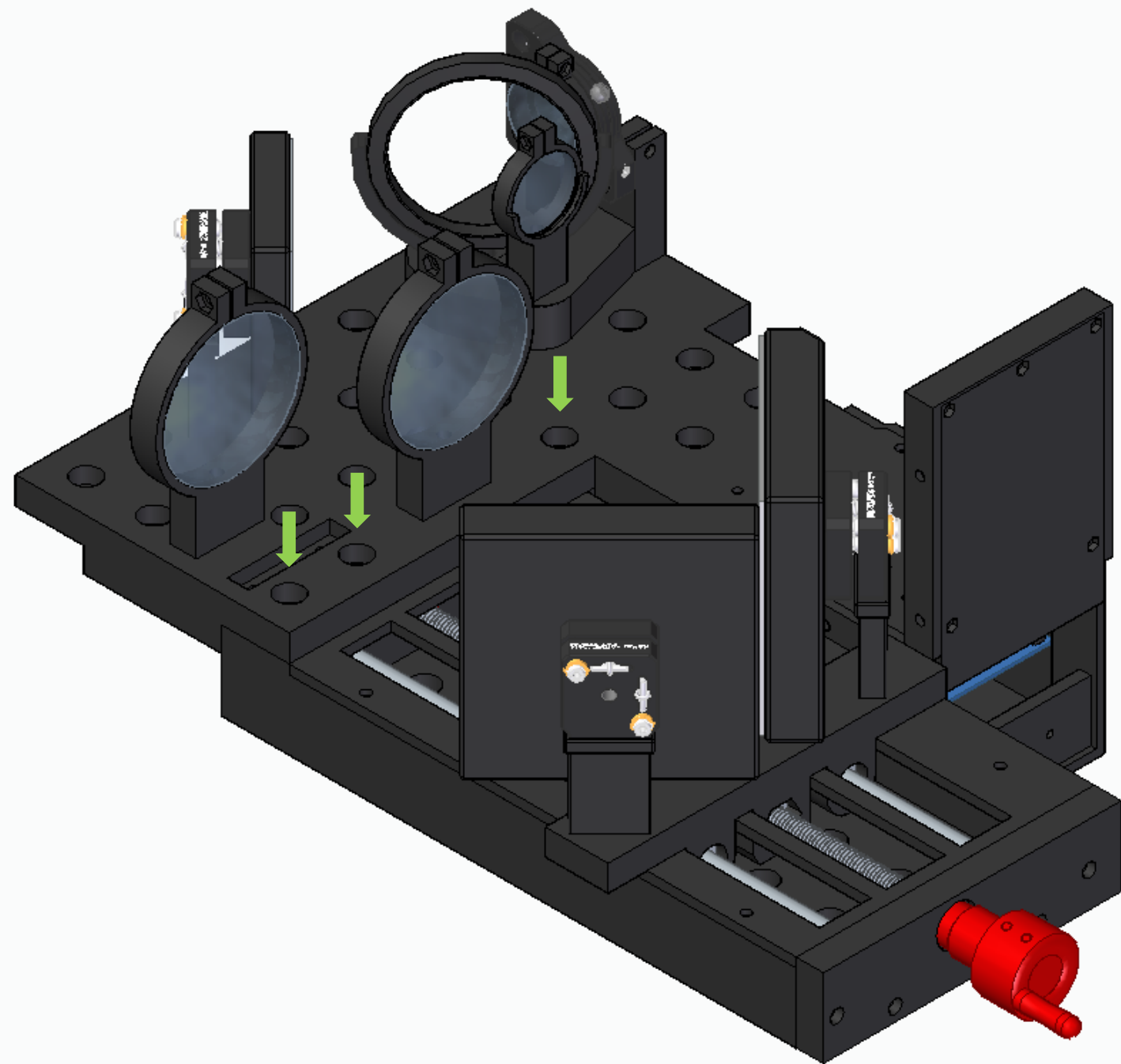
STAGE 2 COMPLETE



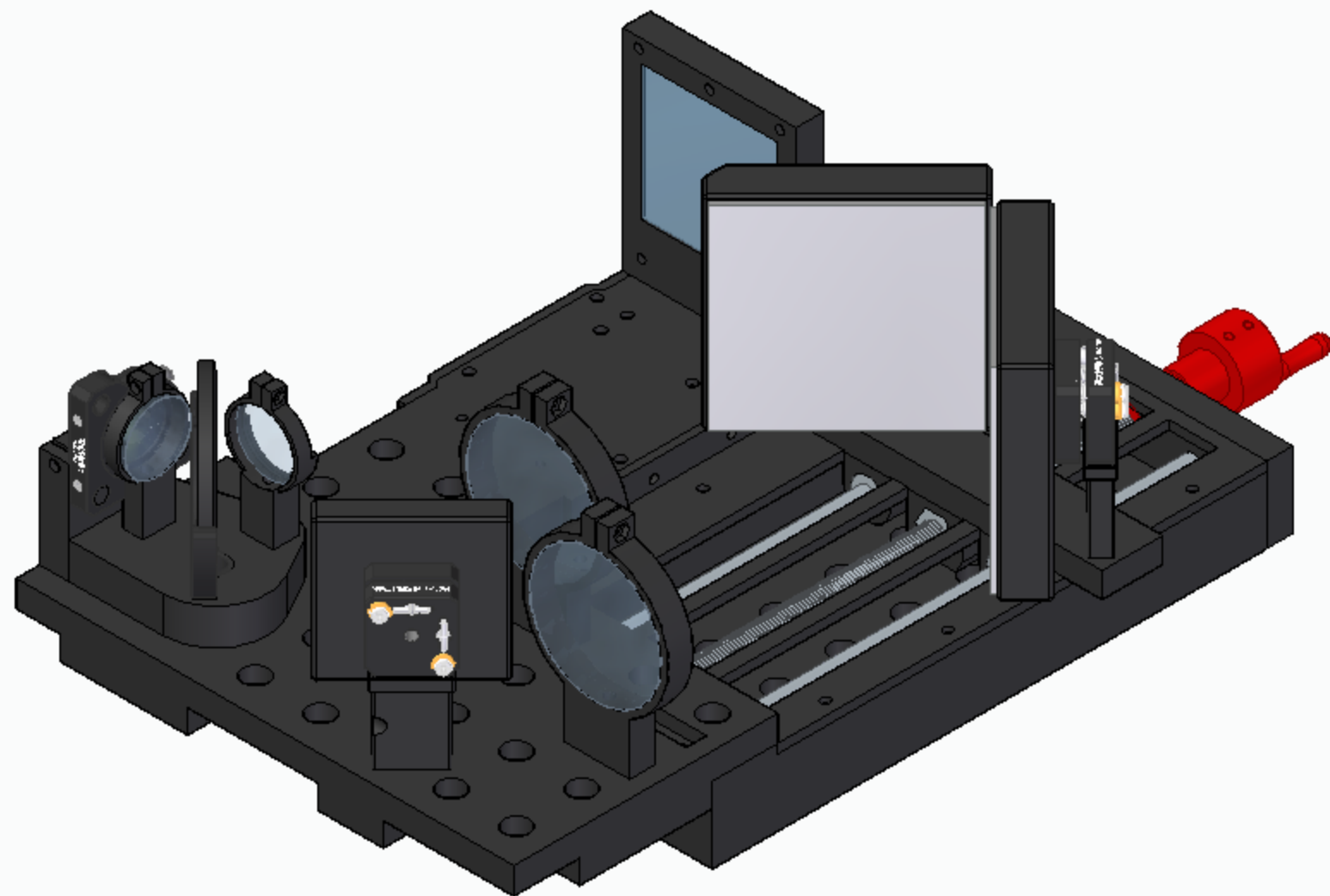
1 × STAGE 1 assembly
1 × STAGE 2 assembly





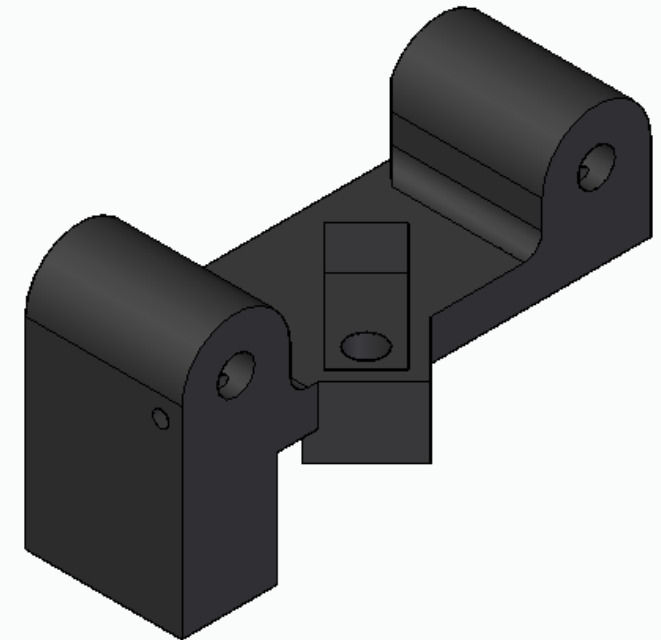
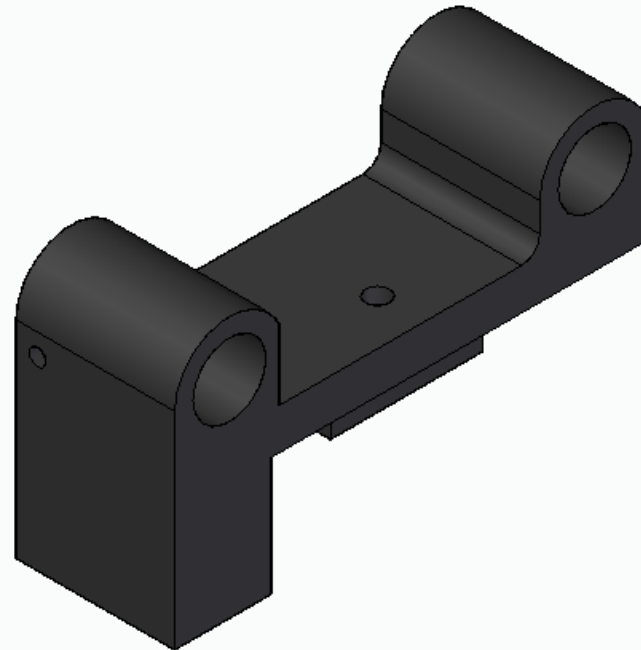
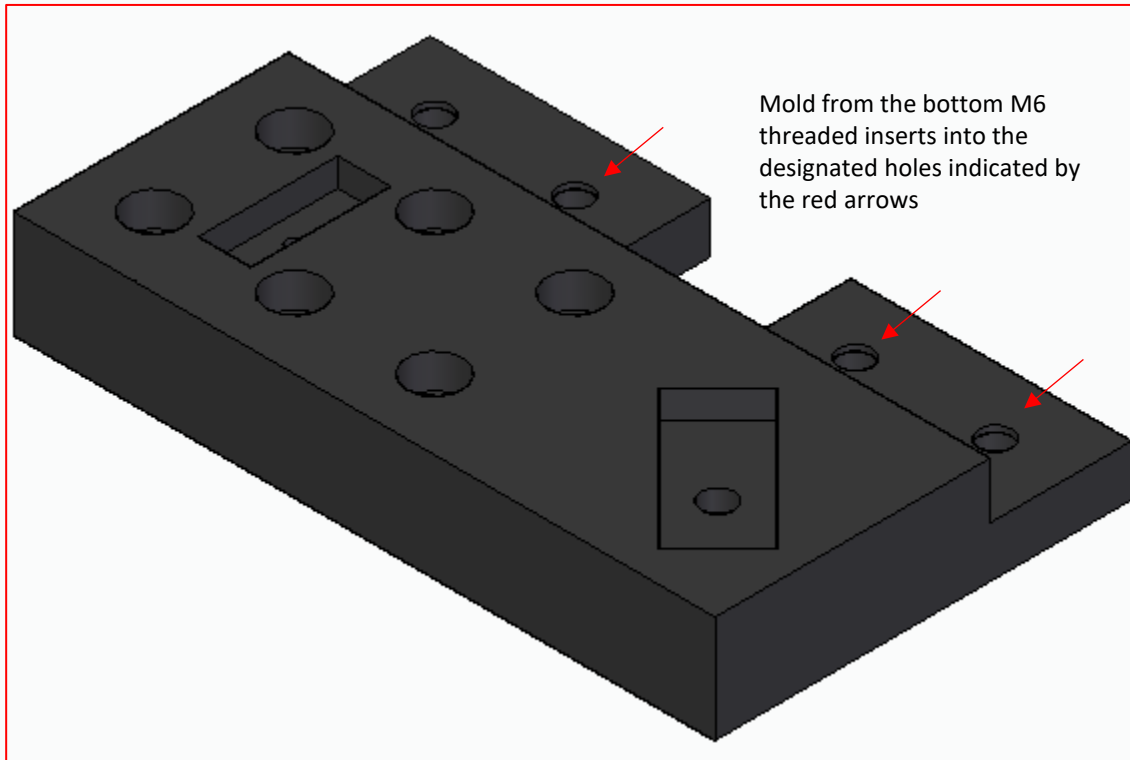
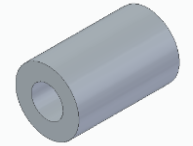
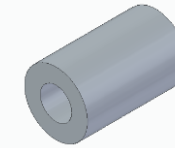
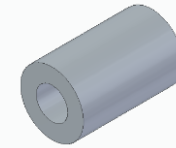
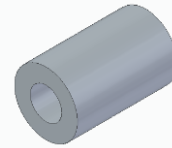
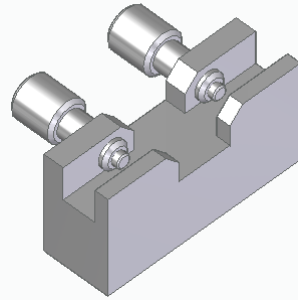


STAGE 1 + STAGE 2 COMPLETE

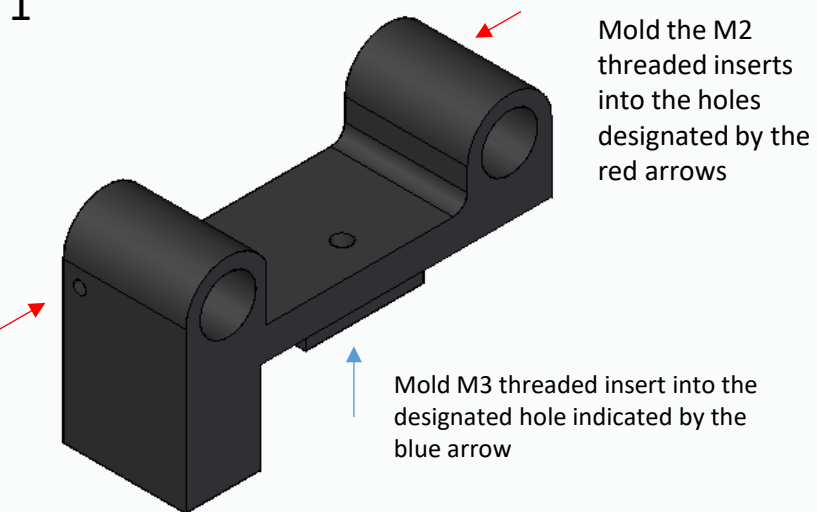


- 1 × 23_FDML_stimfix_dichroic_board
- 1 × 24_FDML_stimfix_dichroic_holder
- 1 × 25_FDML_rods_guiding_addon
- 4 × model_06_linear_ball_bearing_6mm
- 1 × EdmundOptics_54996

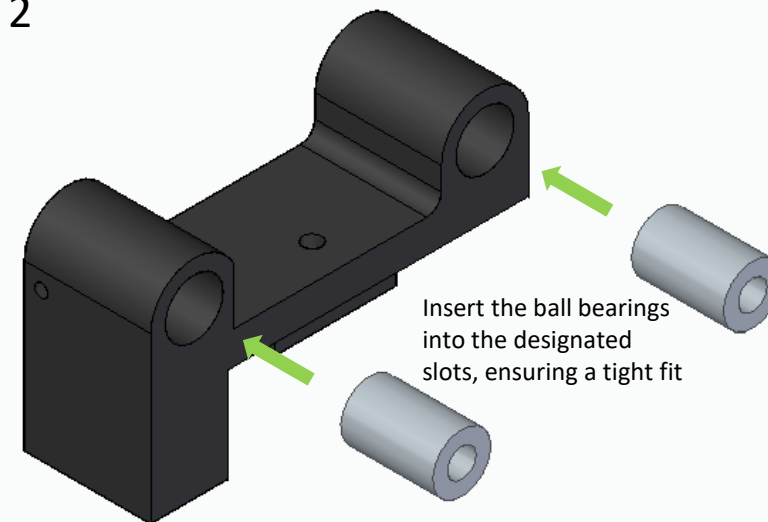
Attach element 23 to the
STAGE 1 + STAGE 2 assembly
first. Then proceed as the
instructions indicate.



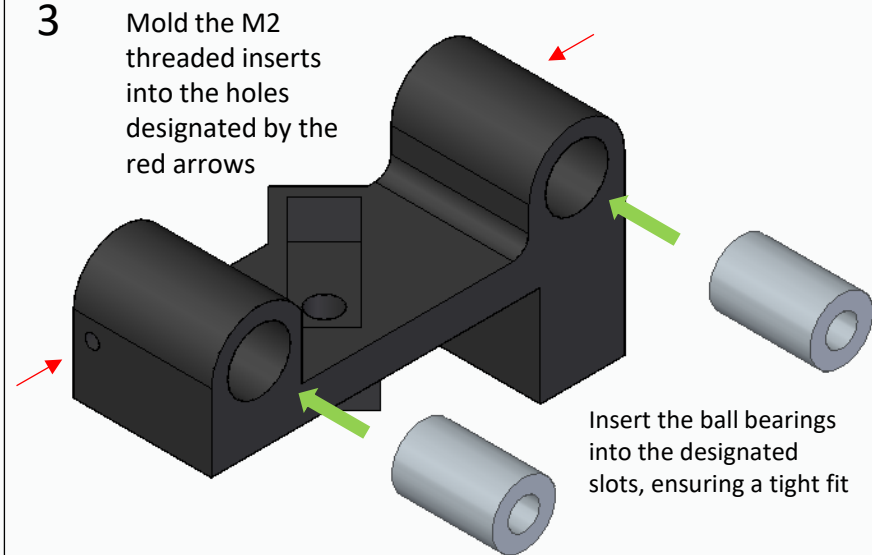
1



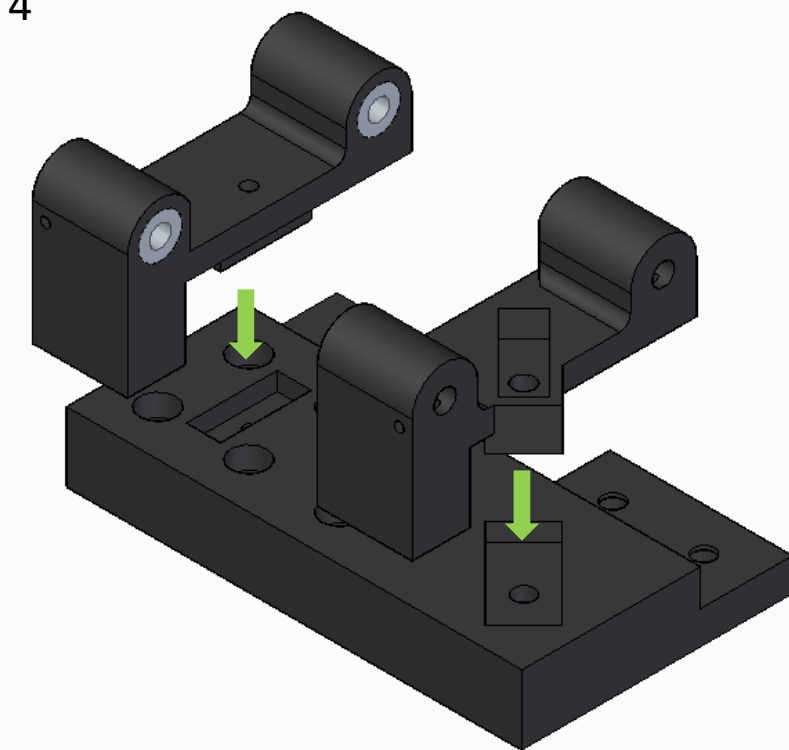
2



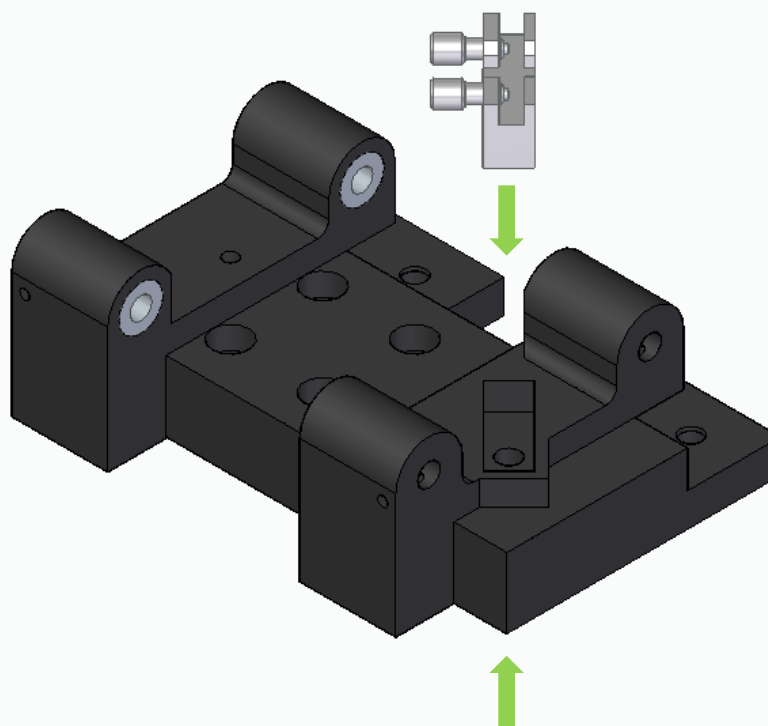
3



4

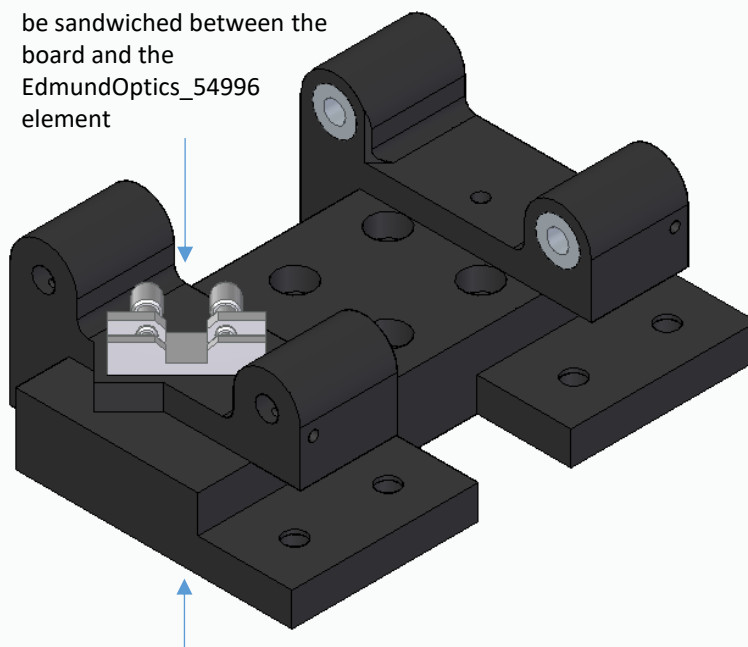


5

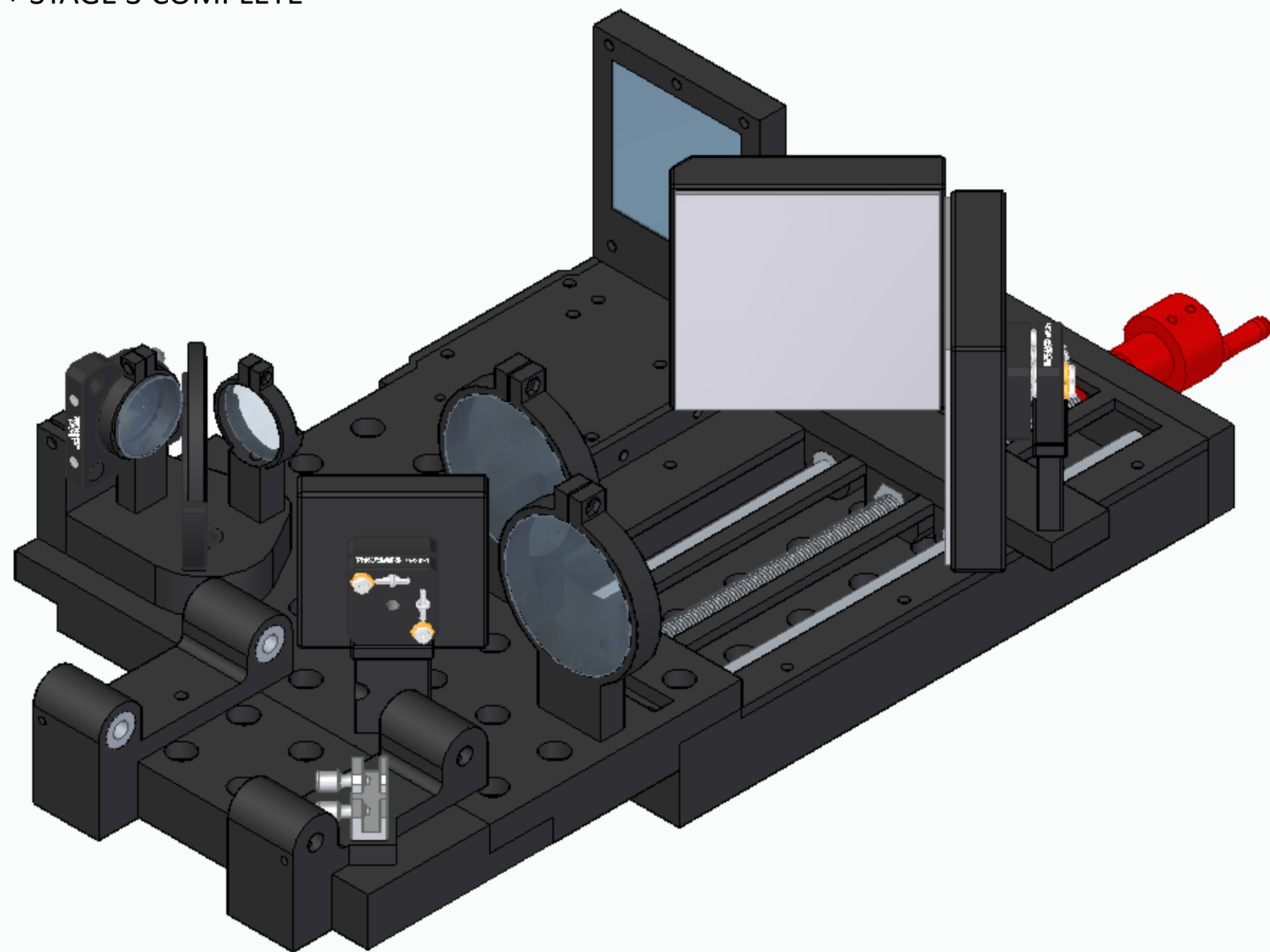


6

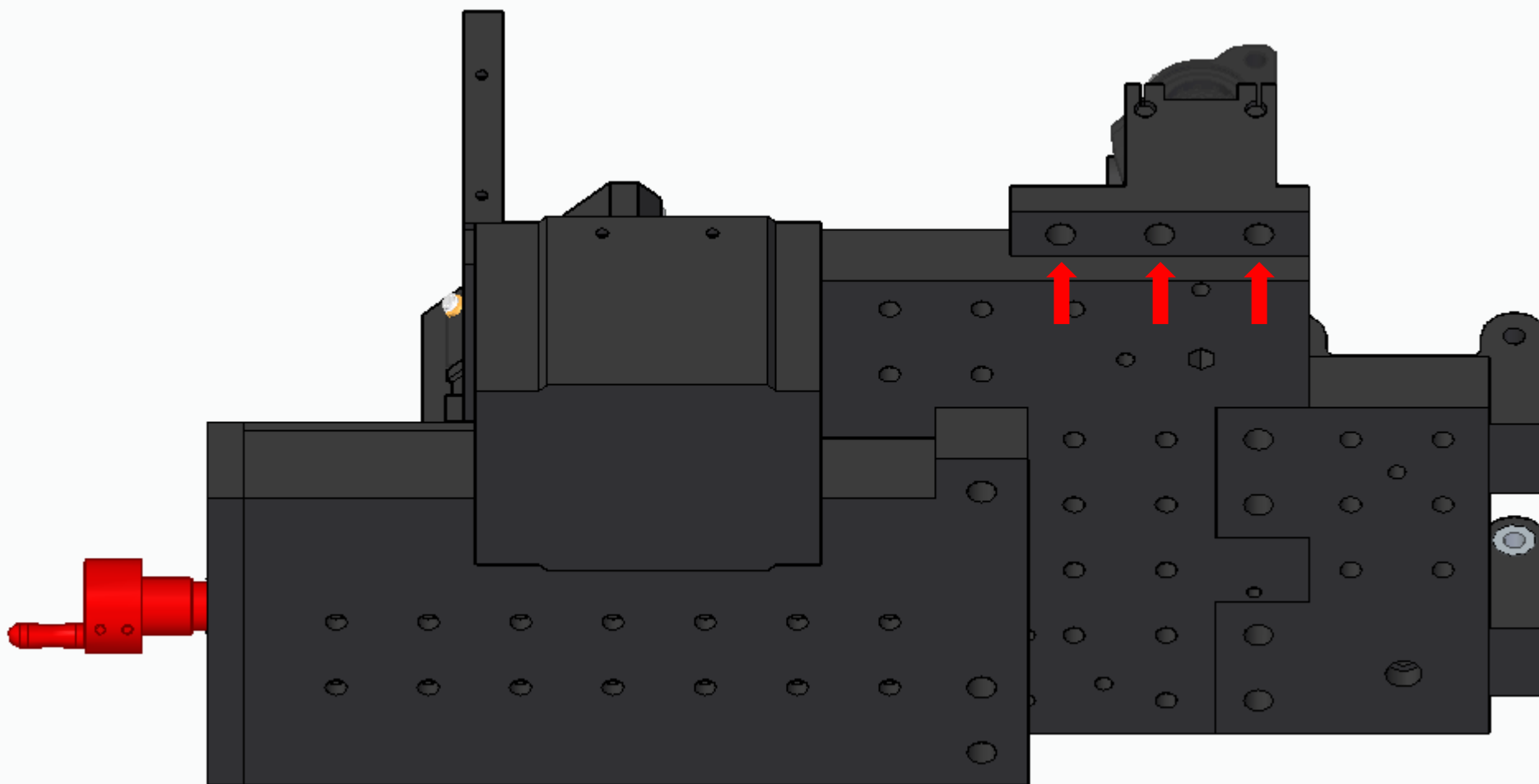
Secure with an M6 cap screw from the bottom. Element 24 will be sandwiched between the board and the EdmundOptics_54996 element



STAGE 1 + STAGE 2 + STAGE 3 COMPLETE

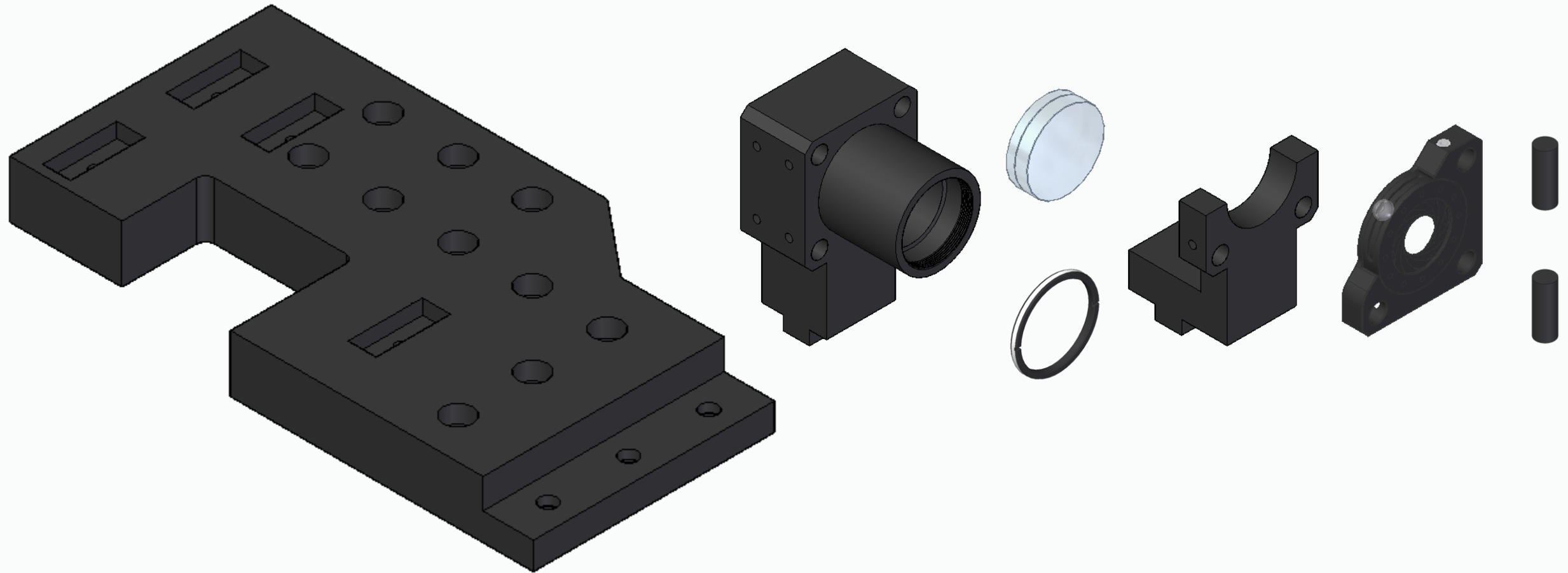


STAGE 1 + STAGE 2 + STAGE 3 COMPLETE

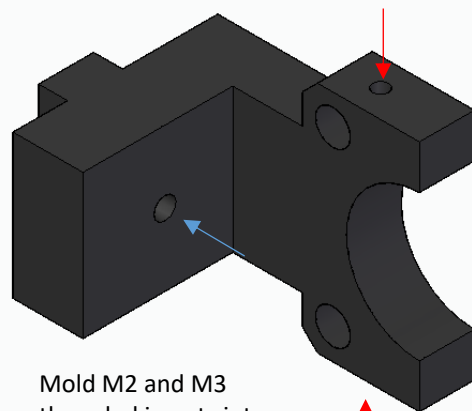
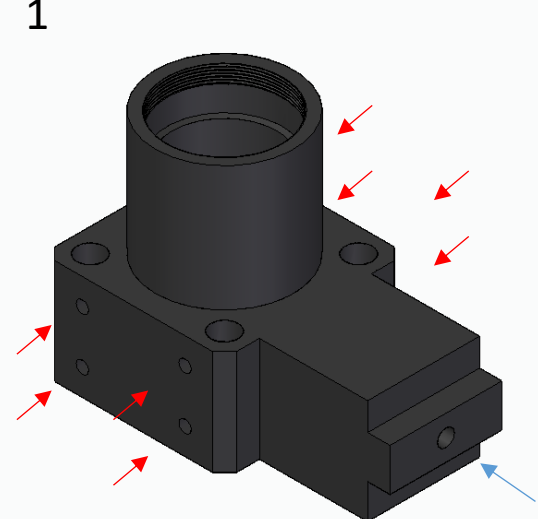


Before proceeding to STAGE 4 ensure to mold M6 threaded inserts into the holes designated by the red arrows

- 1 × 26_FDML_stimfix_stimulus_board
- 1 × 28_FDML_stimfix_stimulus_cage_system_insert
- 1 × Thorlabs_AC254-075-A (<https://www.thorlabs.com/thorproduct.cfm?partnumber=AC254-075-A>)
- 1 × Thorlabs_SM1RR (<https://www.thorlabs.com/thorproduct.cfm?partnumber=SM1RR>)
- 1 × 27_FDML_stimfix_cage_system_insert_iris_holder
- 1 × Thorlabs_CP20D (<https://www.thorlabs.com/thorproduct.cfm?partnumber=CP20D>)
- 2 × 11_FDML_stimfix_iris_mounting_rod

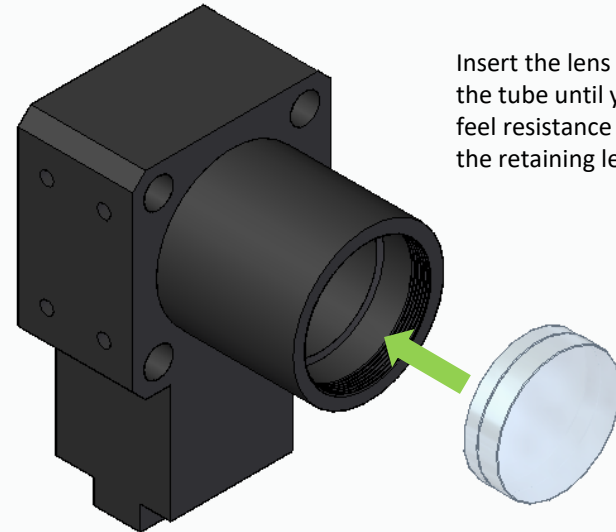


1



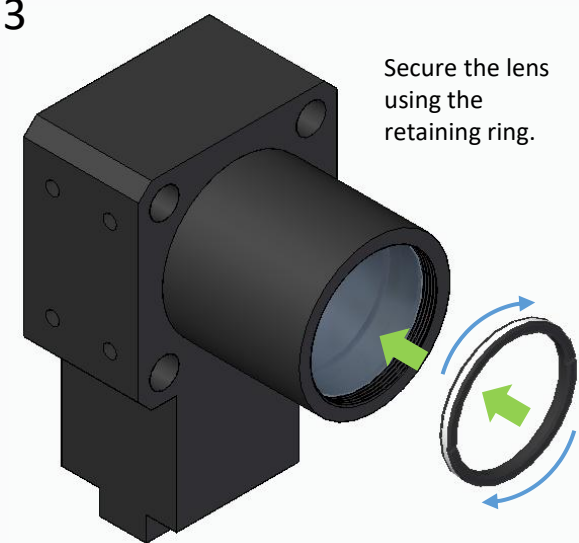
Mold M2 and M3 threaded inserts into the designated holes indicated by red arrows (M2) and blue arrows (M3).

2



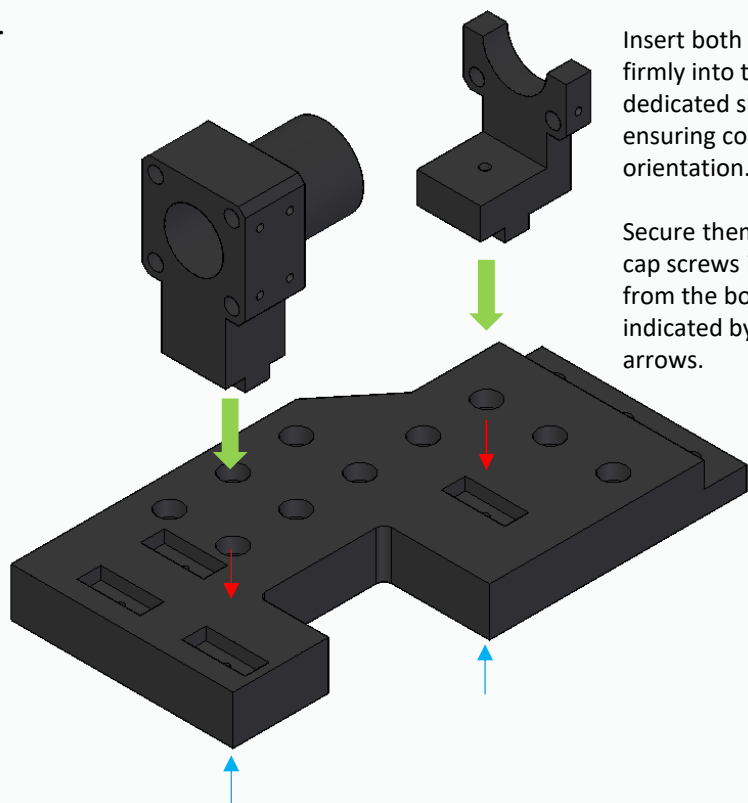
Insert the lens into the tube until you feel resistance from the retaining ledge.

3



Secure the lens using the retaining ring.

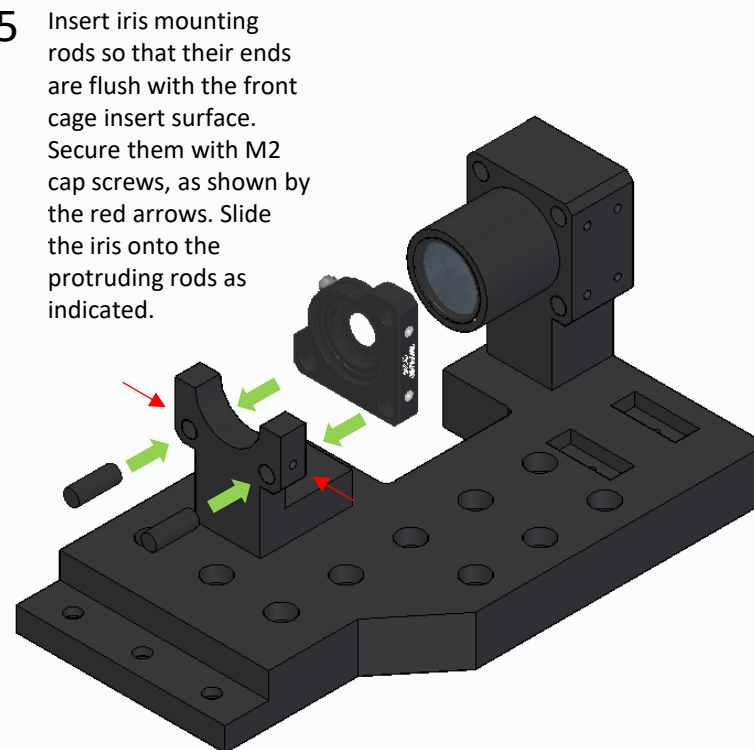
4



Insert both cage inserts firmly into the dedicated slots, ensuring correct orientation.

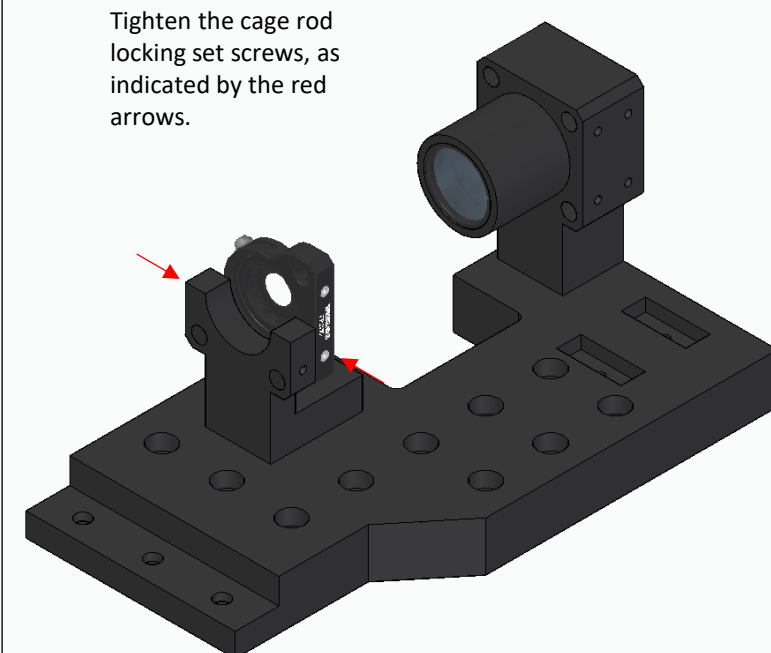
Secure them with M3 cap screws inserted from the bottom, as indicated by the blue arrows.

5



Insert iris mounting rods so that their ends are flush with the front cage insert surface. Secure them with M2 cap screws, as shown by the red arrows. Slide the iris onto the protruding rods as indicated.

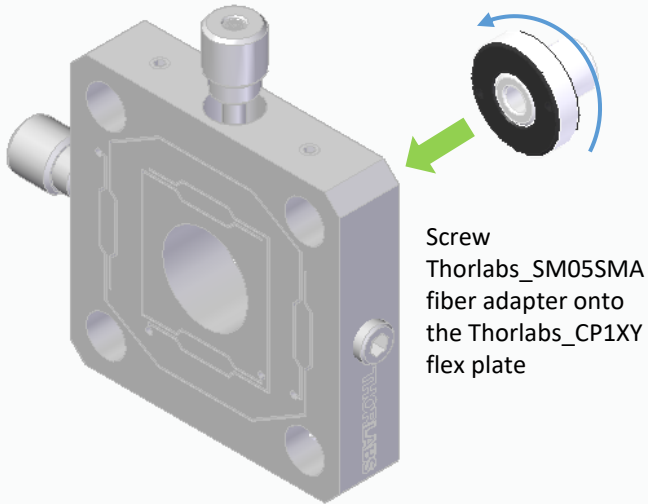
6



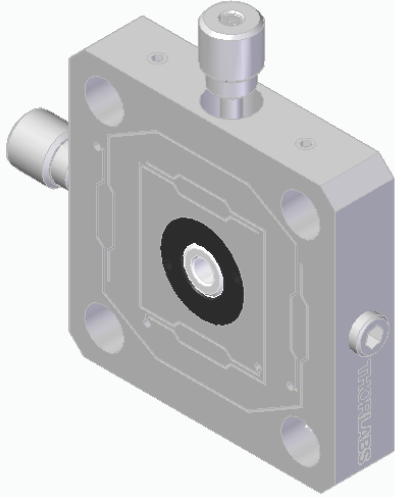
Tighten the cage rod locking set screws, as indicated by the red arrows.

[illegible]

1

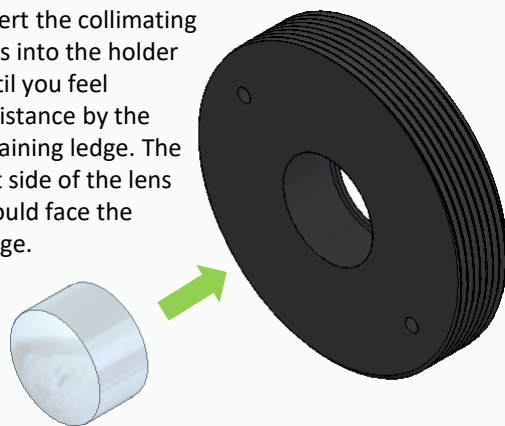


2

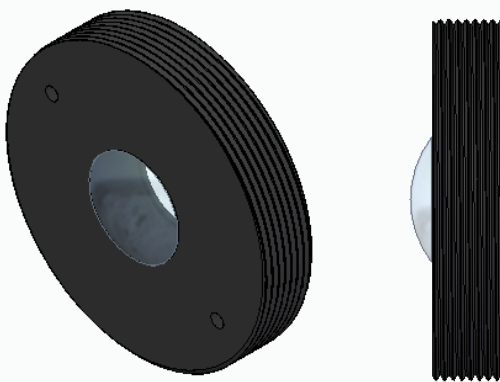


3

Insert the collimating lens into the holder until you feel resistance by the retaining ledge. The flat side of the lens should face the ledge.

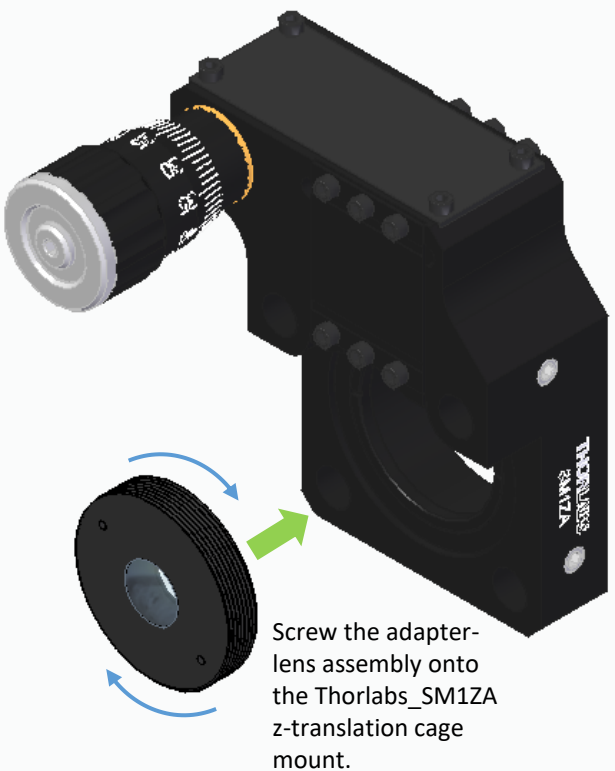


4

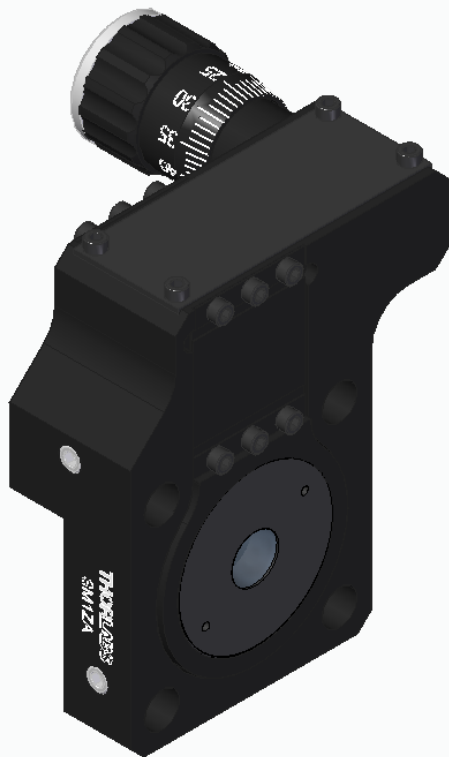


Confirm the lens is fully seated.

5

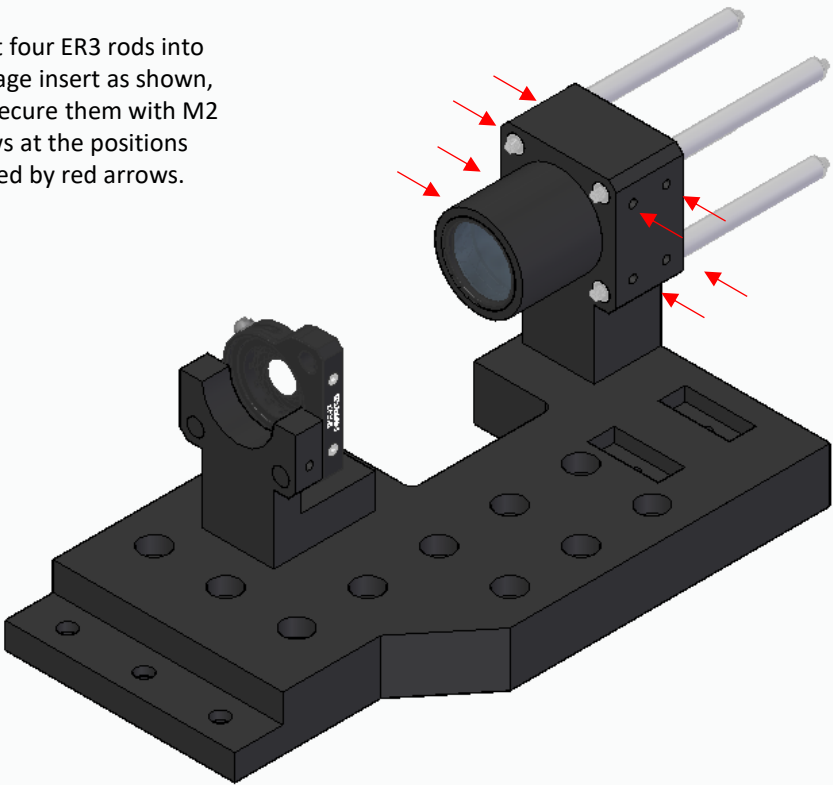


6



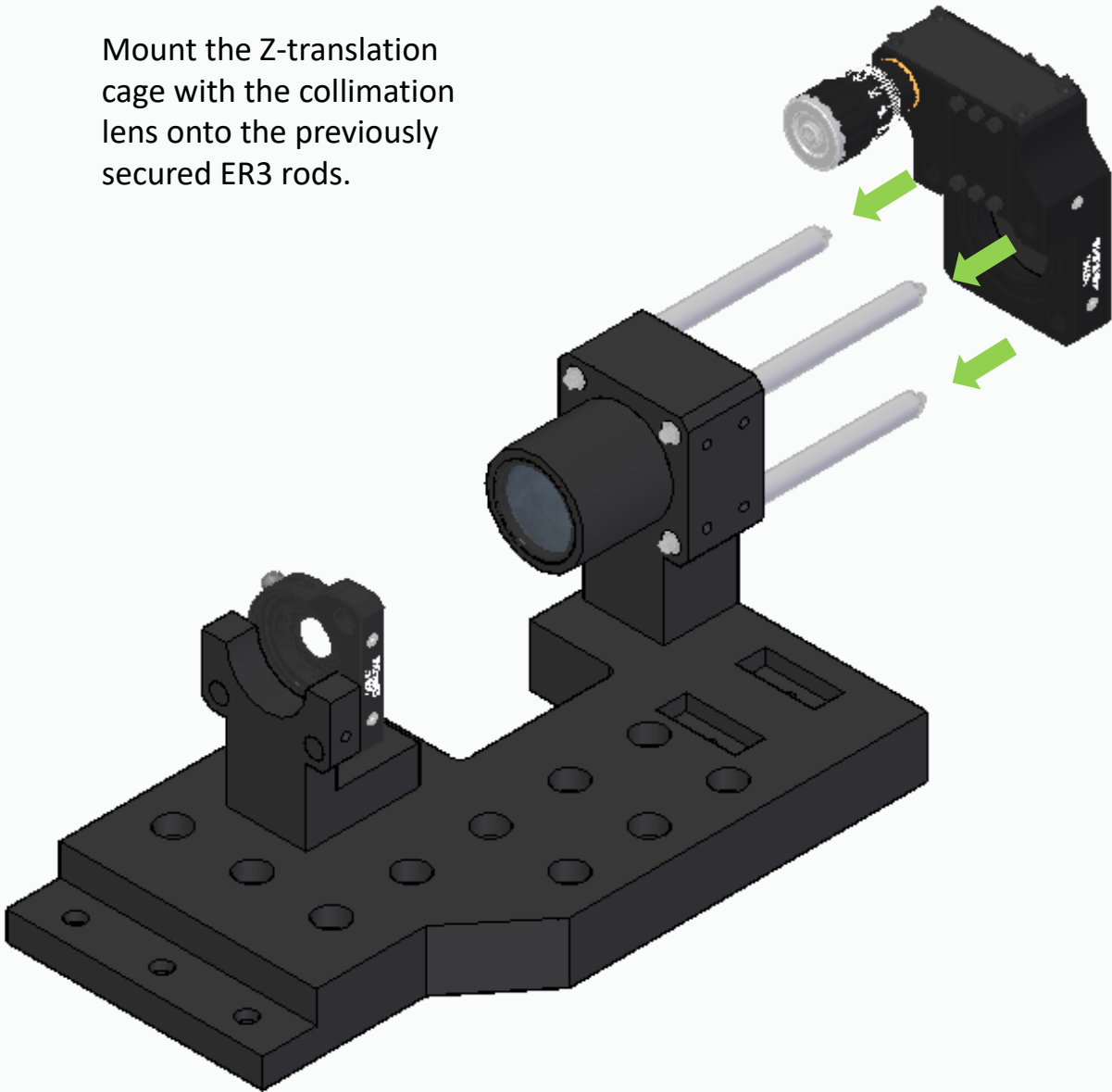
7

Insert four ER3 rods into the cage insert as shown, and secure them with M2 screws at the positions marked by red arrows.



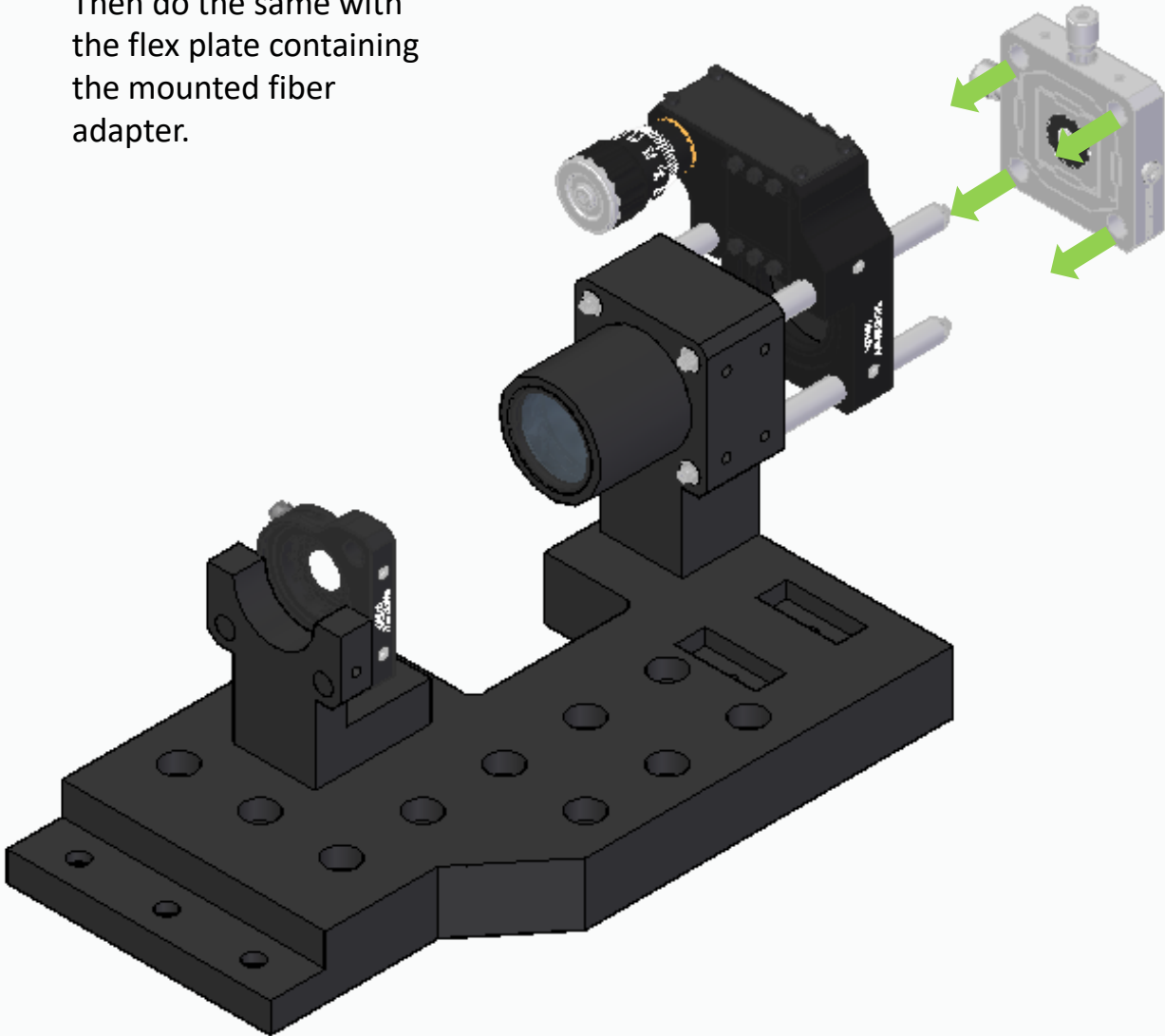
1

Mount the Z-translation cage with the collimation lens onto the previously secured ER3 rods.

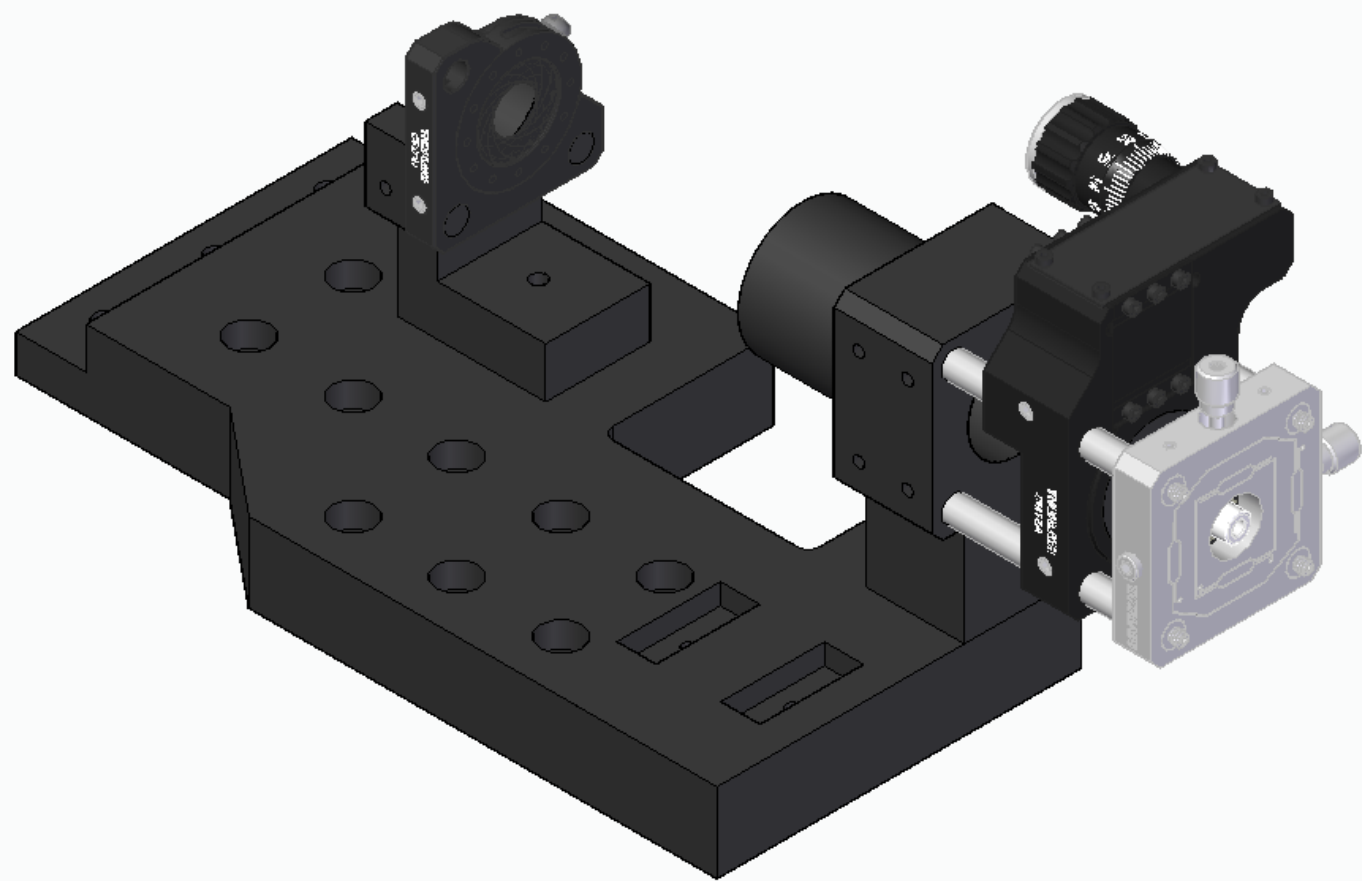
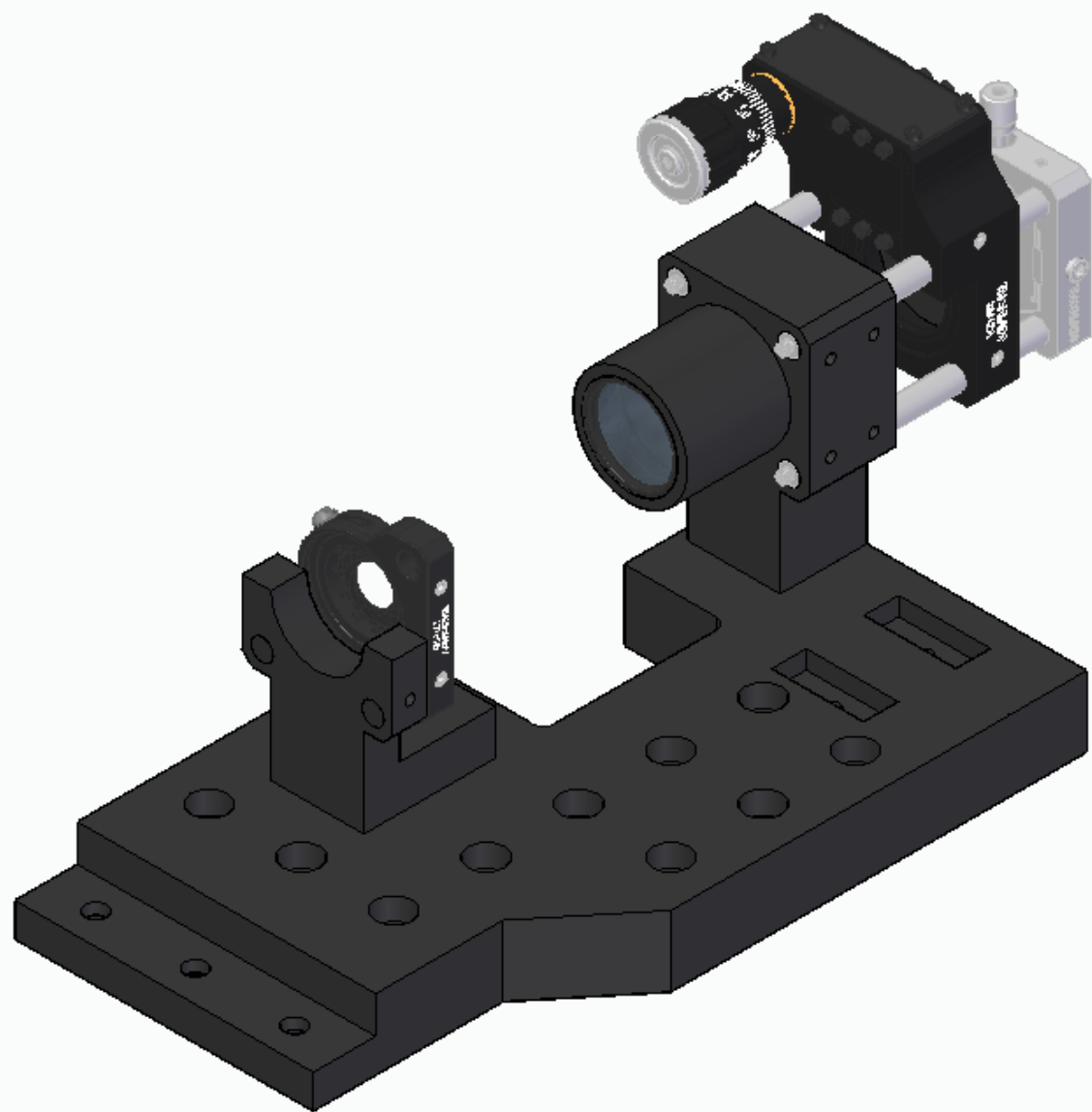


2

Then do the same with the flex plate containing the mounted fiber adapter.

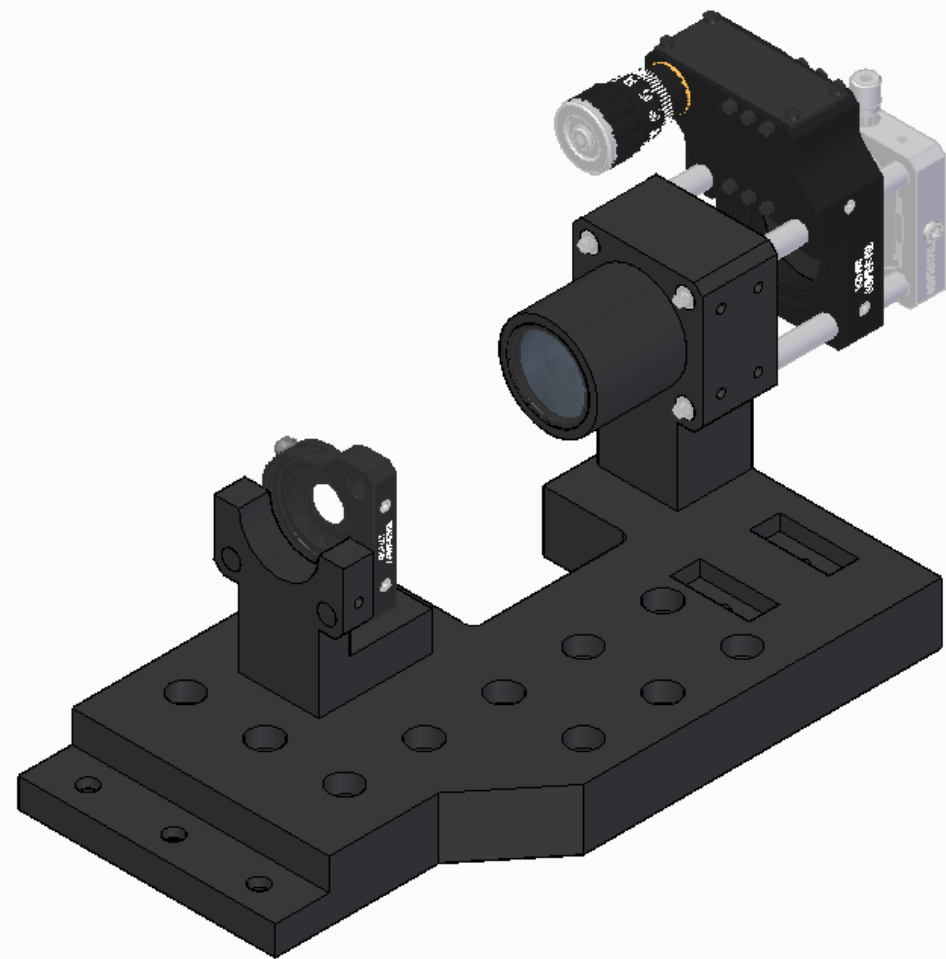
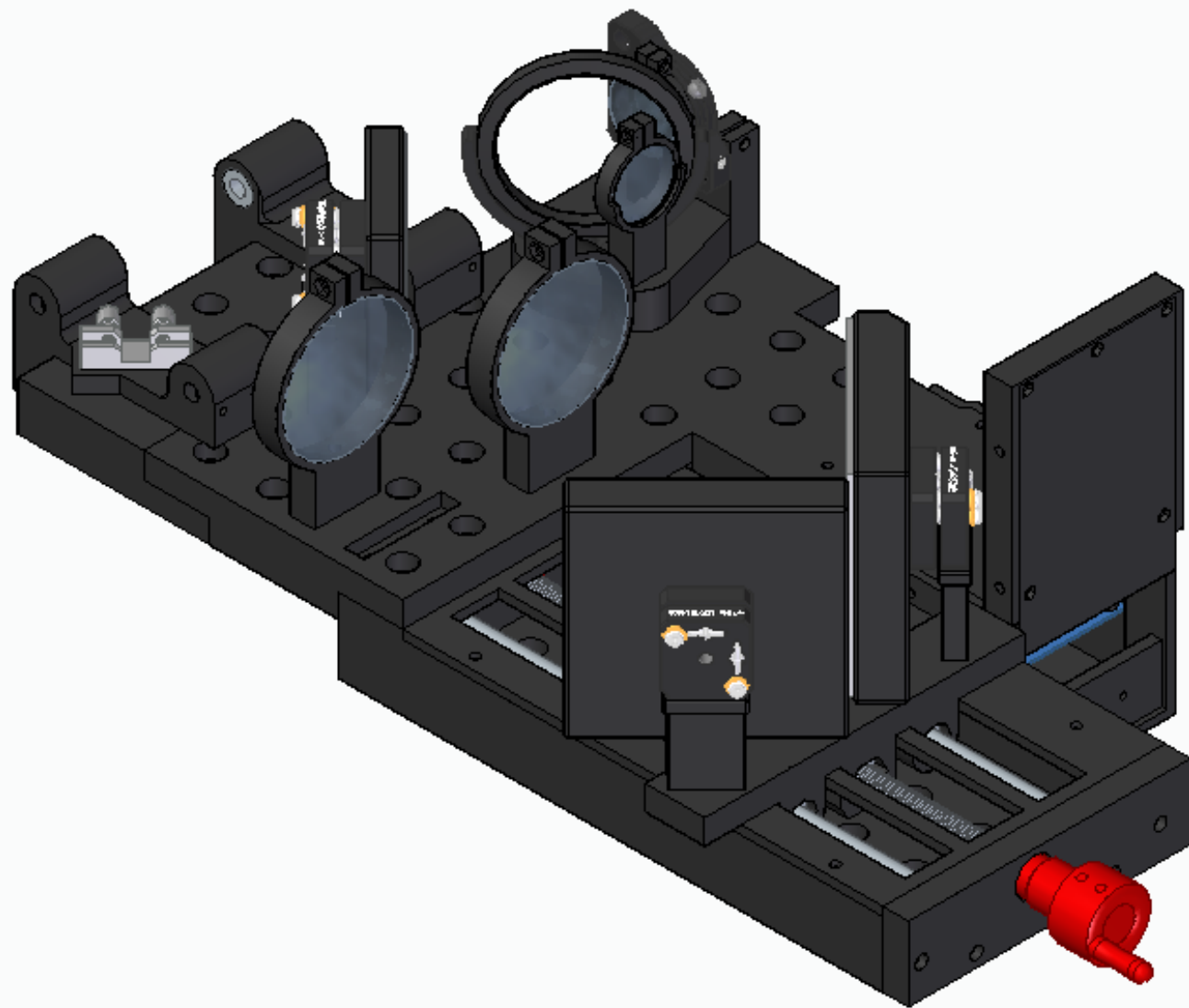


STAGE 4 COMPLETE

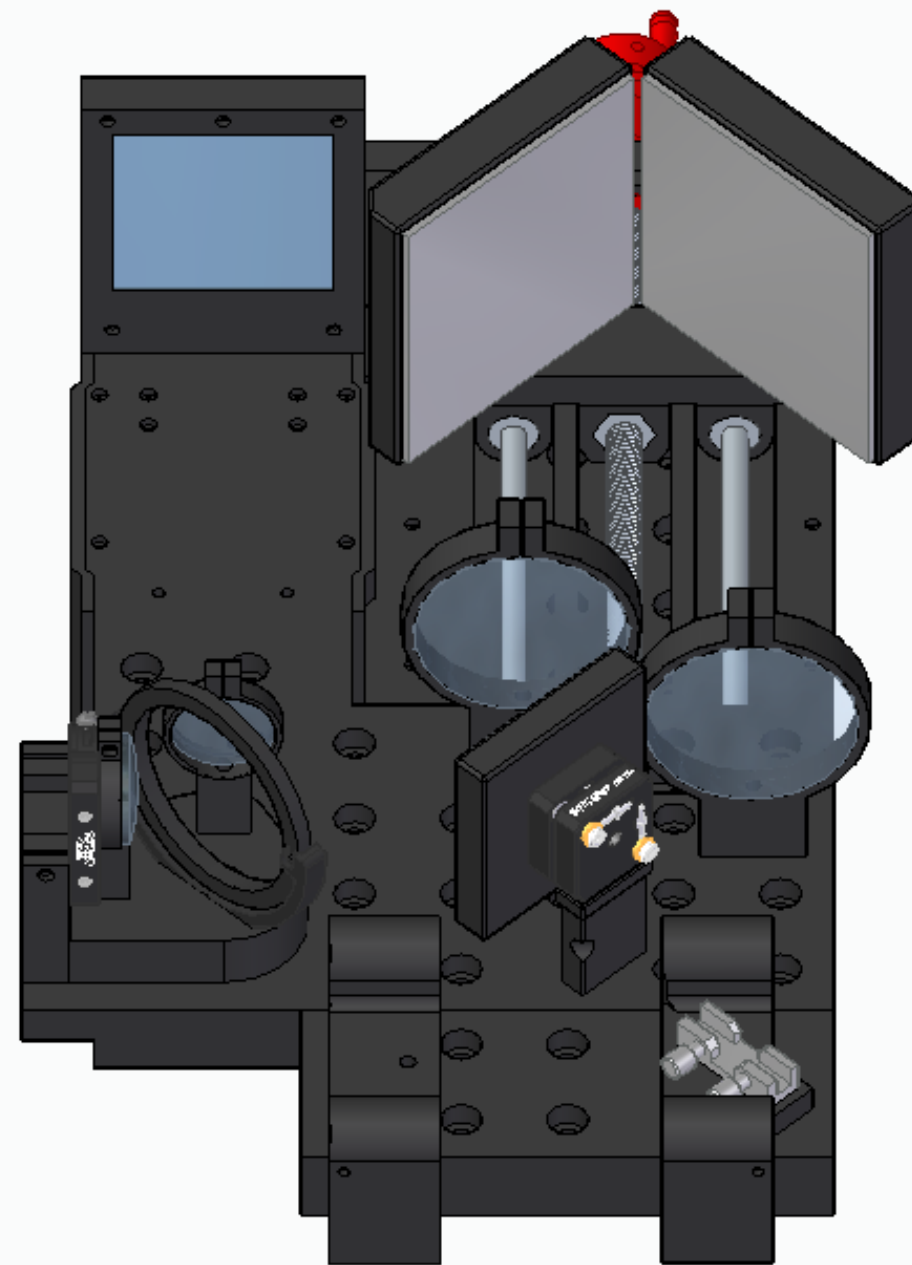
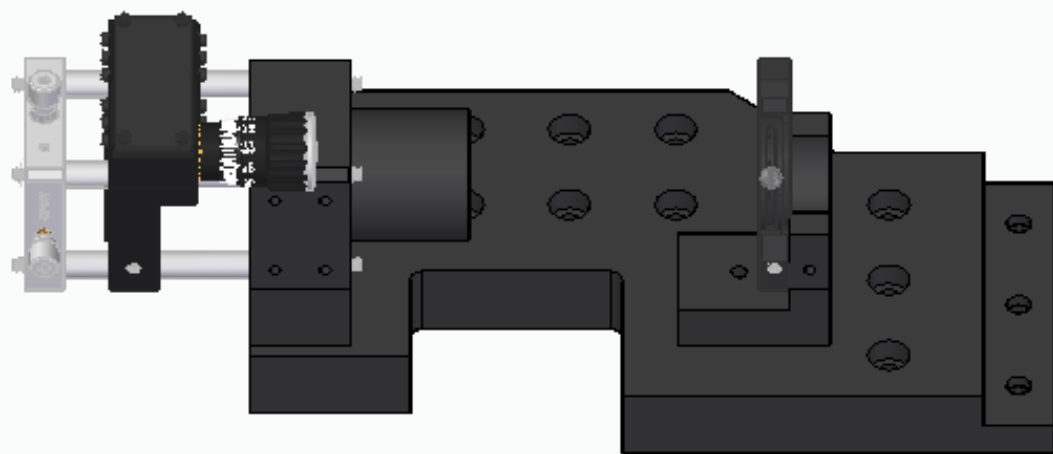


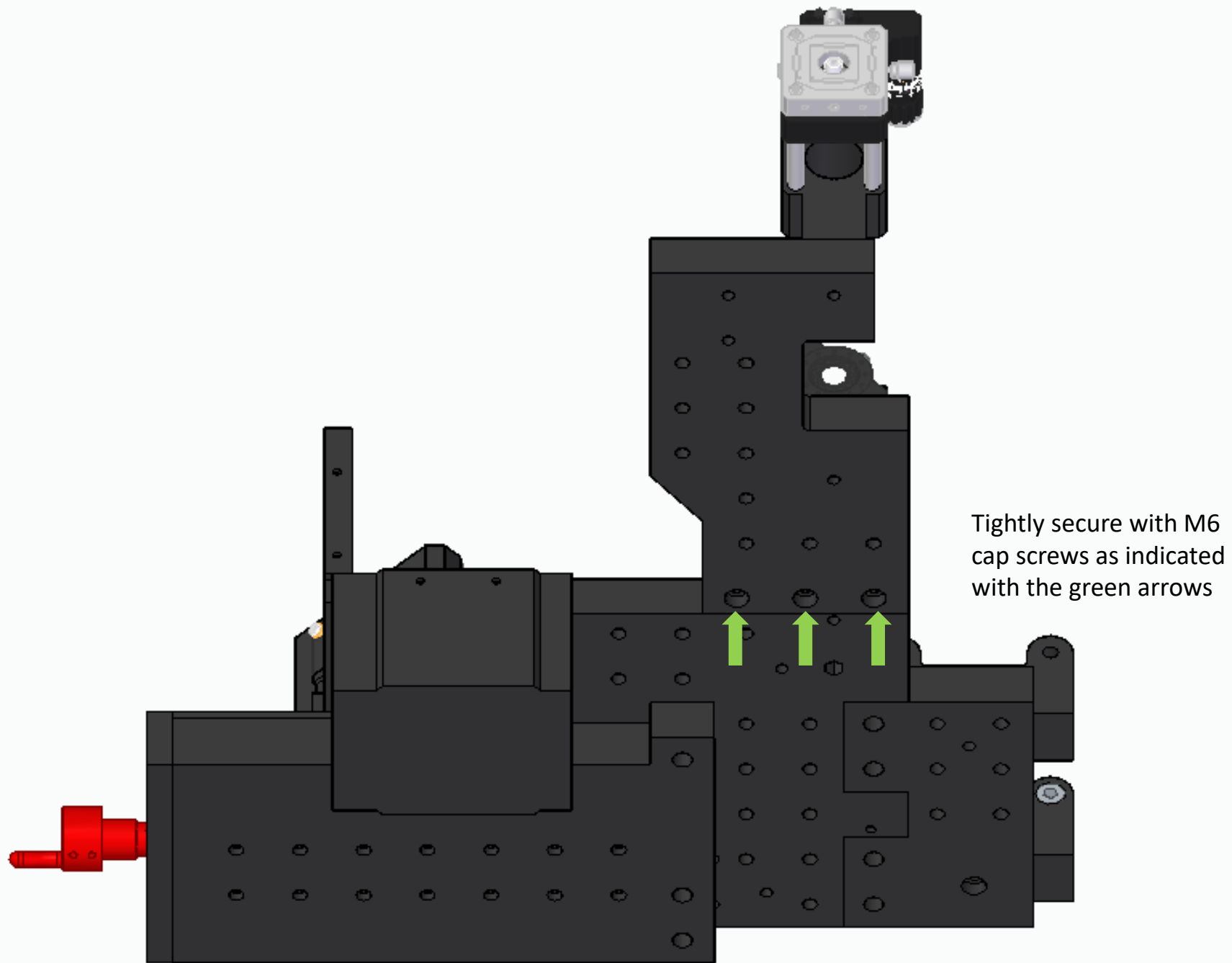
1 × STAGE 1 + STAGE 2 + STAGE 3 assembly

1 × STAGE 4 assembly

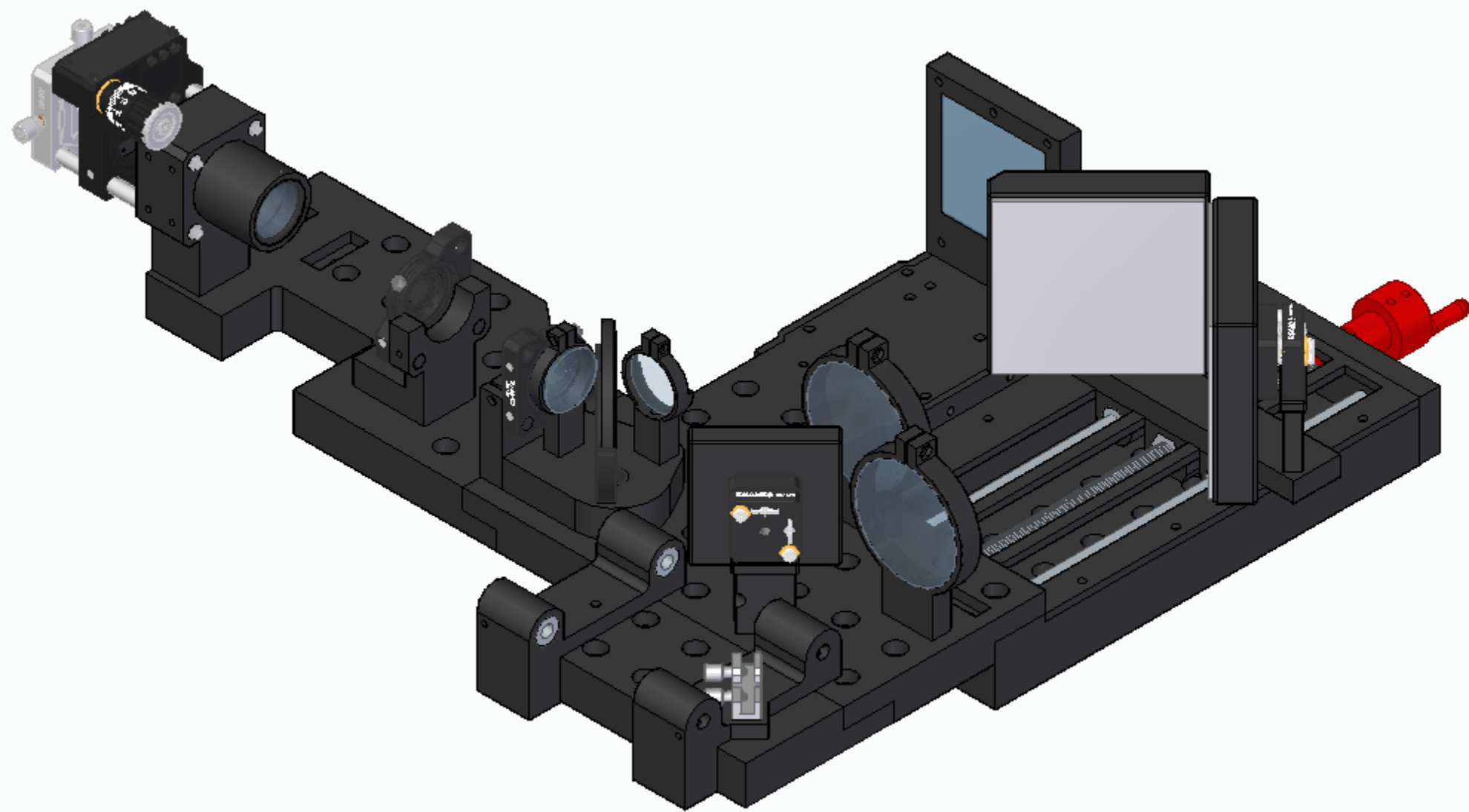


Slide the STAGE 4 assembly into the rest of the setup, aligning it with the matching slots as indicated.

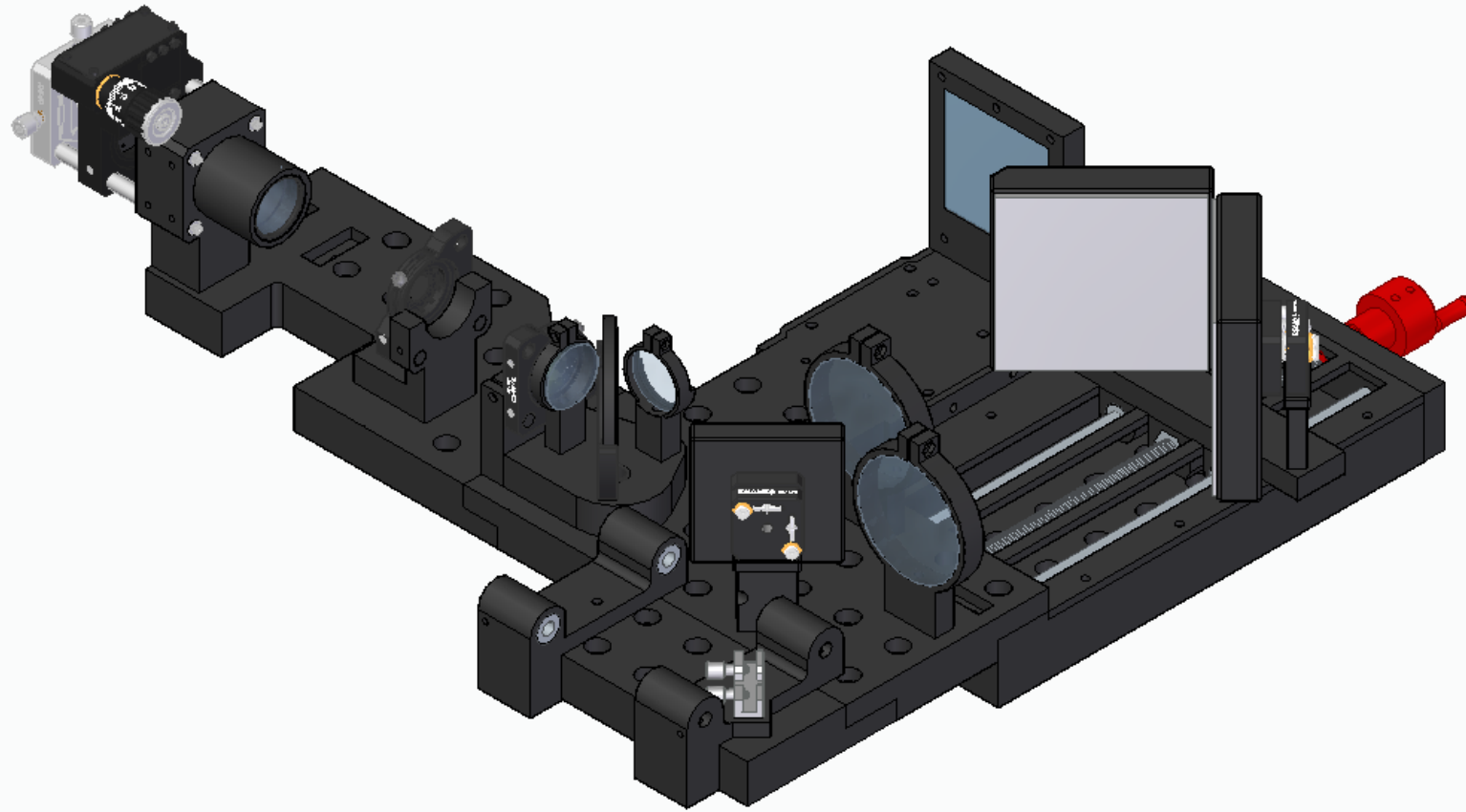




STAGE 1 + STAGE 2 + STAGE 3 + STAGE 4 COMPLETE



STAGE 1 + STAGE 2 + STAGE 3 + STAGE 4
30_FDML_stimfix_mounting_post_long
31_FDML_stimfix_mounting_post_short

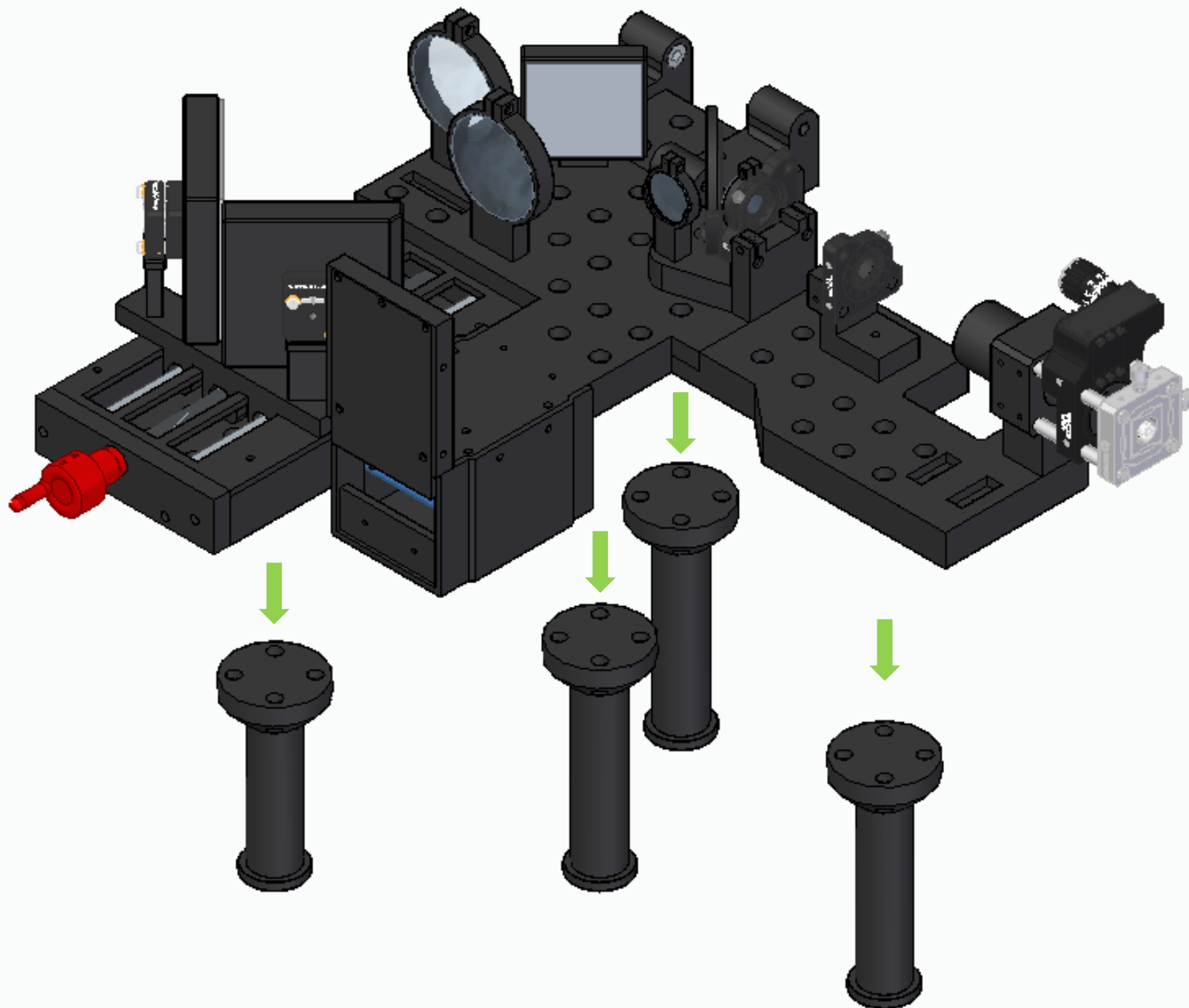


Mold M6 threaded inserts into each hole at the top of each post (insert from the top side).

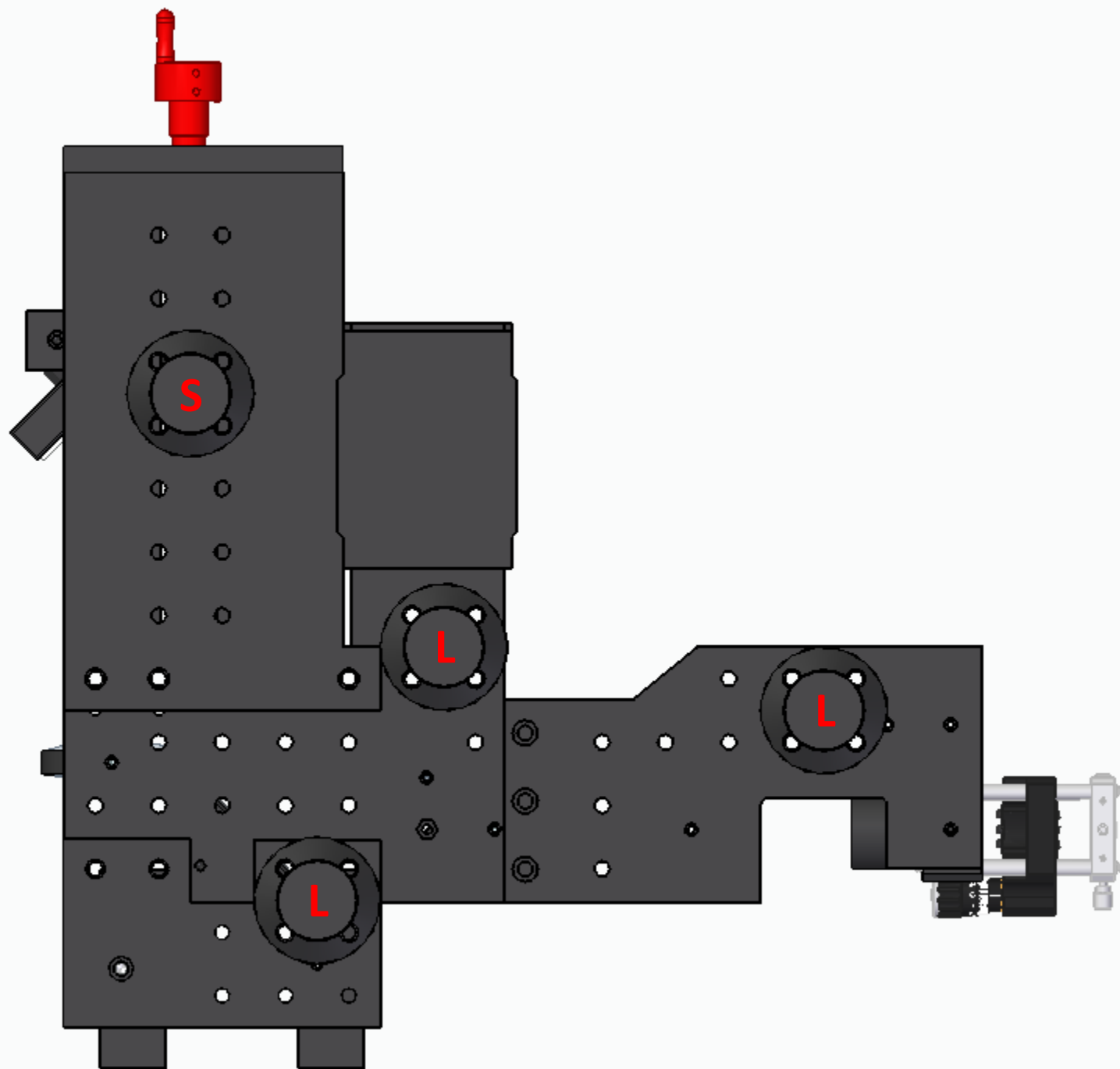


The full assembly is supported by three longer posts and one shorter post (supporting the Badal stage). Posts can be secured to the optical table using standard optical forks. Secure the assembly to the posts using M6 cap screws inserted through the holes in the assembly, threading into the molded inserts at the top of each post.

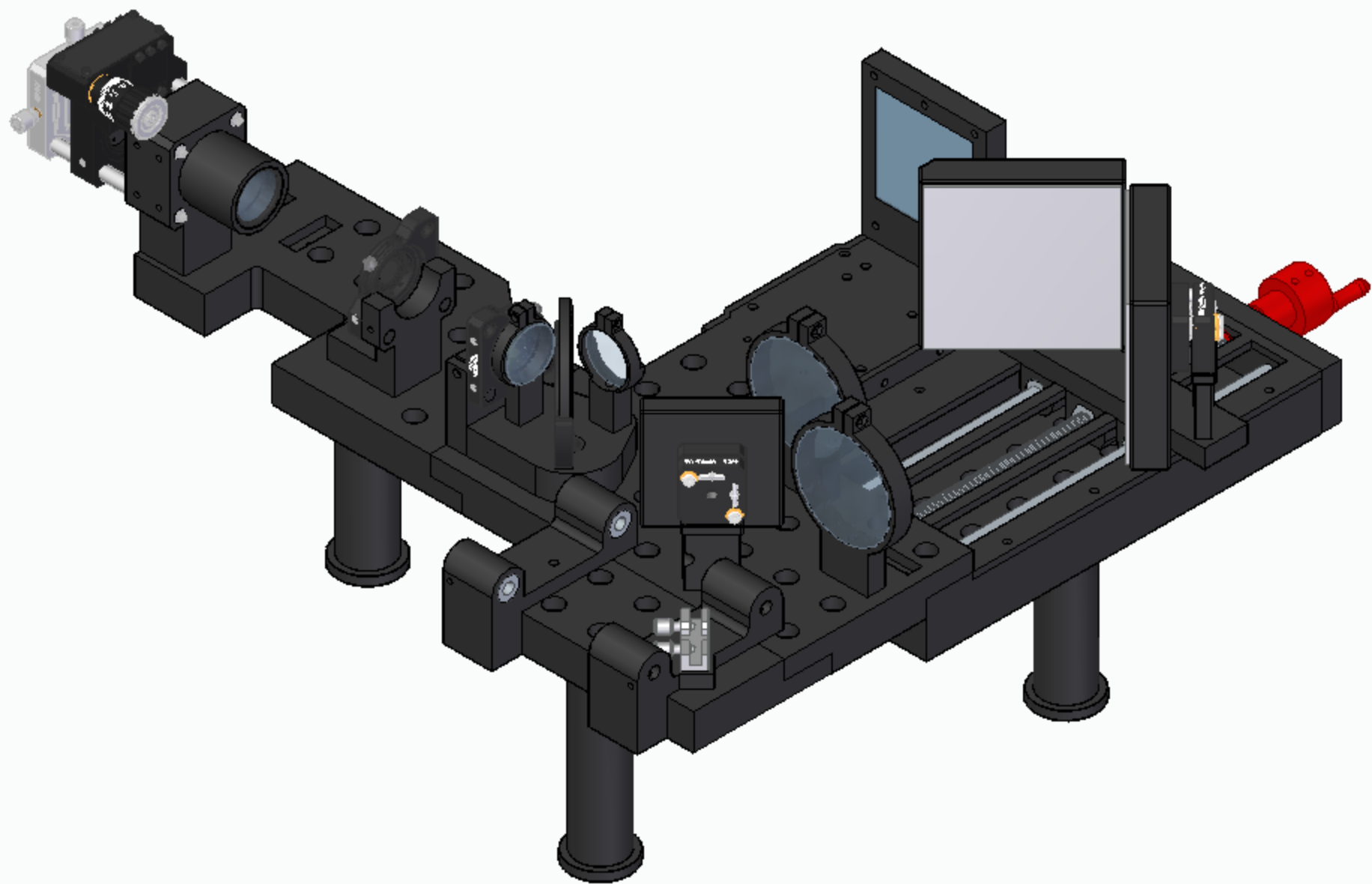
Suggested post positions are shown on the next slide, which presents a bottom view of the assembly with precise post placement labeled (L = long, S = short).



L = long post
S = short post



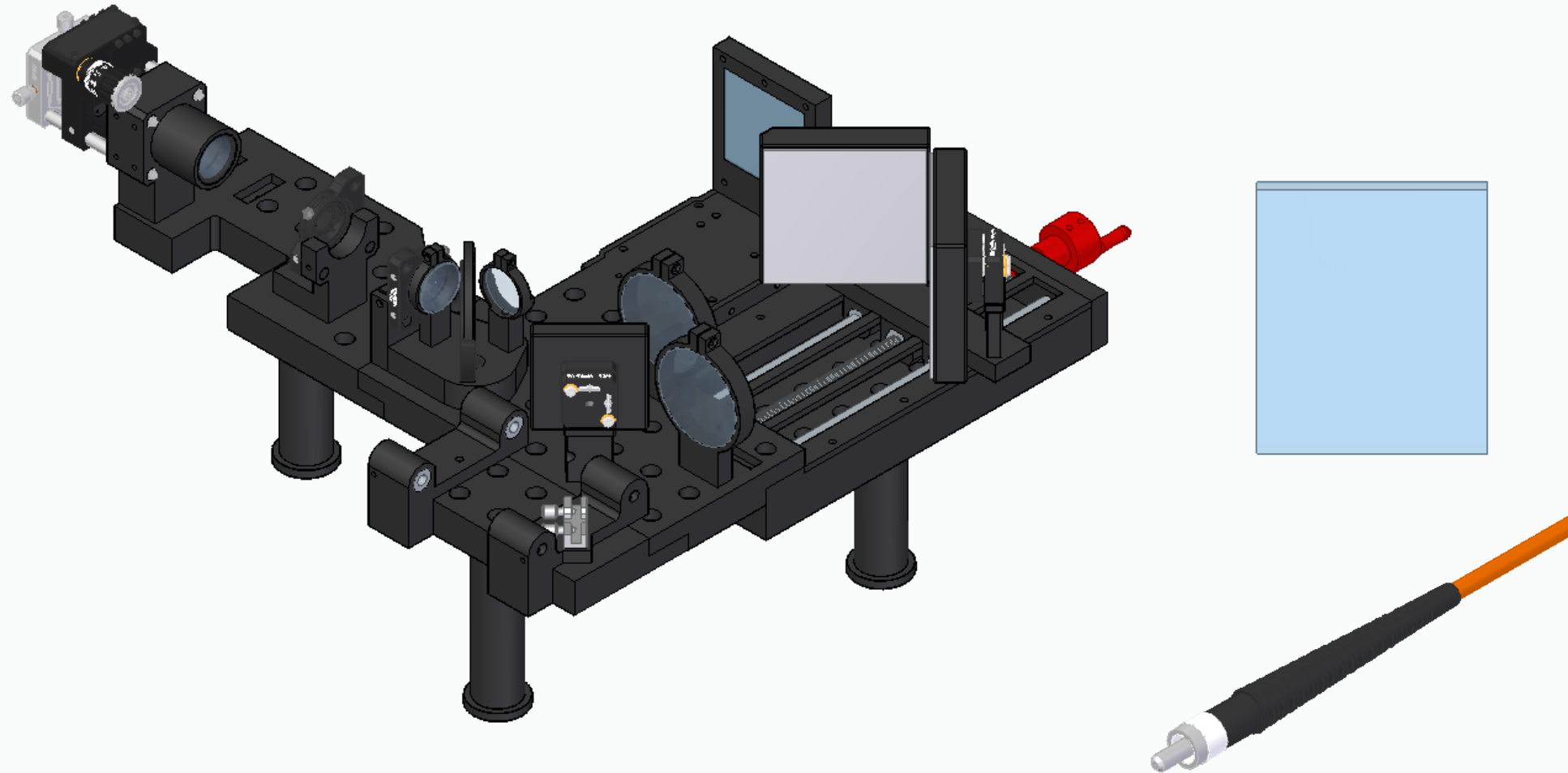
STAGE 1 + STAGE 2 + STAGE 3 + STAGE 4 + STAGE 5 COMPLETE



STAGE 1 + STAGE 2 + STAGE 3 + STAGE 4 + STAGE 5 assembly

1 × Thorlabs_M45L01 (<https://www.thorlabs.com/thorproduct.cfm?partnumber=M45L01>)

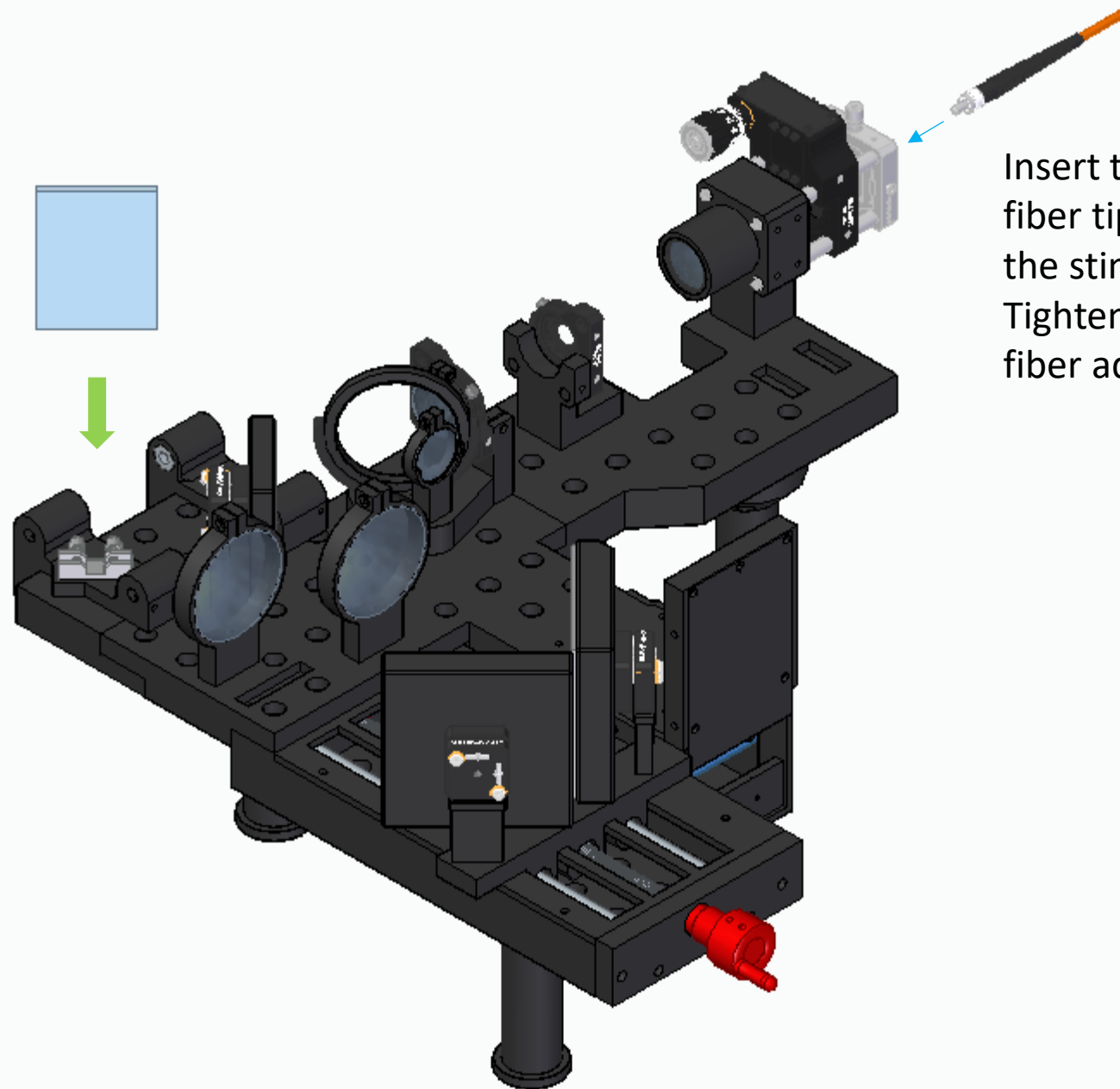
1 × Chroma_dichroic_mirror (**PART NUMBER MISSING**)



Insert the dichroic mirror into designated holder and secure using the screws in the holder.

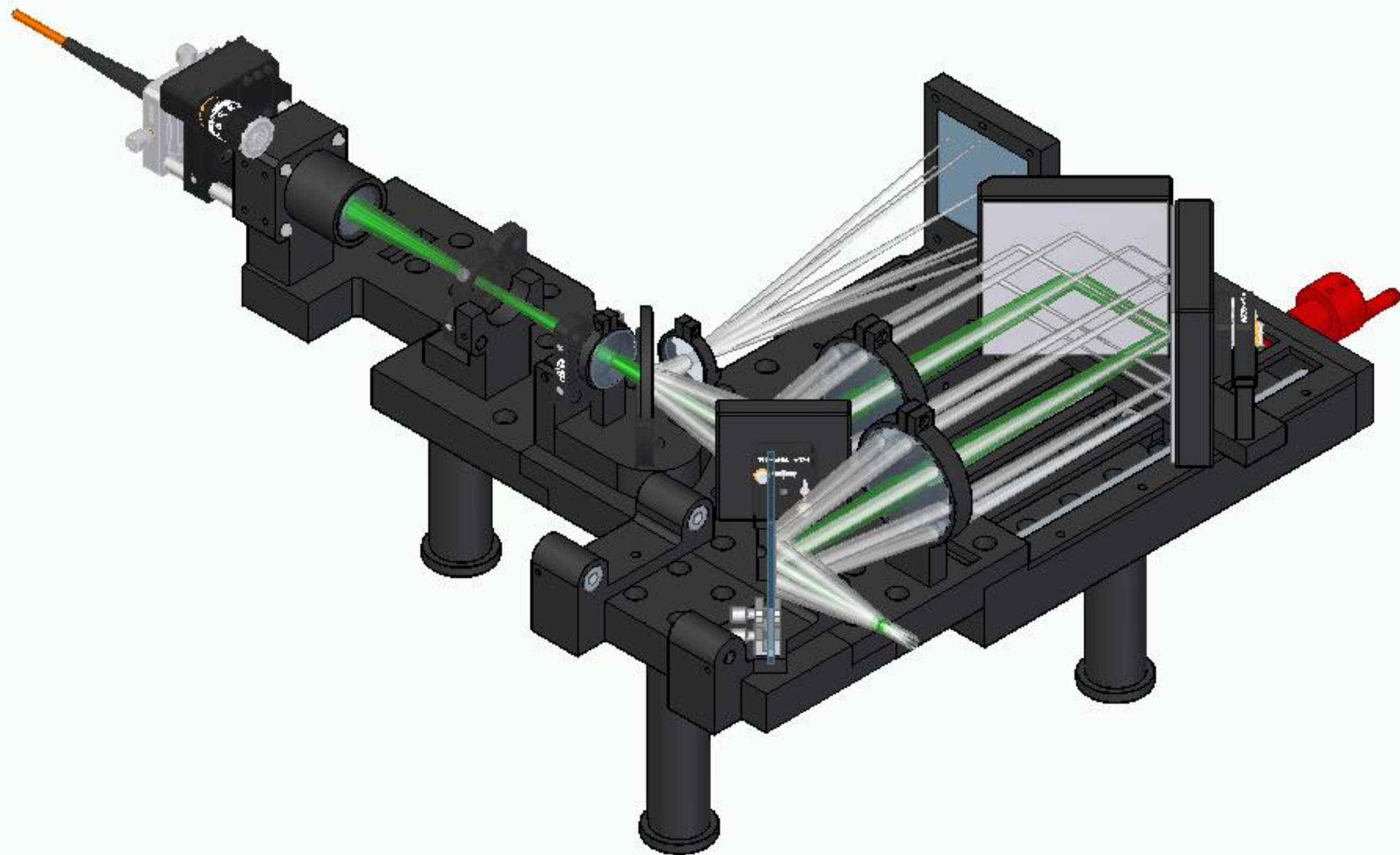


After inserting the fiber and the dichroic mirror you may proceed to final alignment of the optics. The stimulus-fixation channel is now ready.

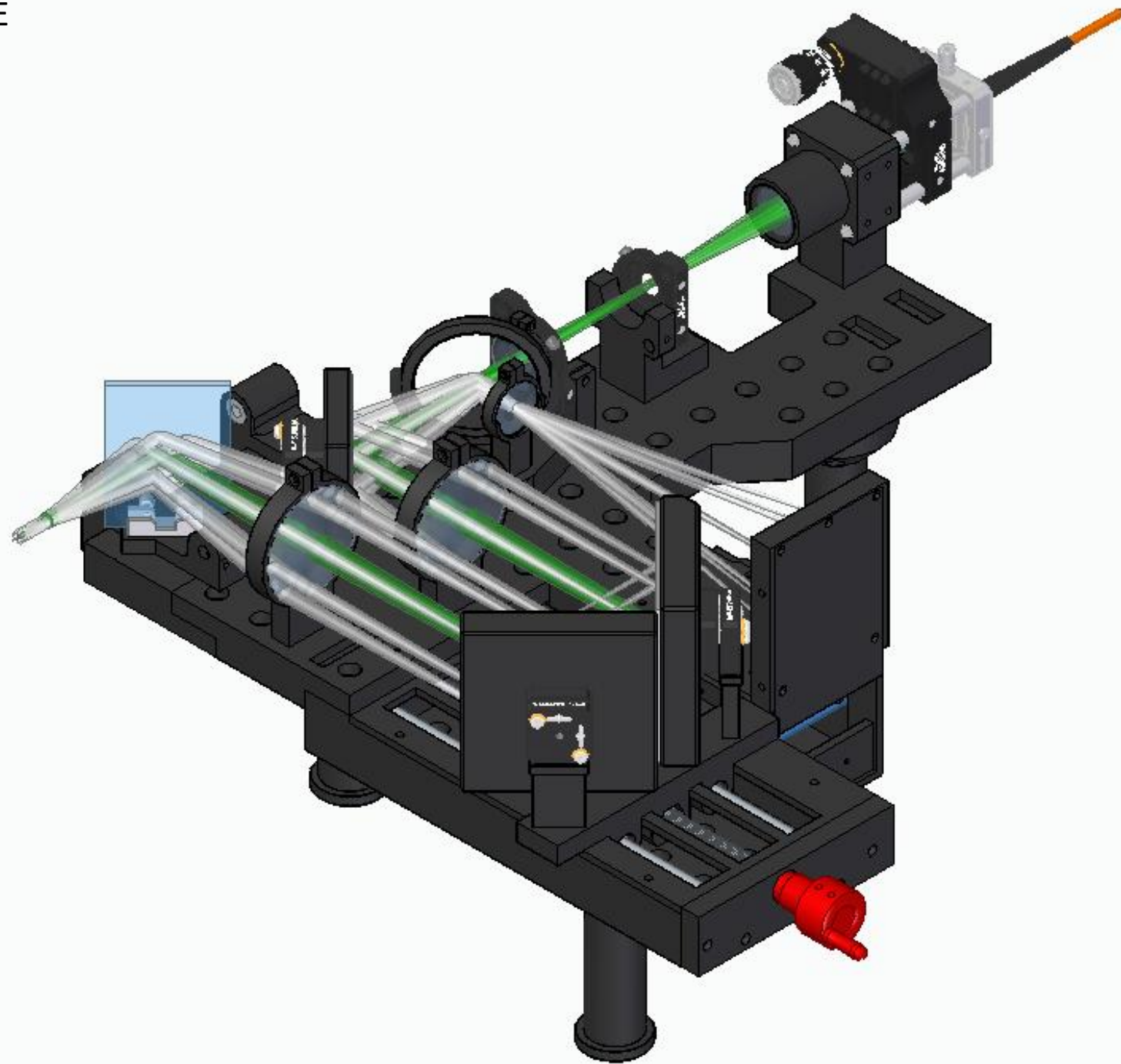


Insert the multimode fiber tip into the entry of the stimulus channel. Tighten the screw to the fiber adapter.

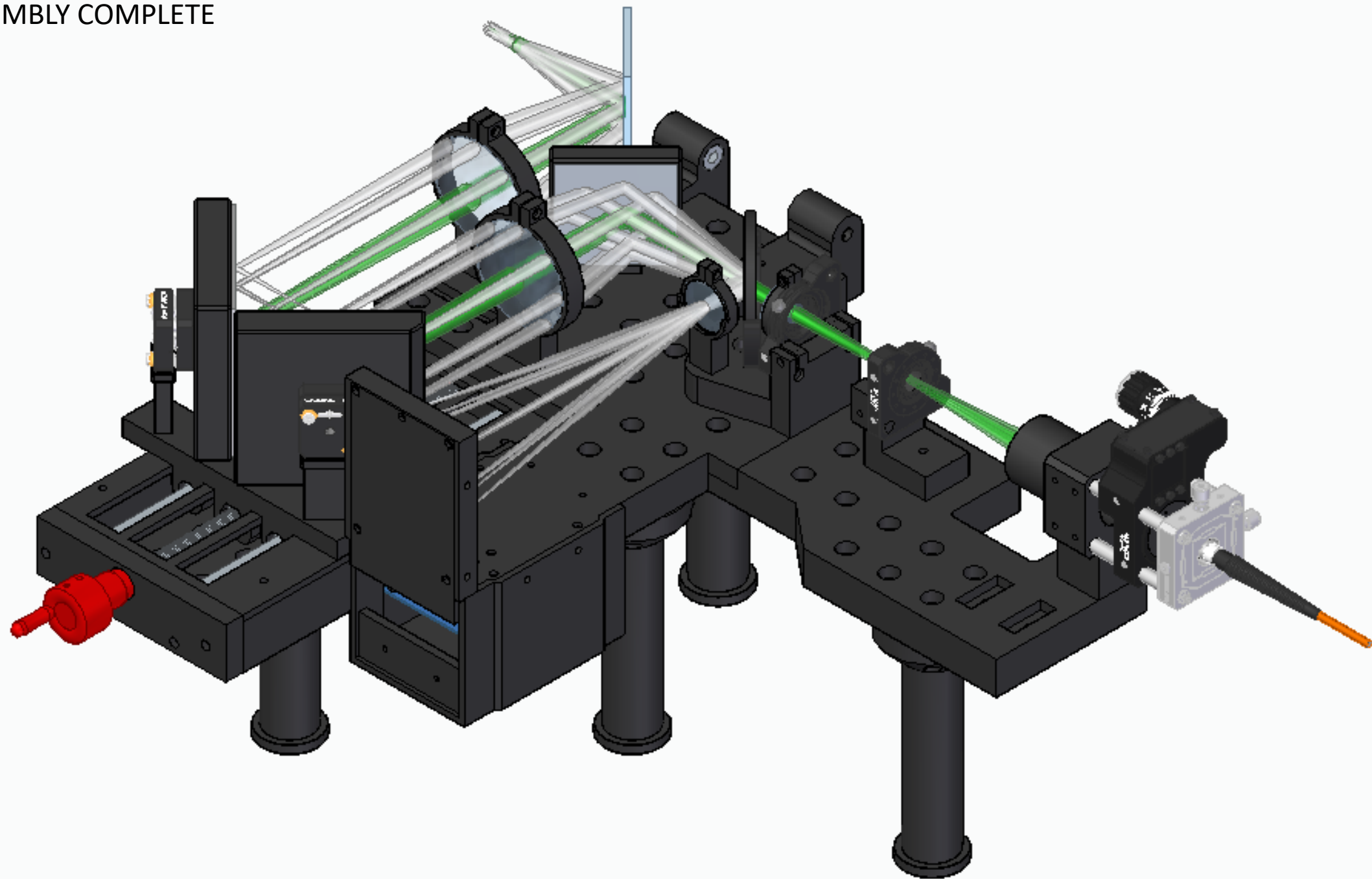
FULL ASSEMBLY COMPLETE



FULL ASSEMBLY COMPLETE



FULL ASSEMBLY COMPLETE



FULL ASSEMBLY COMPLETE

